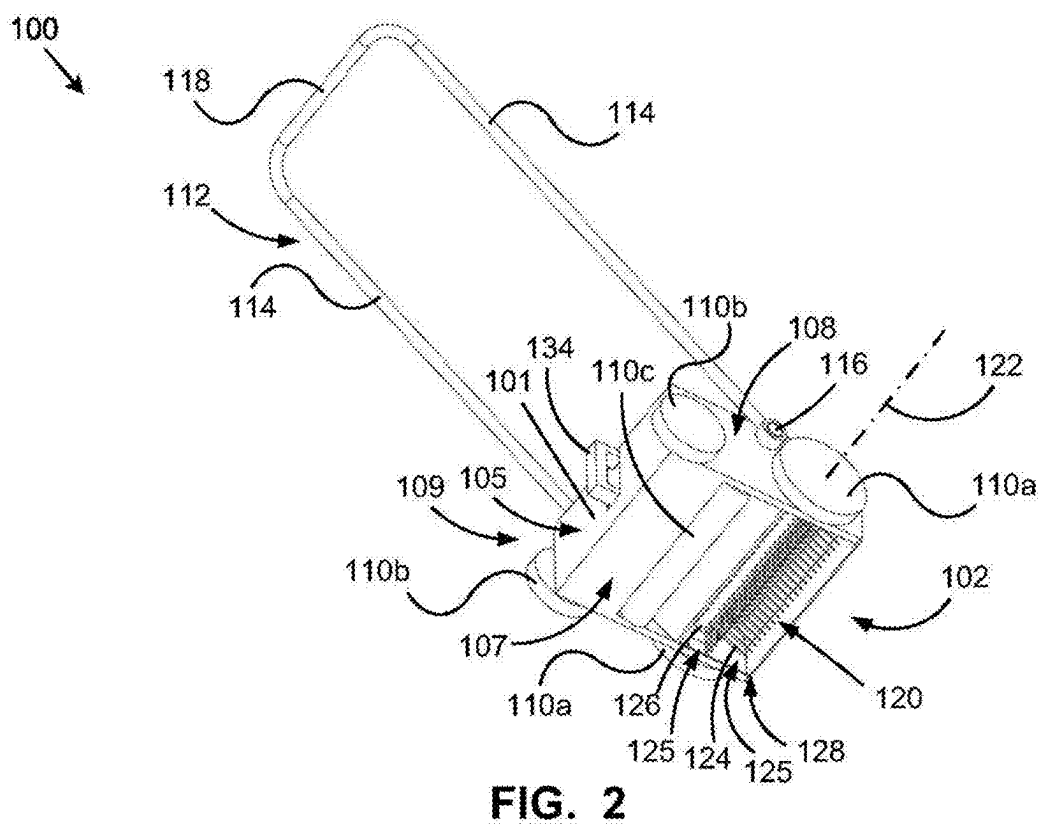
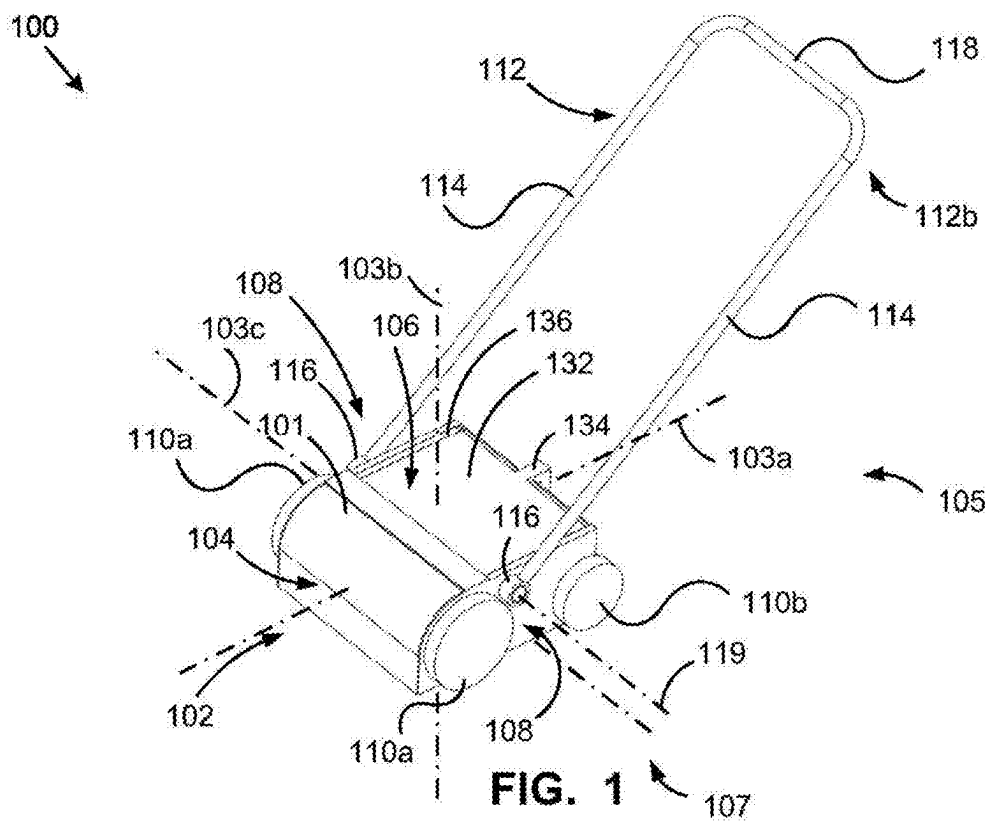




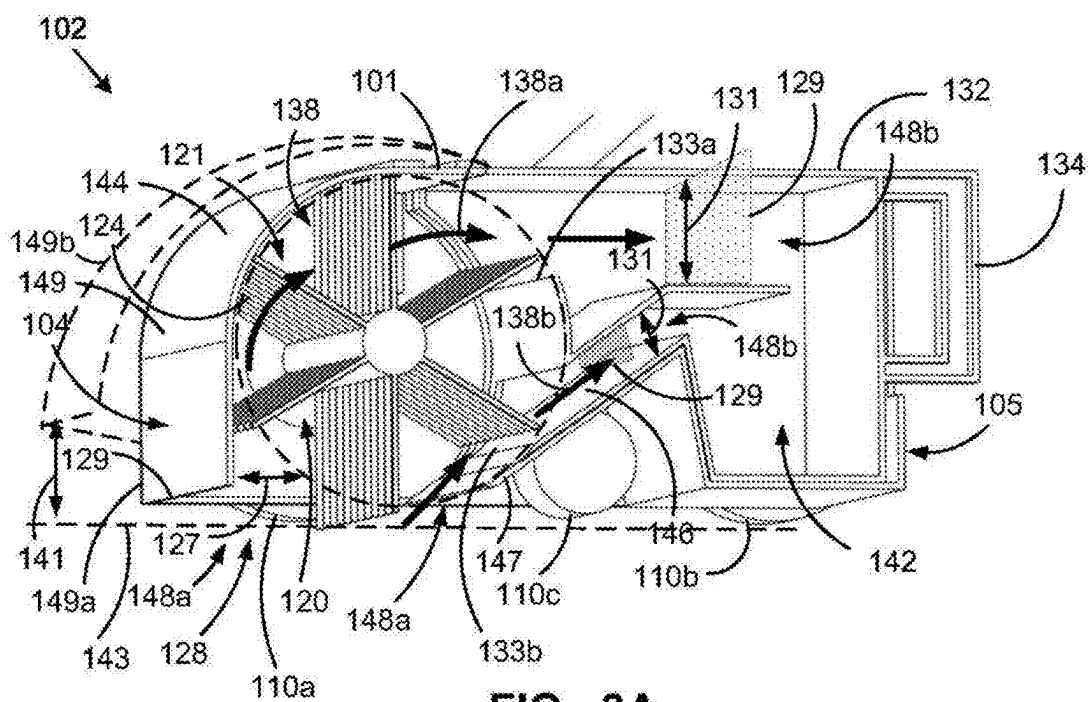


FIG. 3A

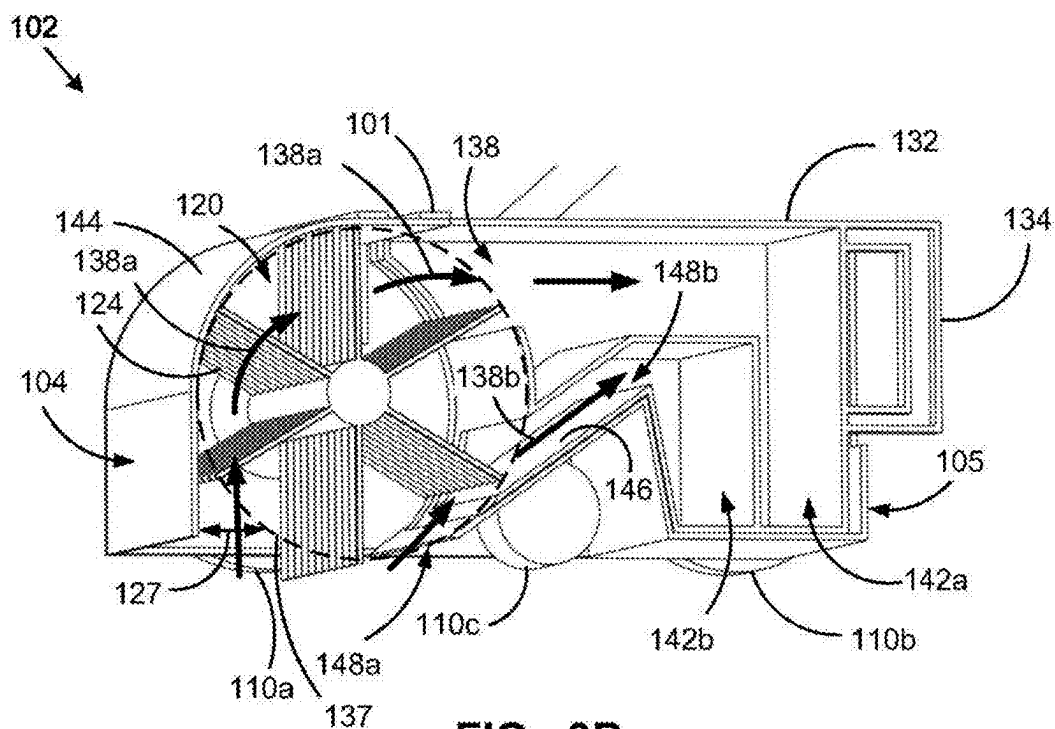


FIG. 3B

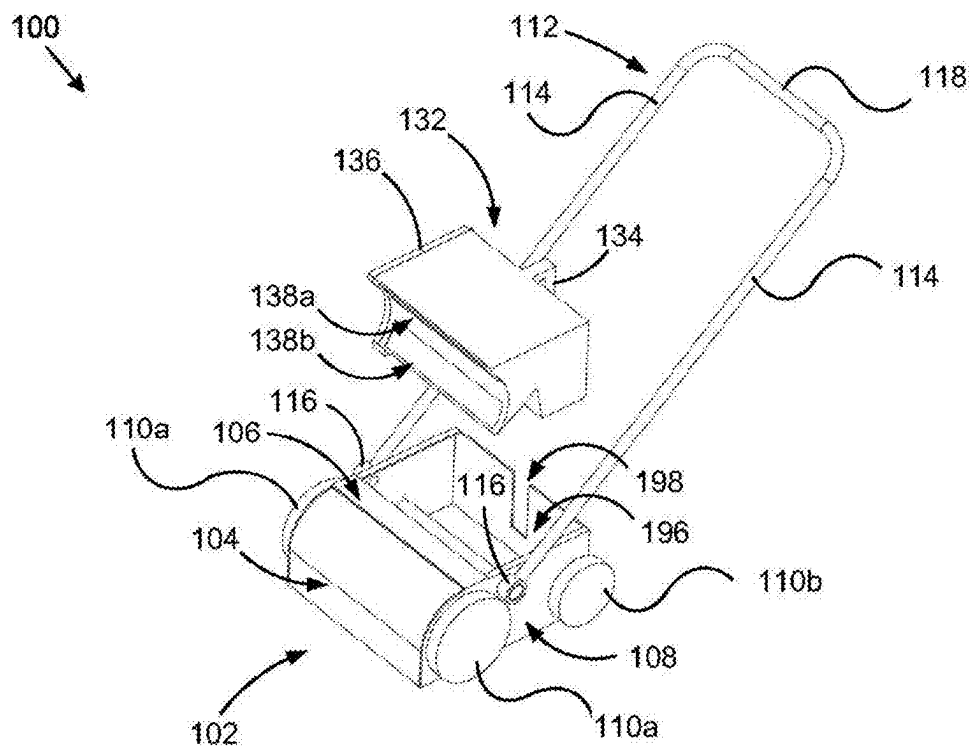


FIG. 4

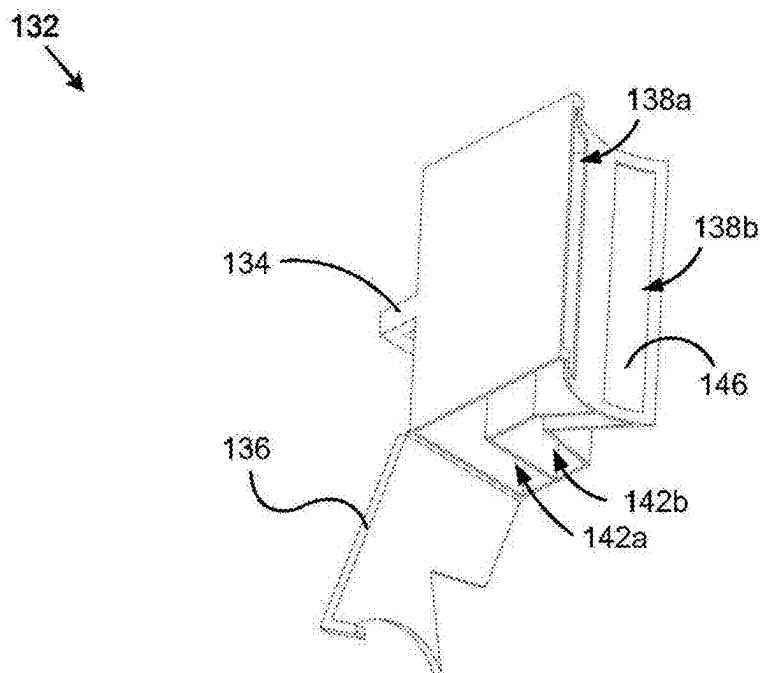


FIG. 5

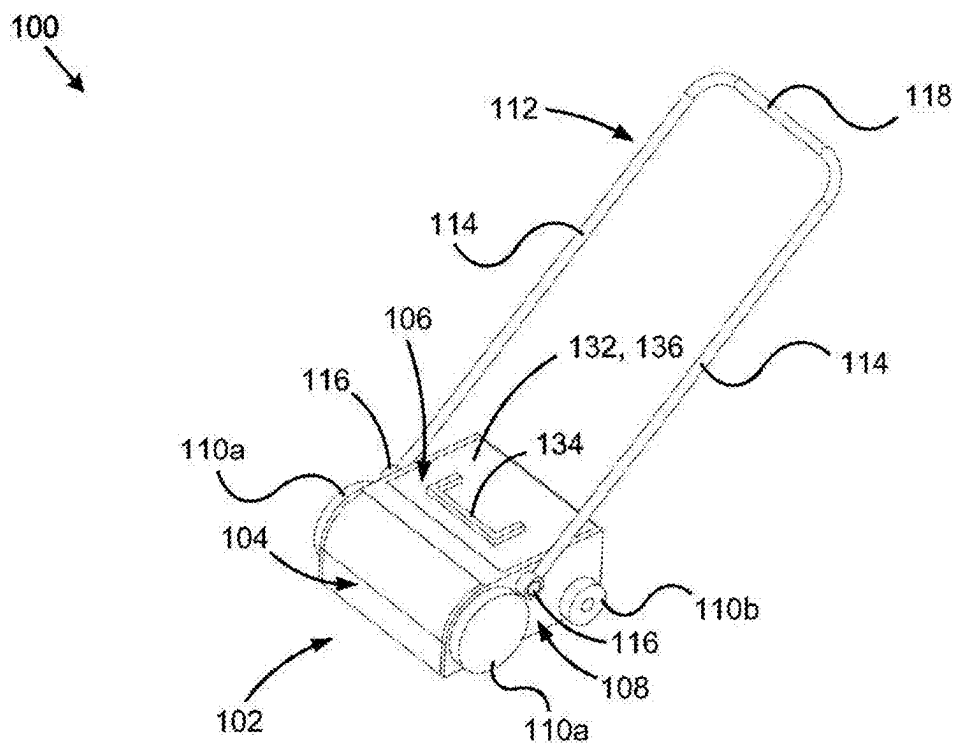


FIG. 6

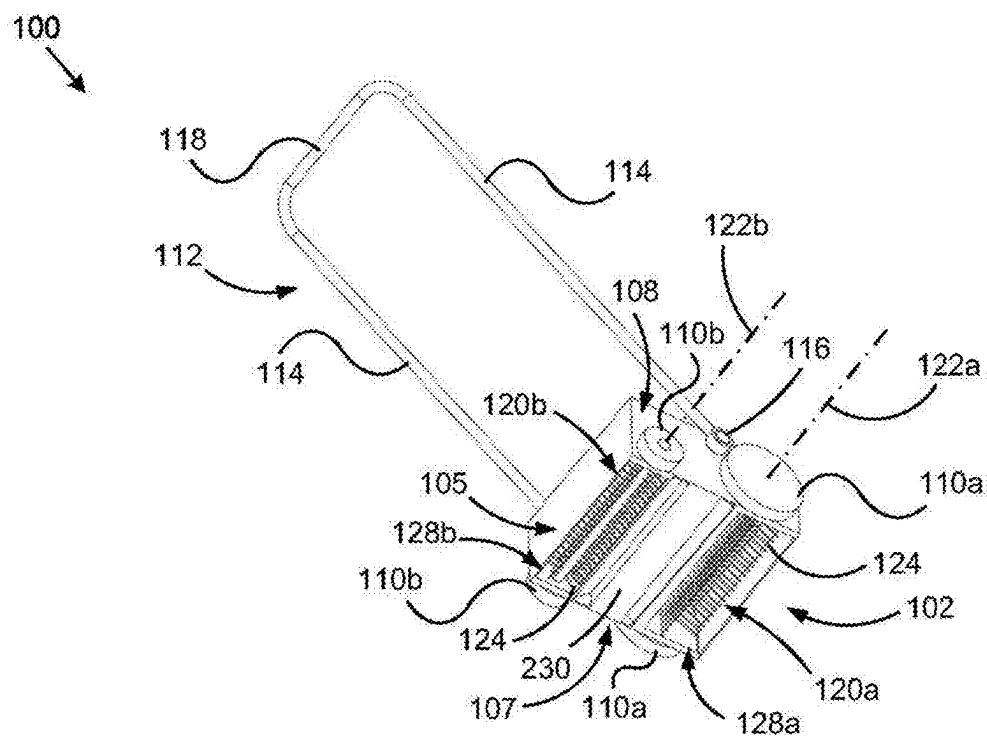
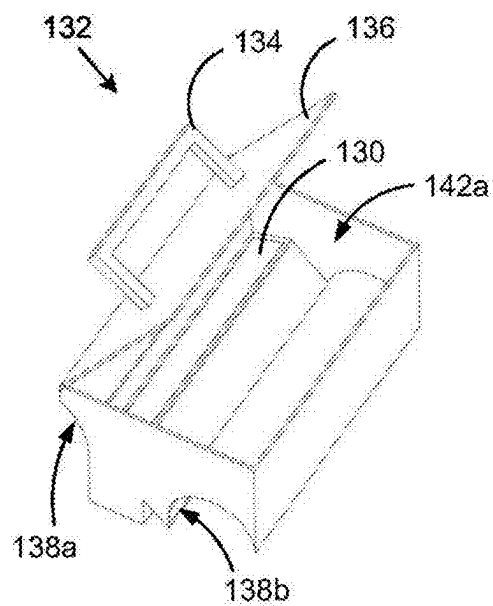
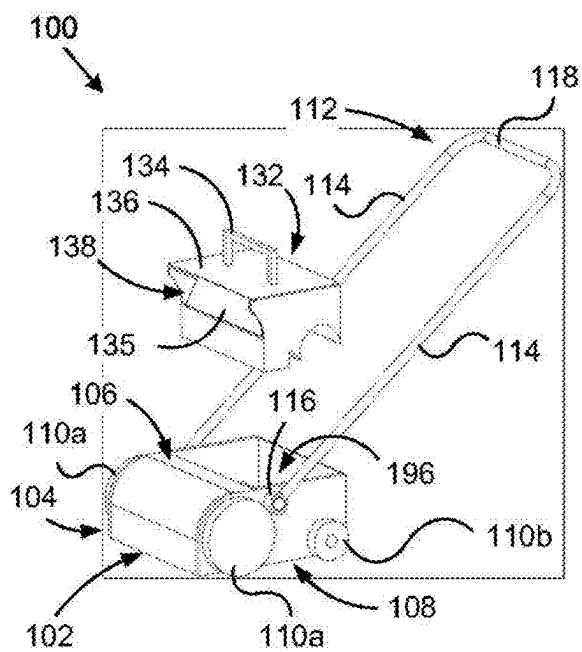
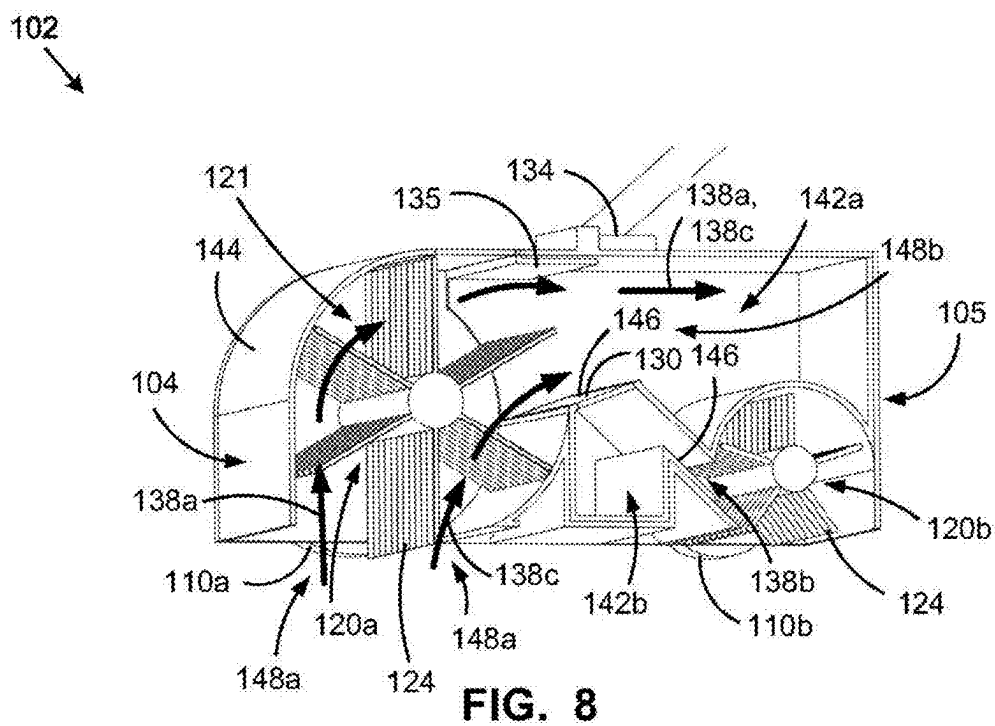


FIG. 7



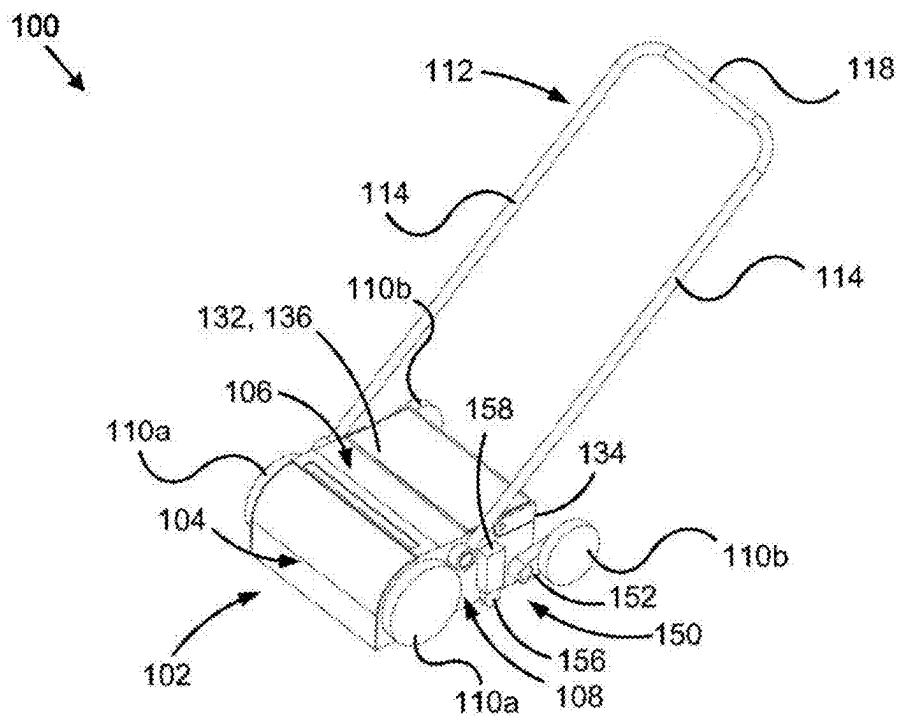


FIG. 11

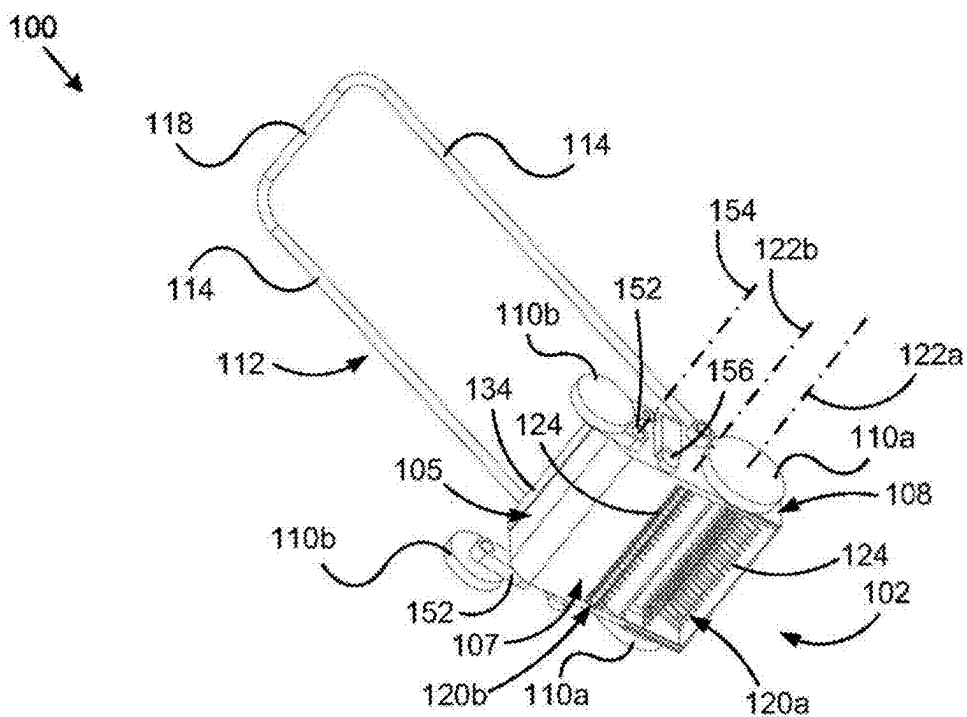


FIG. 12

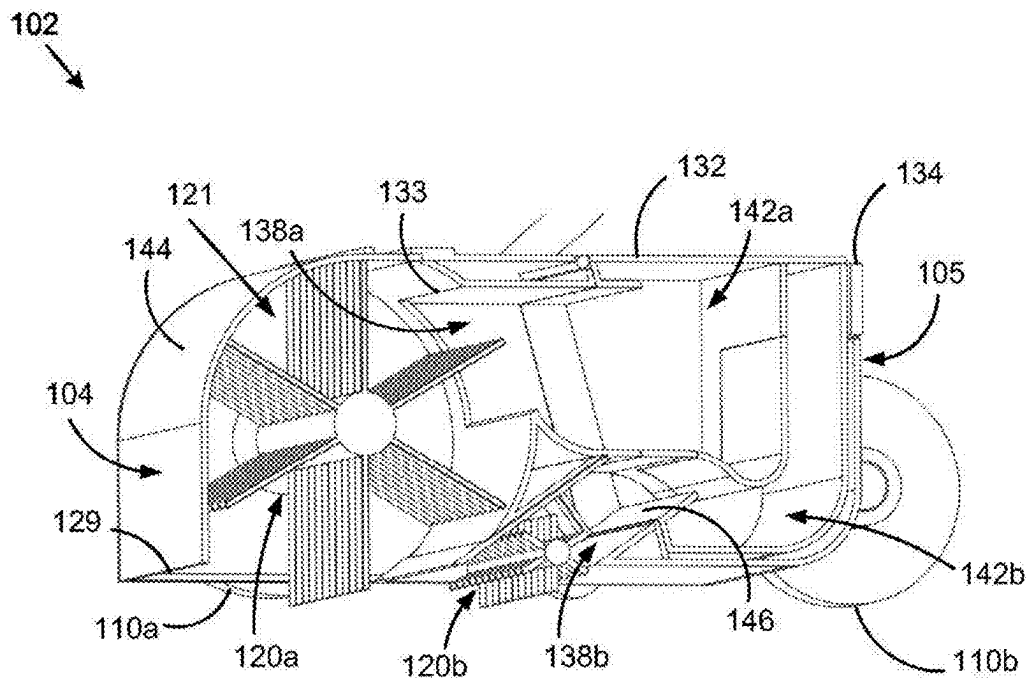


FIG. 13

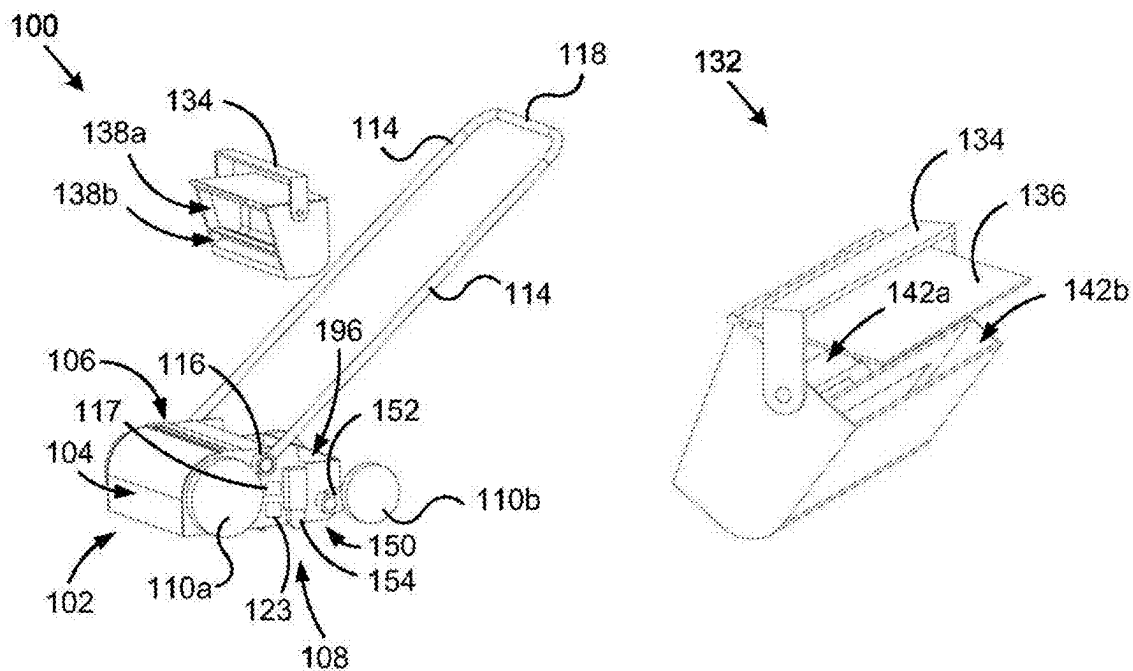


FIG. 14

FIG. 15

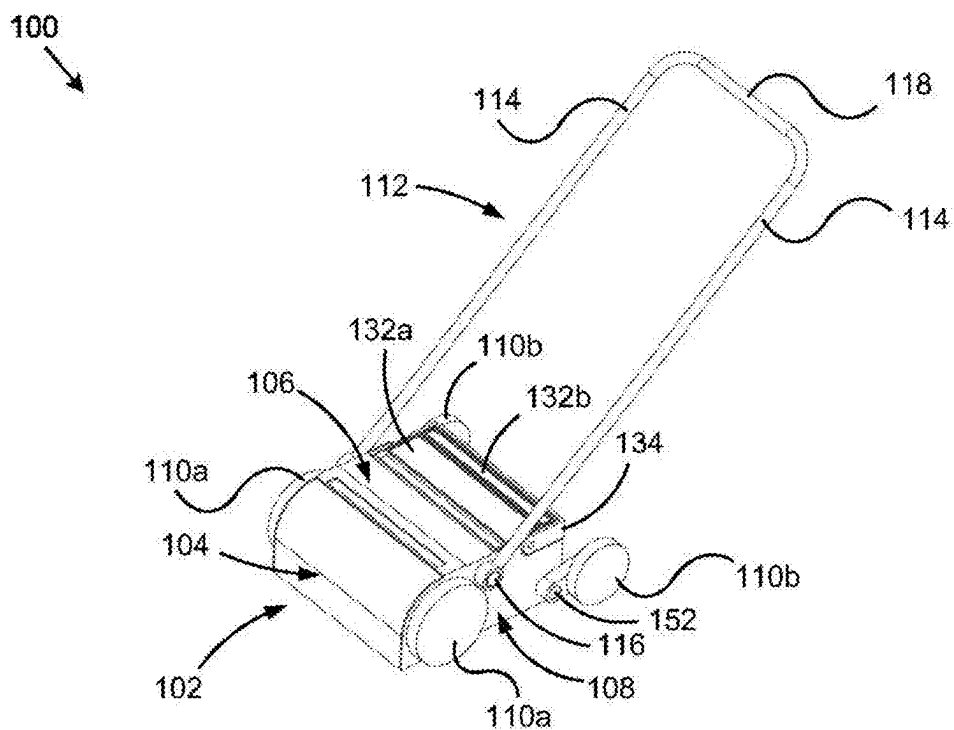


FIG. 16

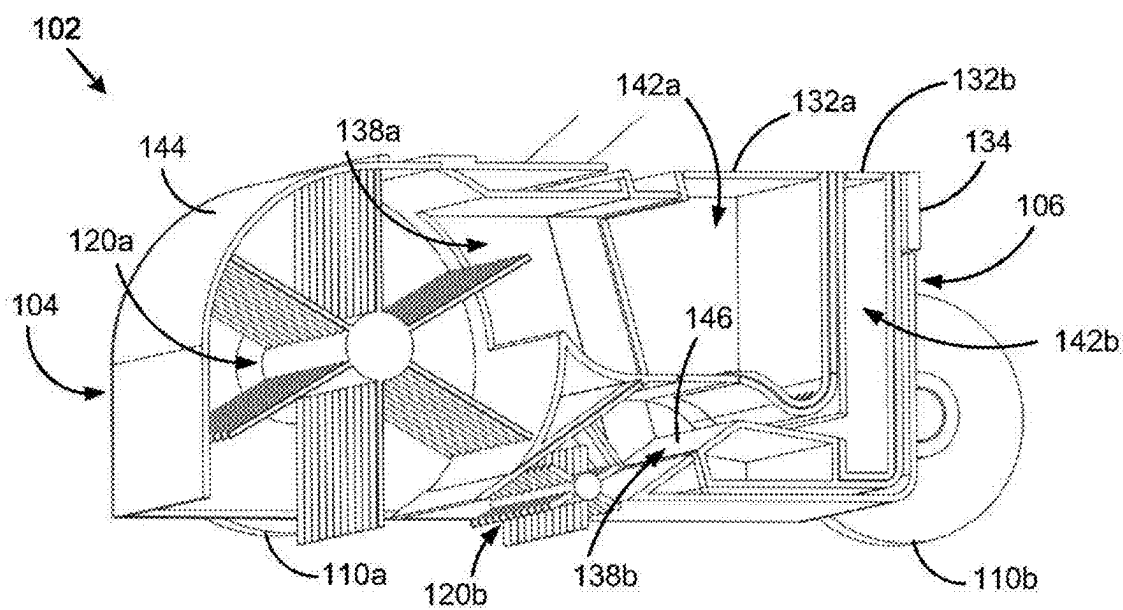


FIG. 17

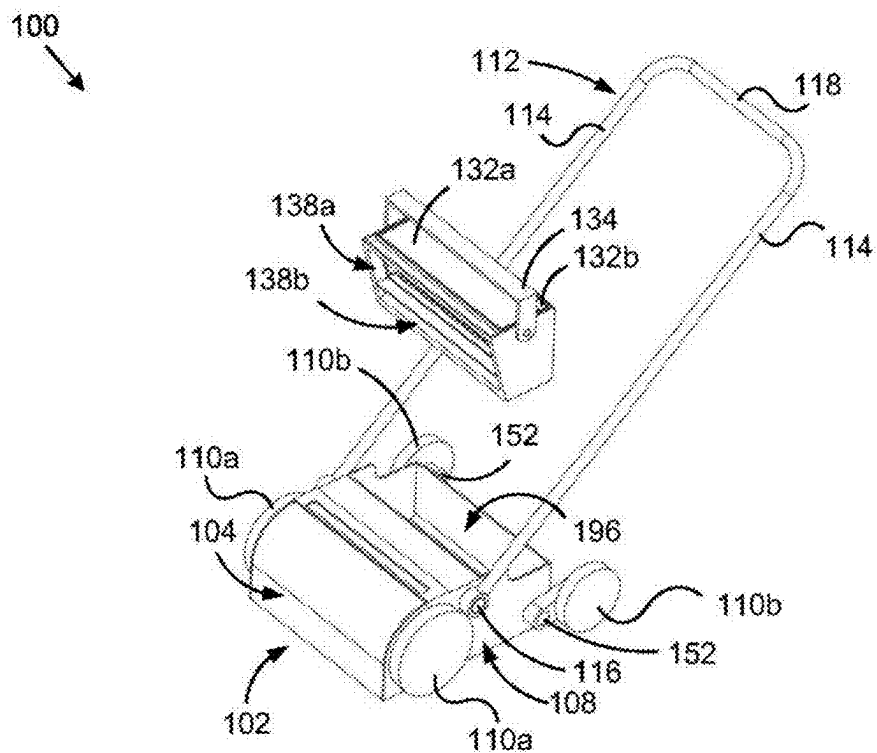


FIG. 18A

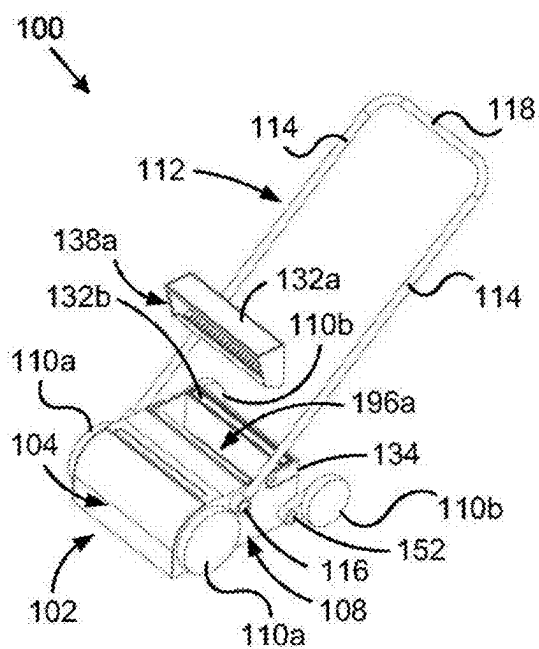


FIG. 18B

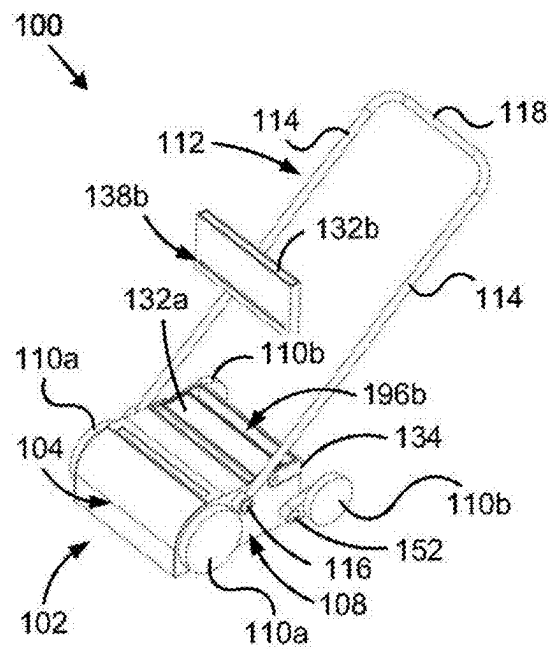


FIG. 18C

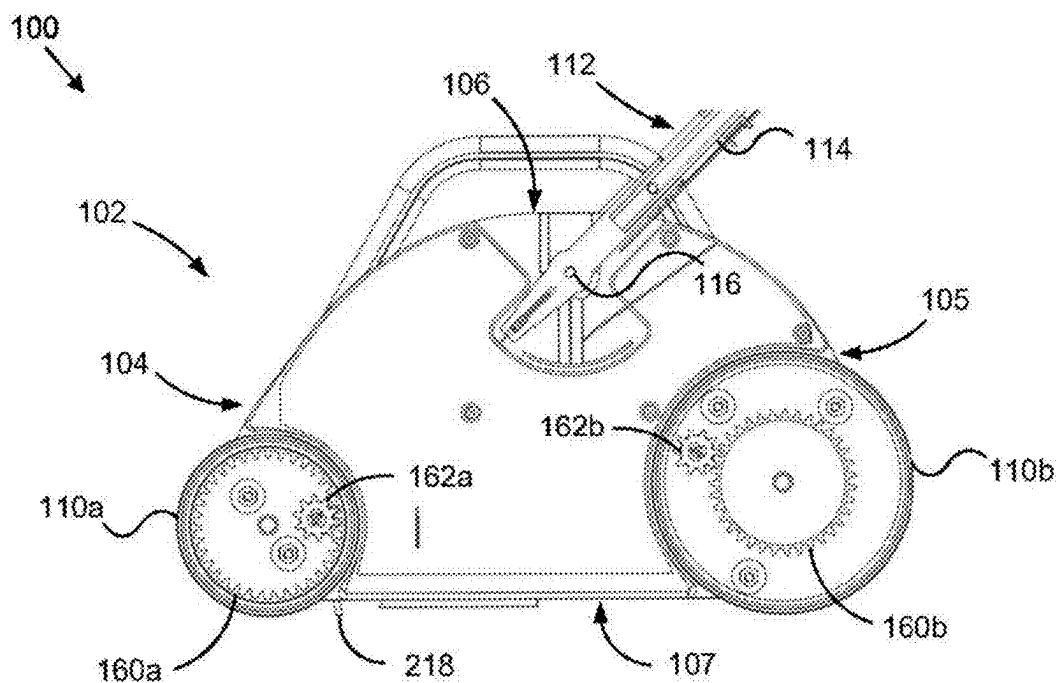


FIG. 19

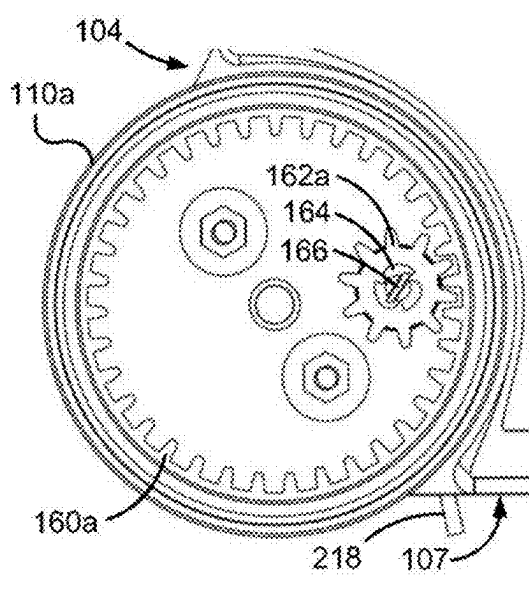


FIG. 20A

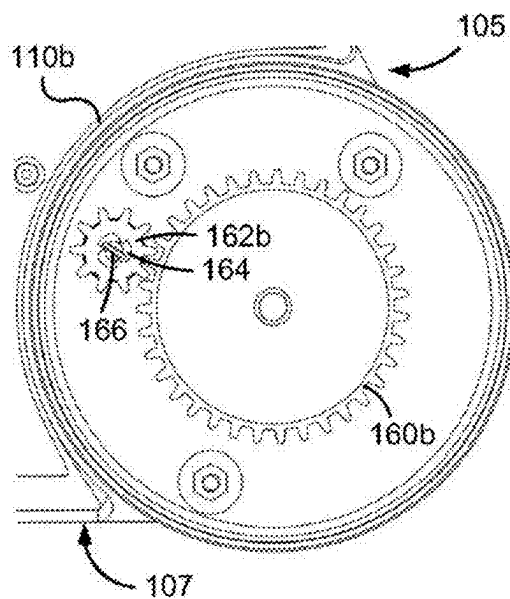


FIG. 20B

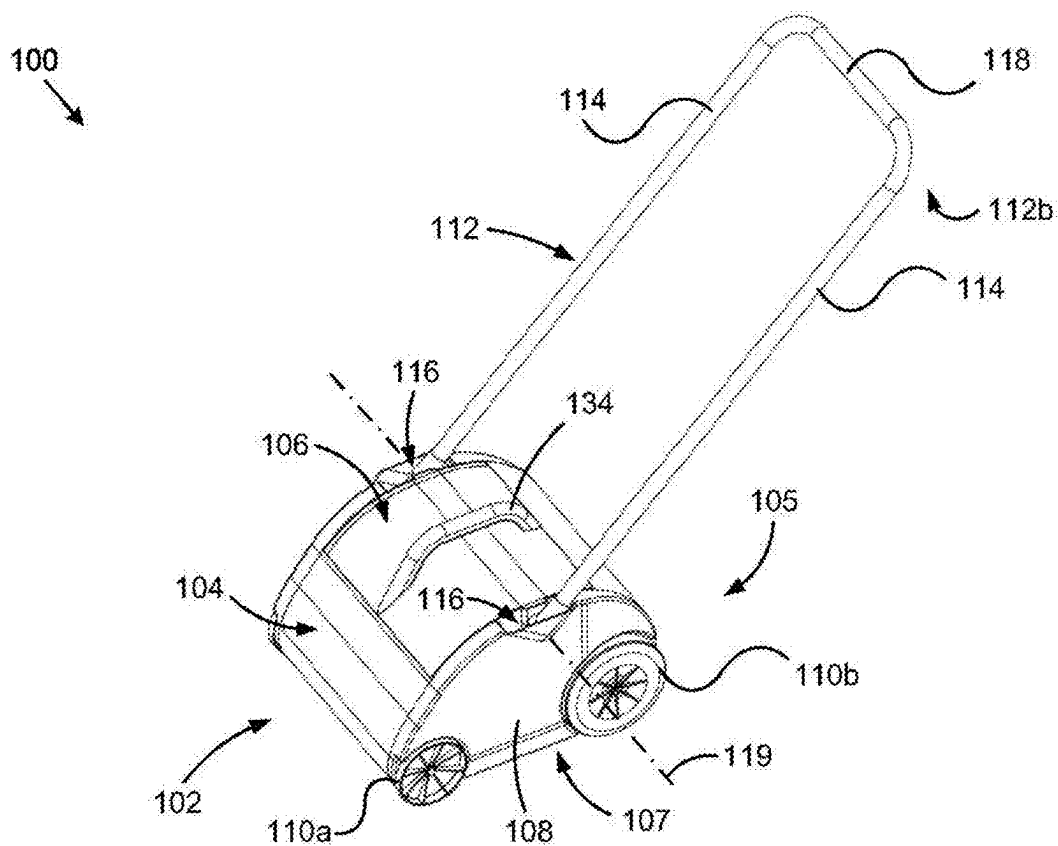


FIG. 21

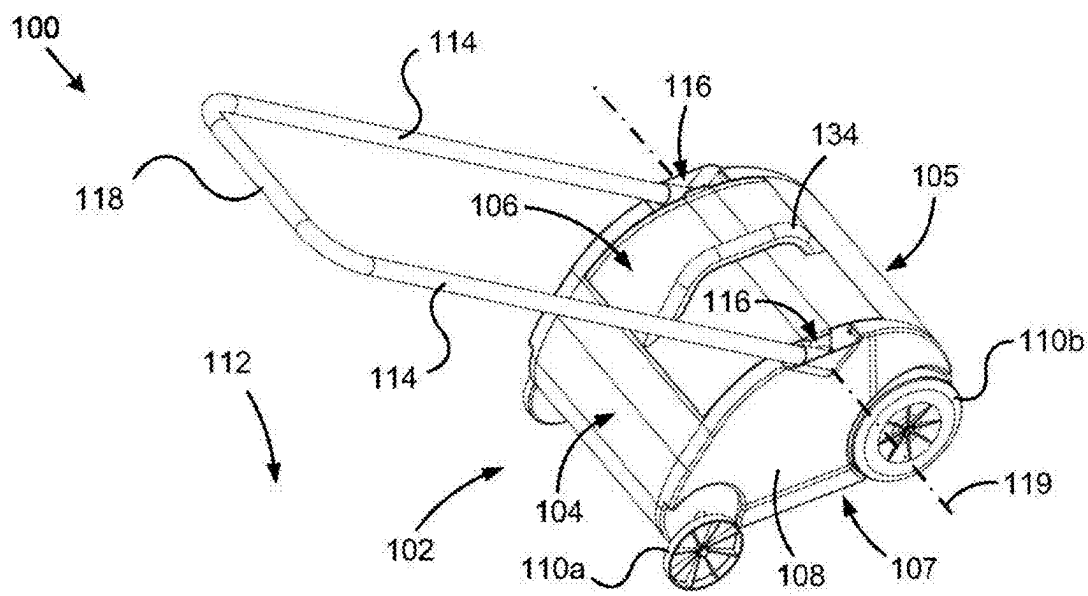


FIG. 22

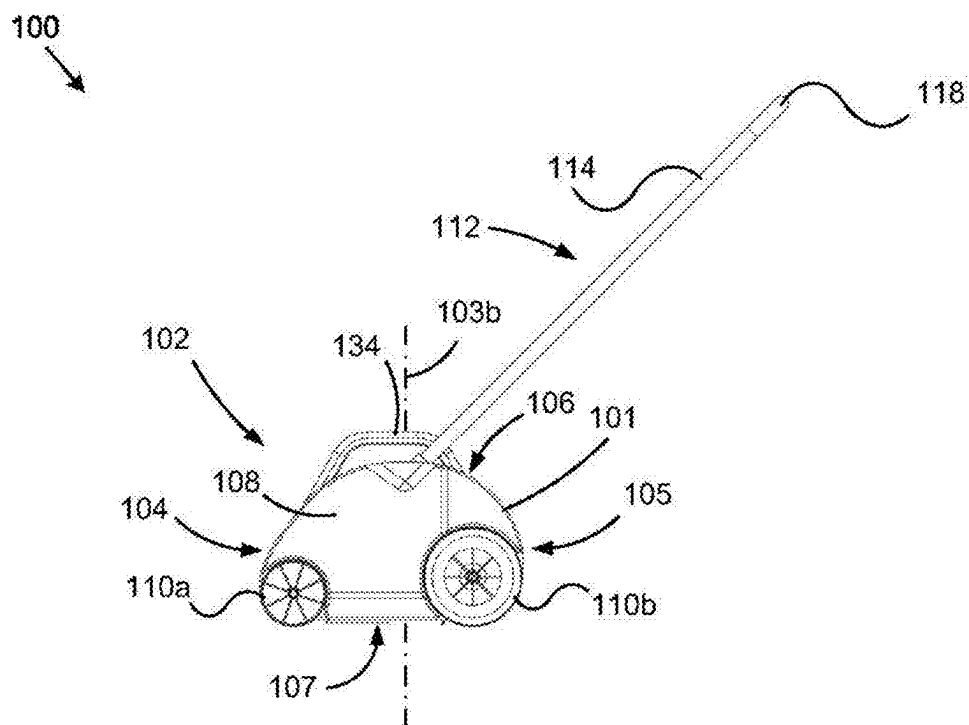


FIG. 23

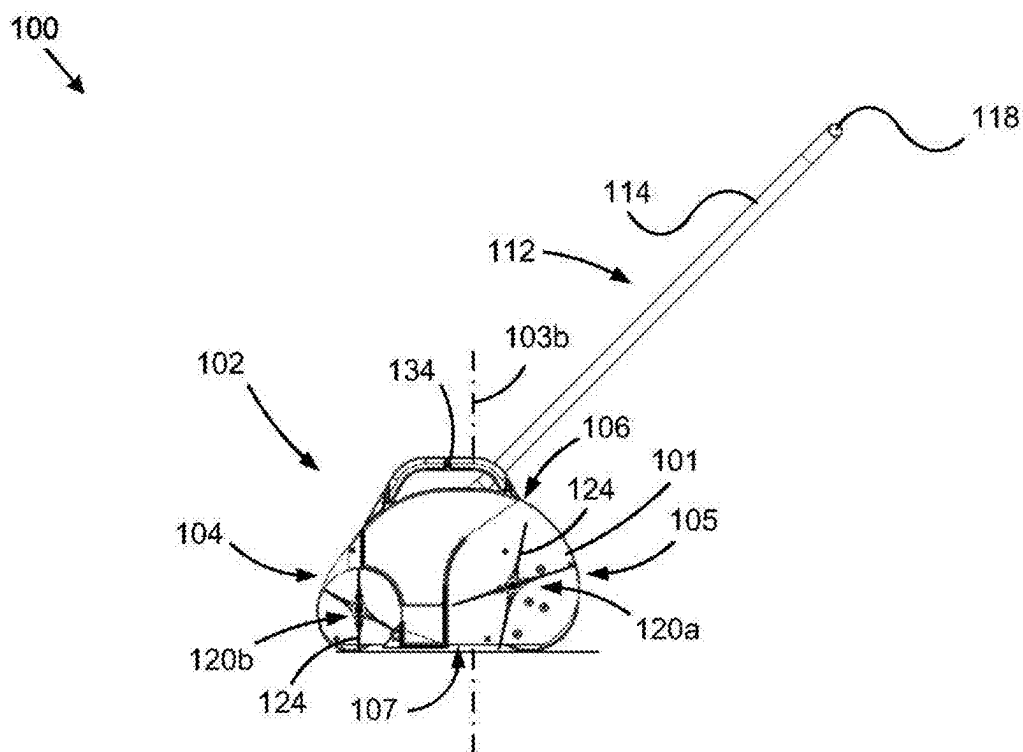


FIG. 24

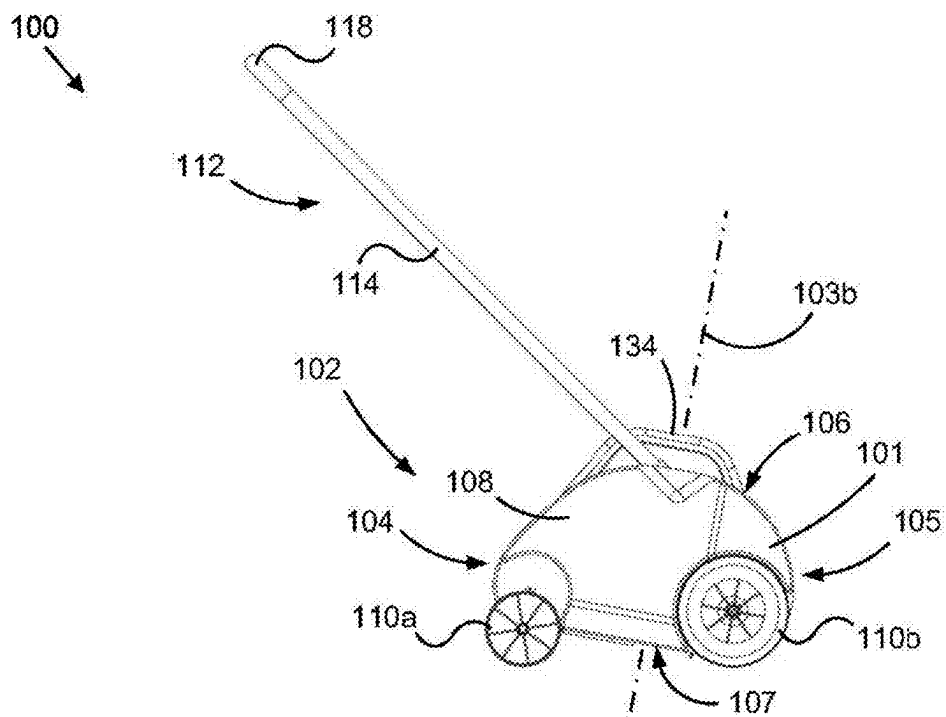


FIG. 25

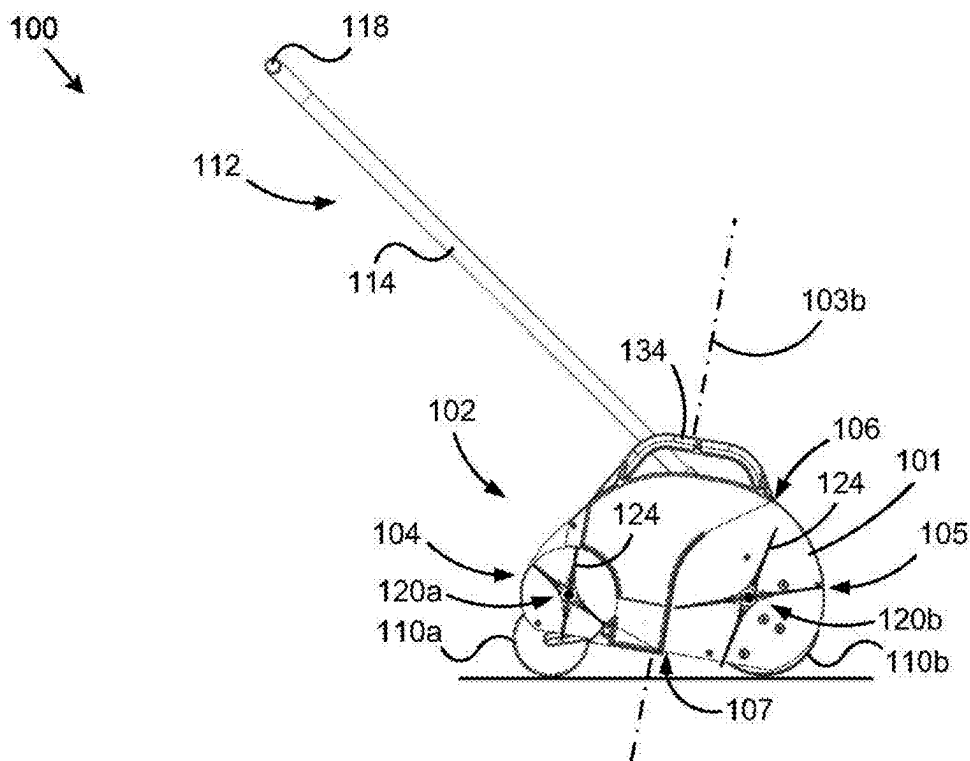


FIG. 26

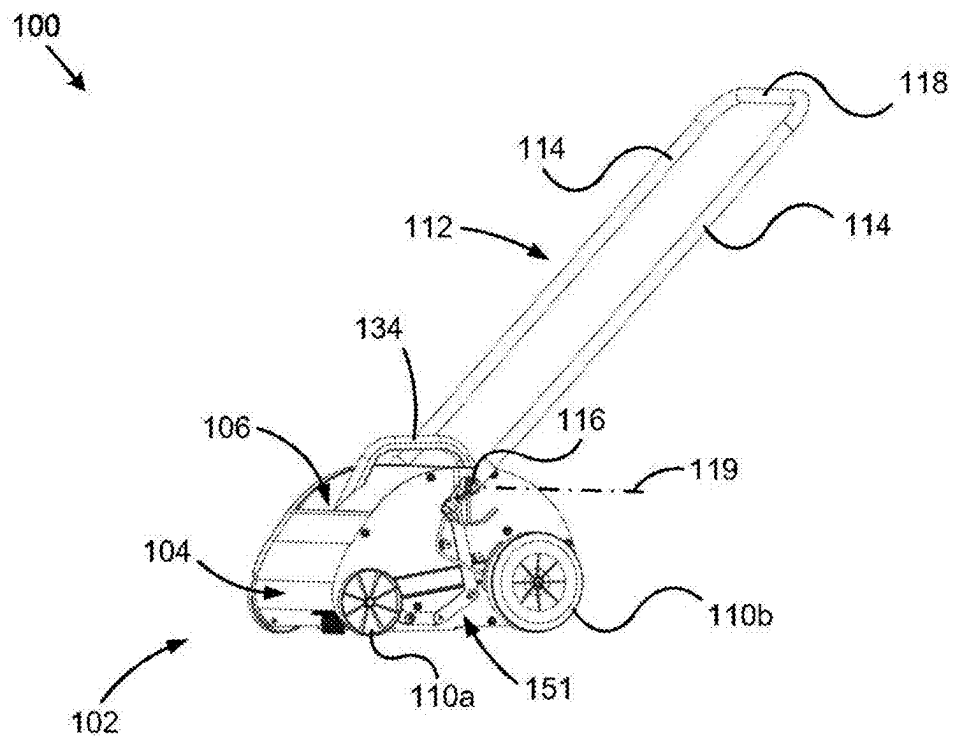


FIG. 27

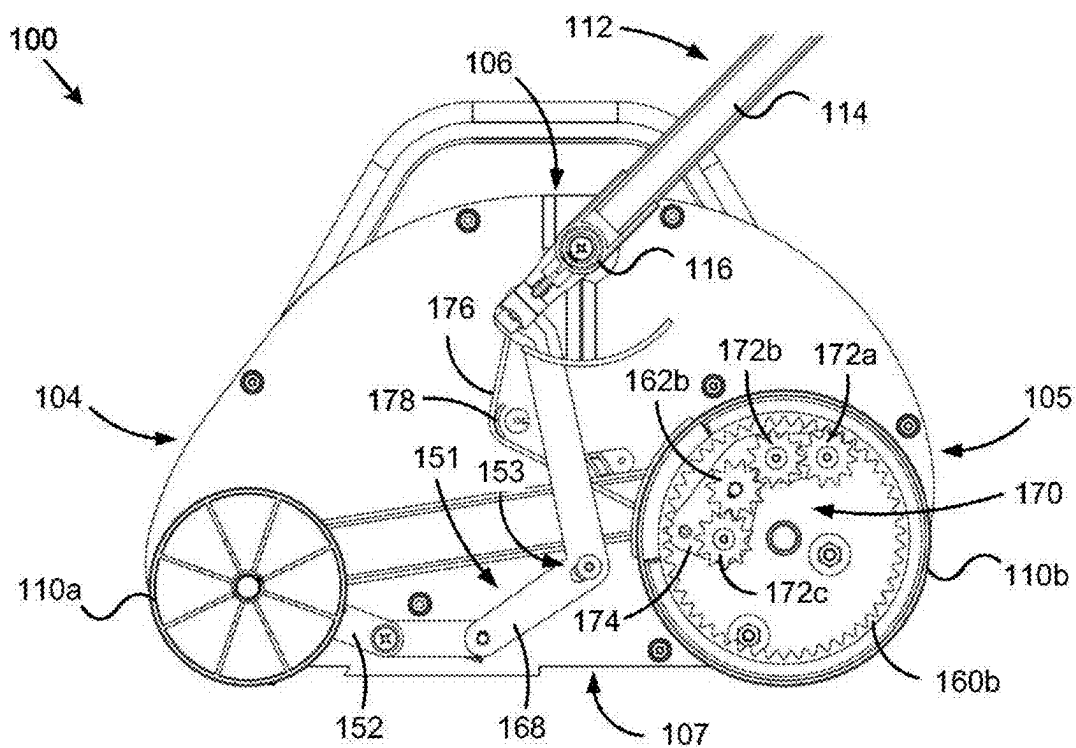


FIG. 28

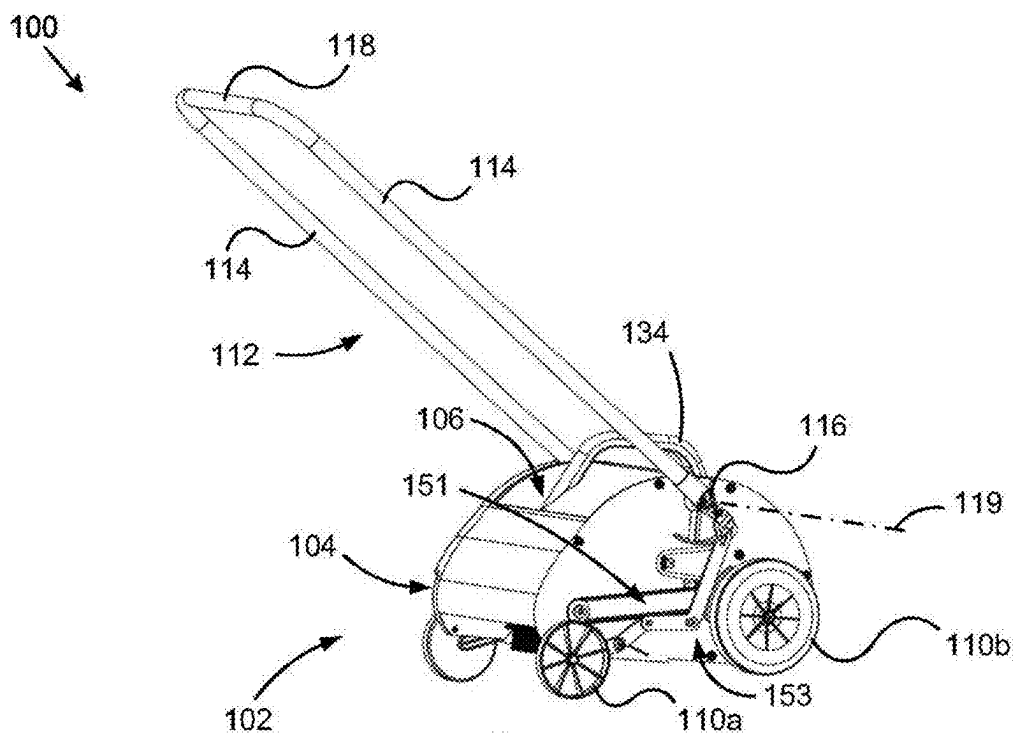


FIG. 29

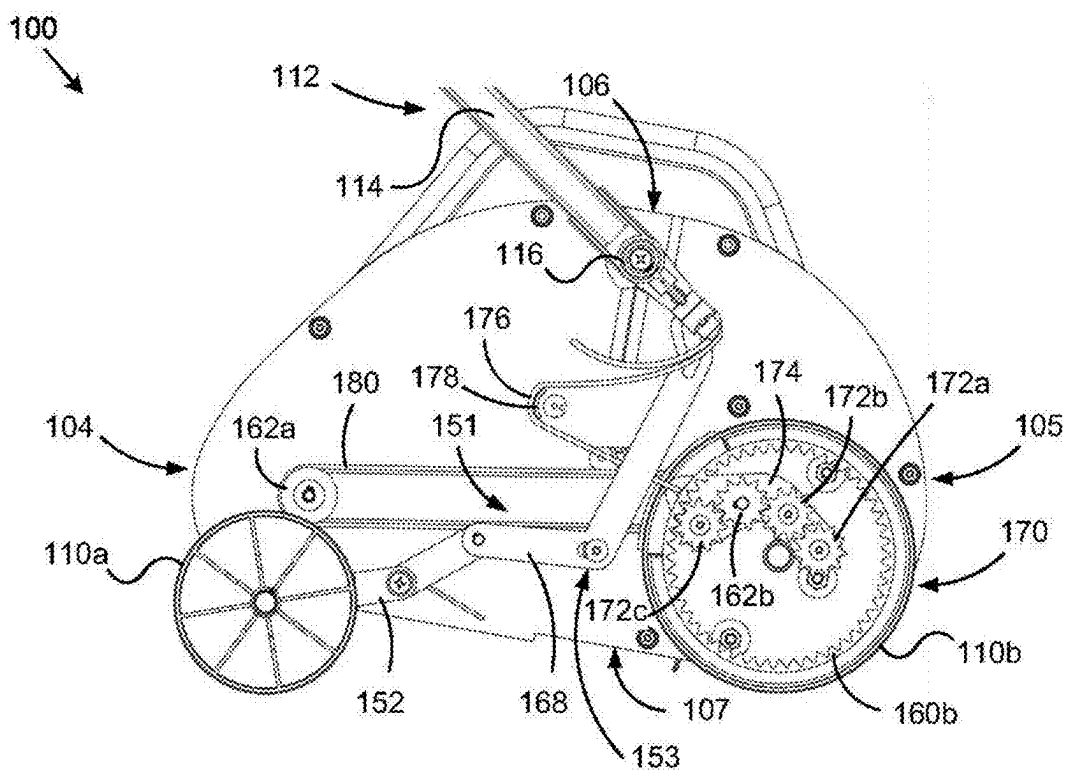


FIG. 30

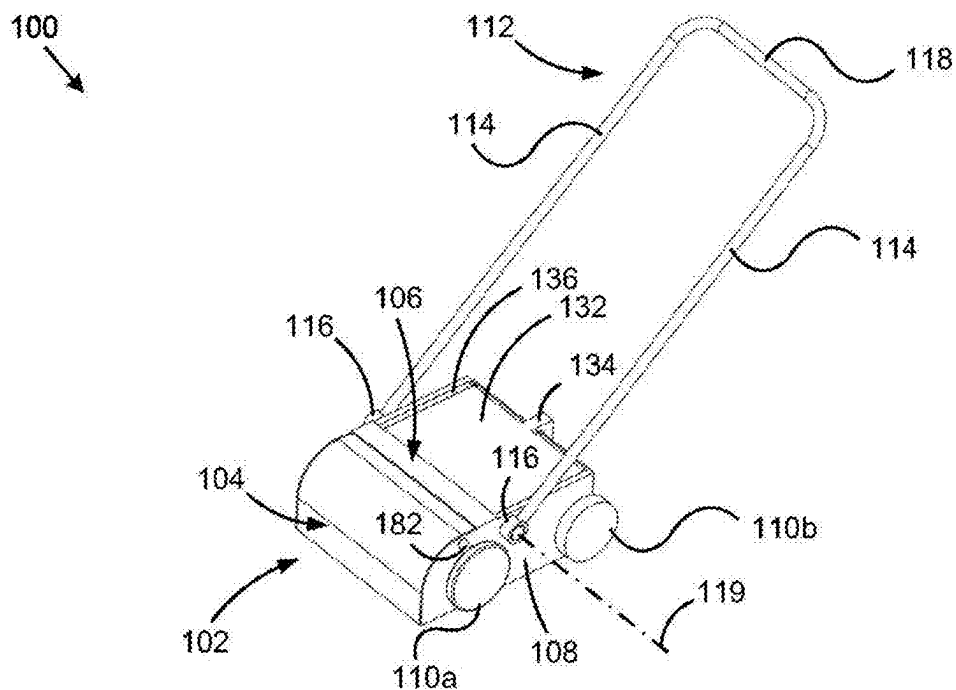


FIG. 31

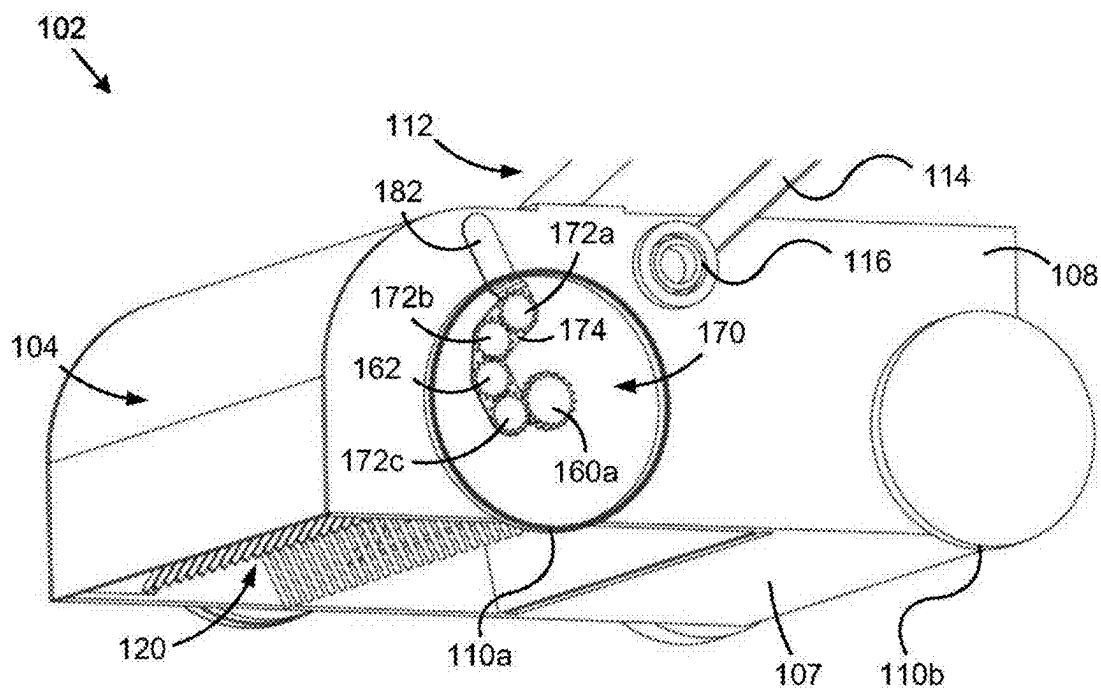


FIG. 32

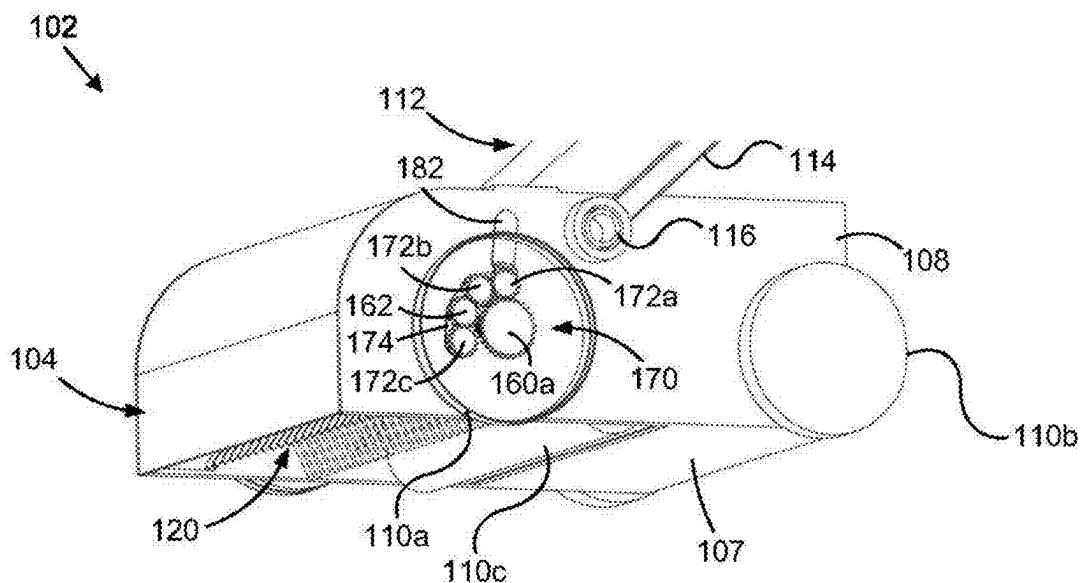


FIG. 33

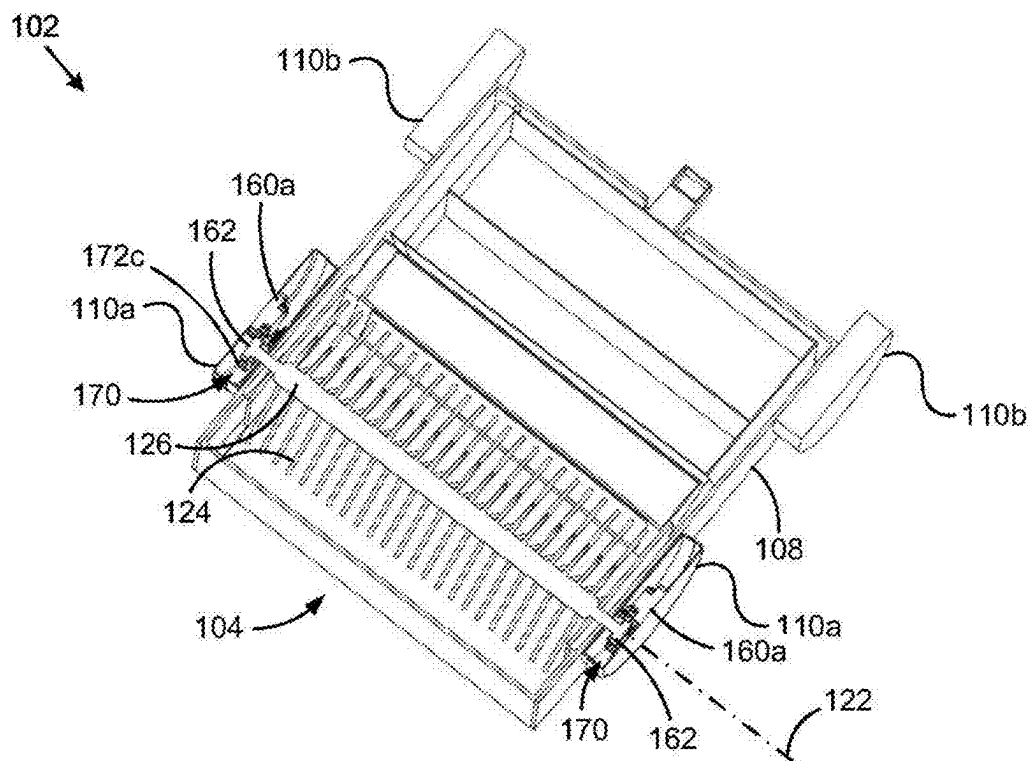


FIG. 34

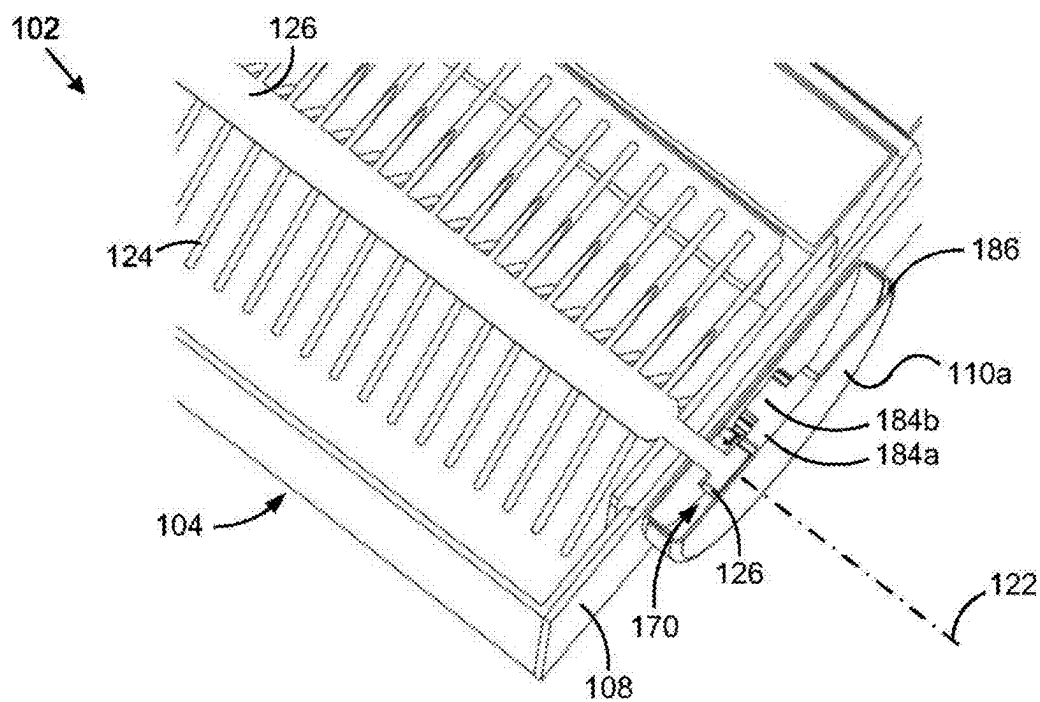


FIG. 35

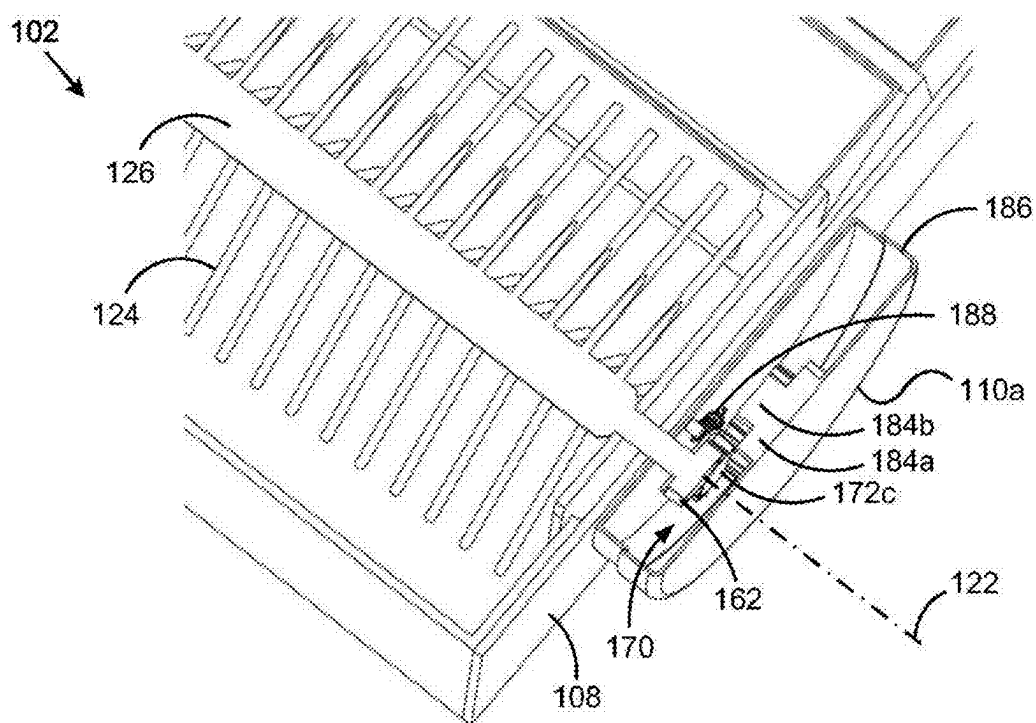


FIG. 36

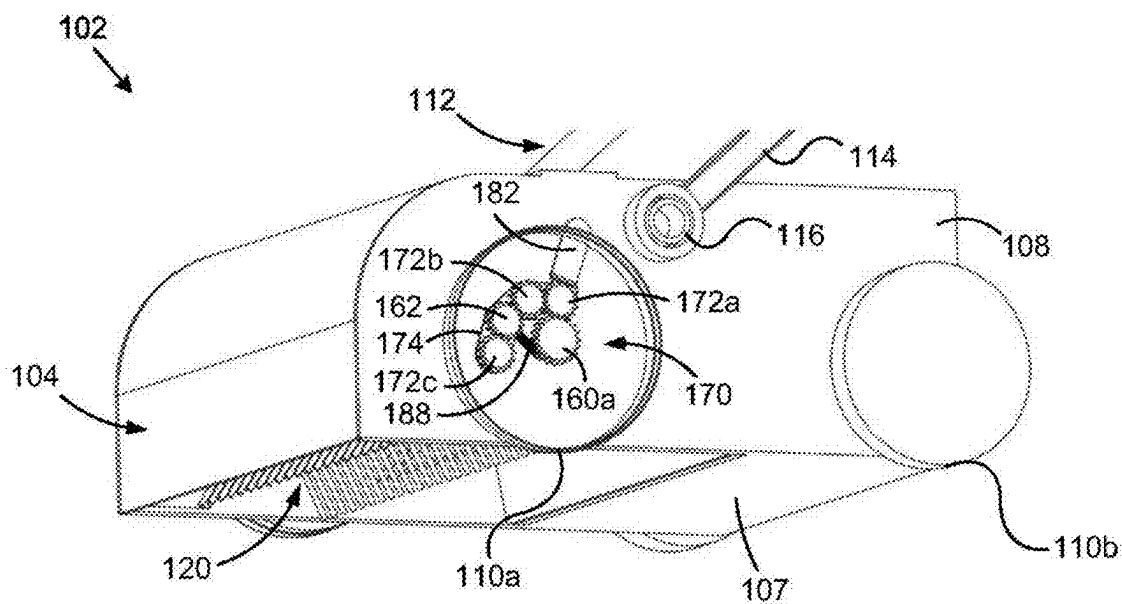


FIG. 37

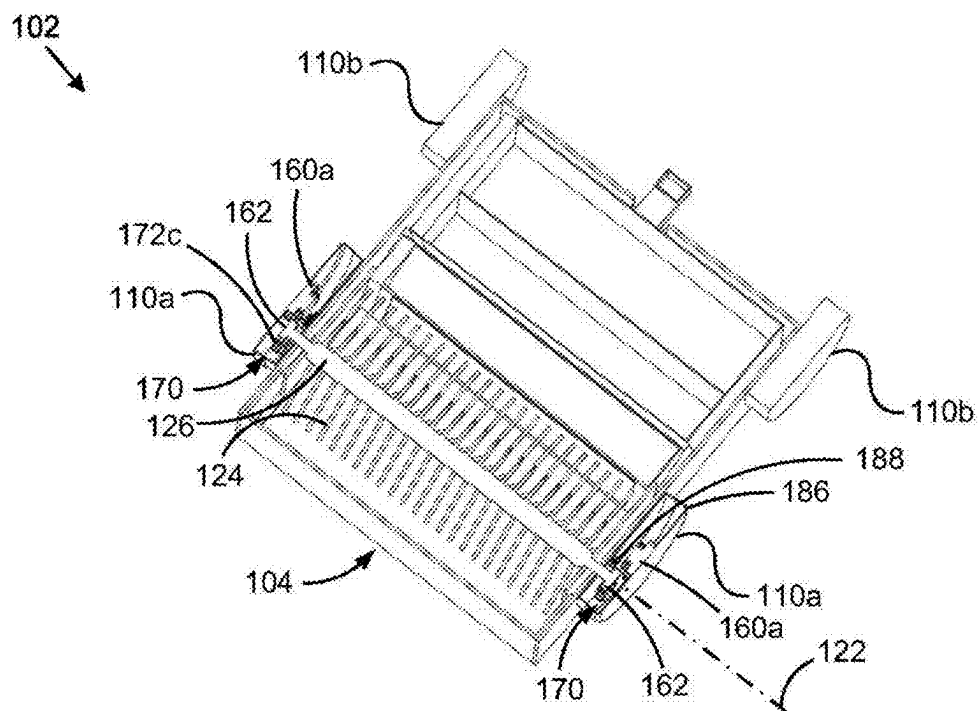


FIG. 38

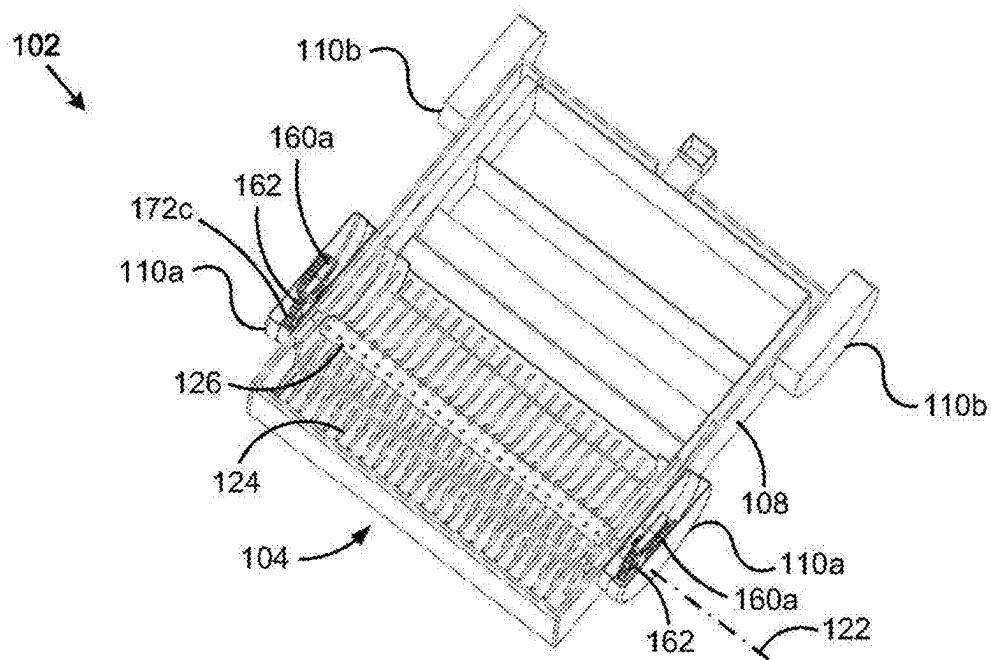


FIG. 39

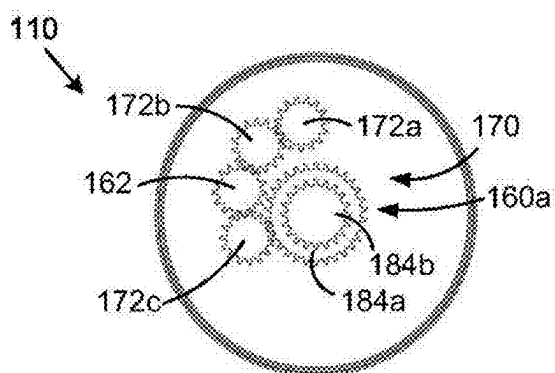


FIG. 40A

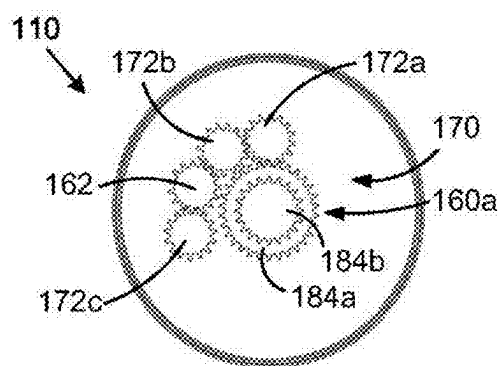


FIG. 40B

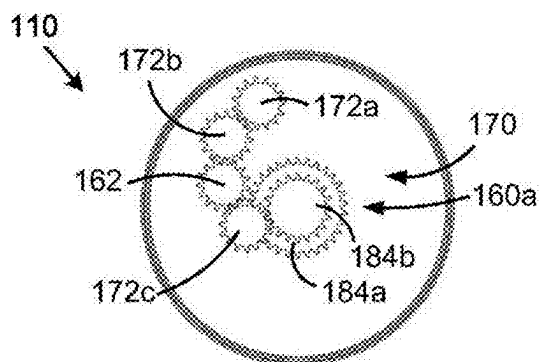


FIG. 40C

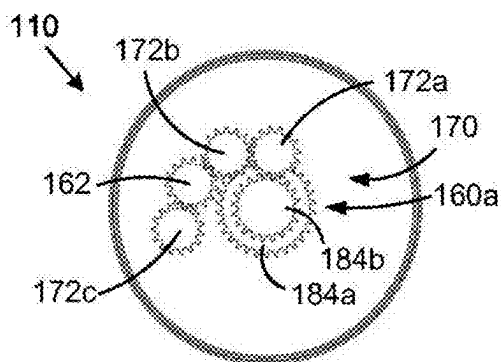


FIG. 40D

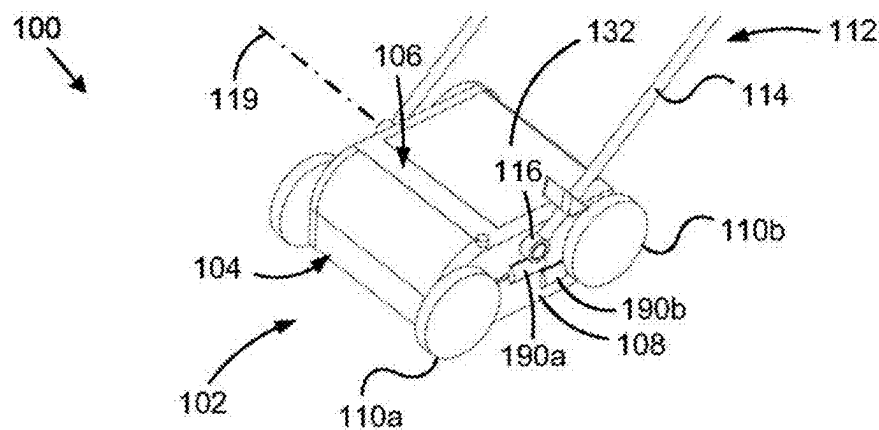


FIG. 41A

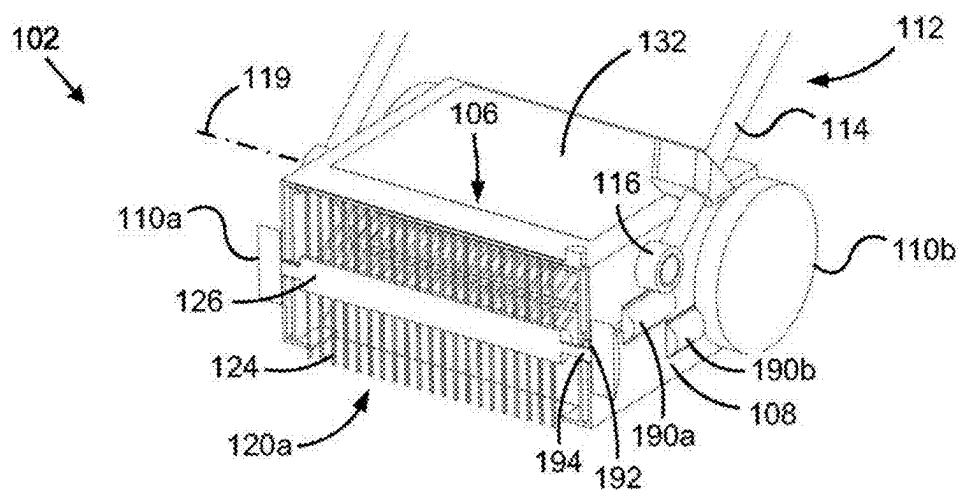


FIG. 41B

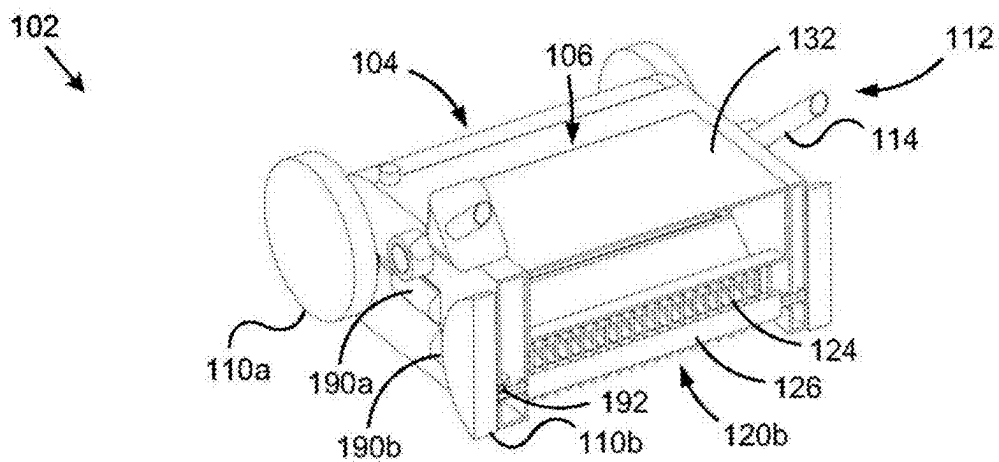


FIG. 41C

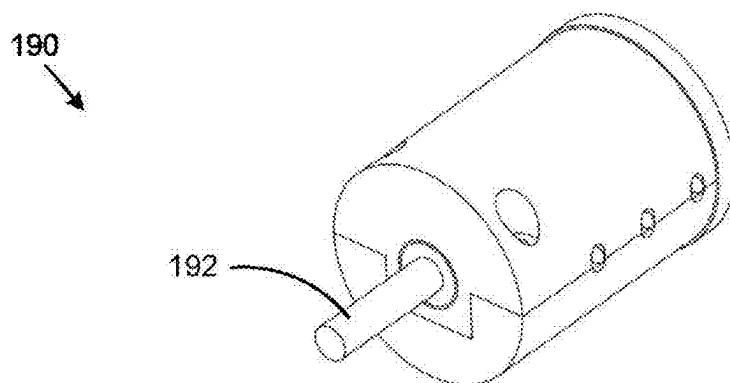


FIG. 42A

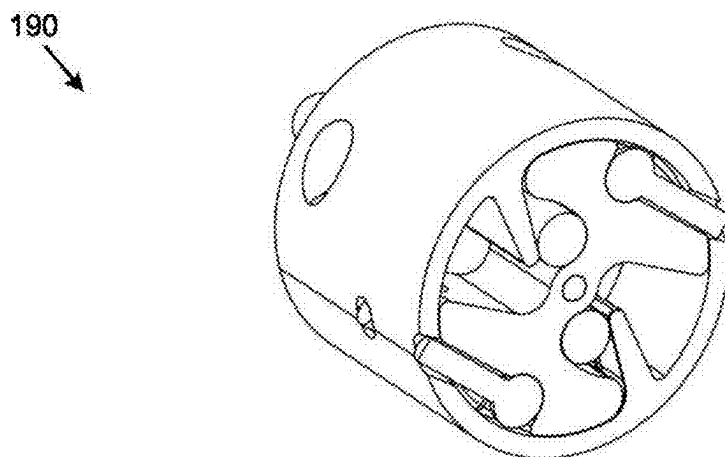


FIG. 42B

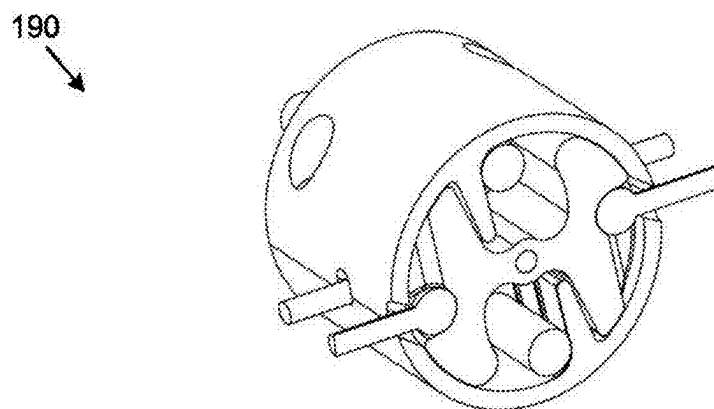


FIG. 42C

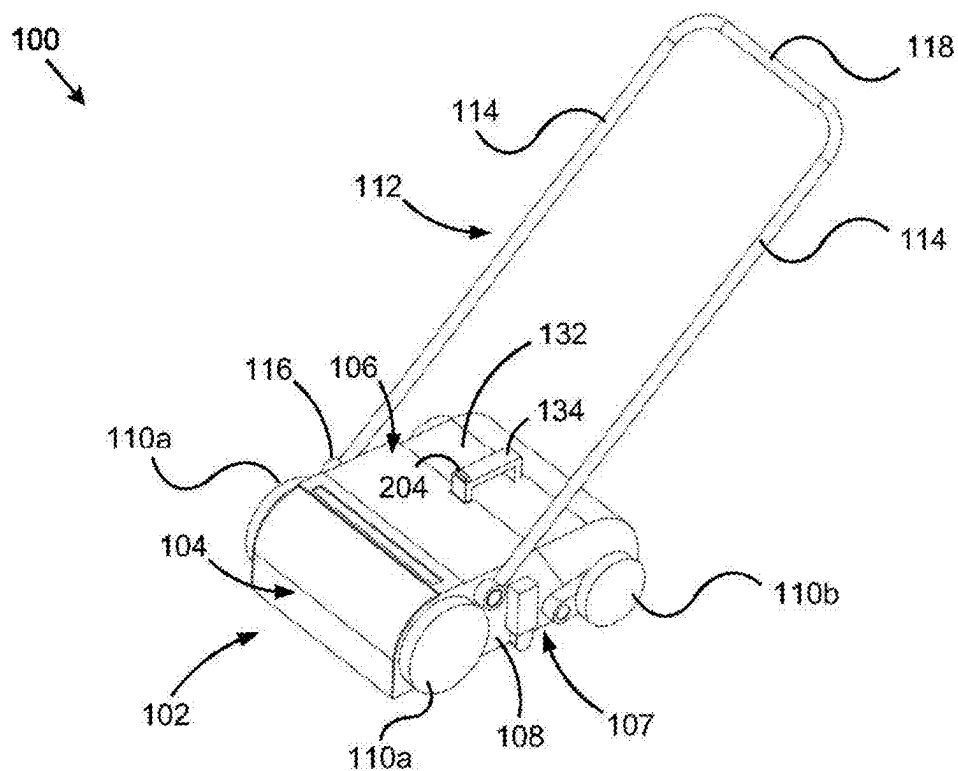


FIG. 43

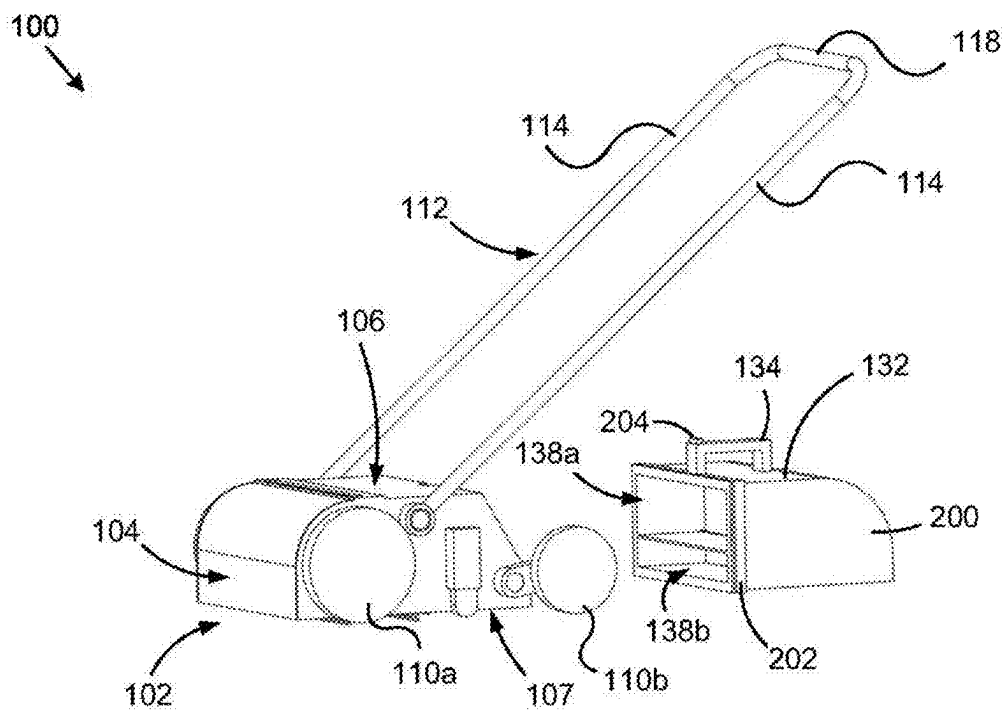


FIG. 44

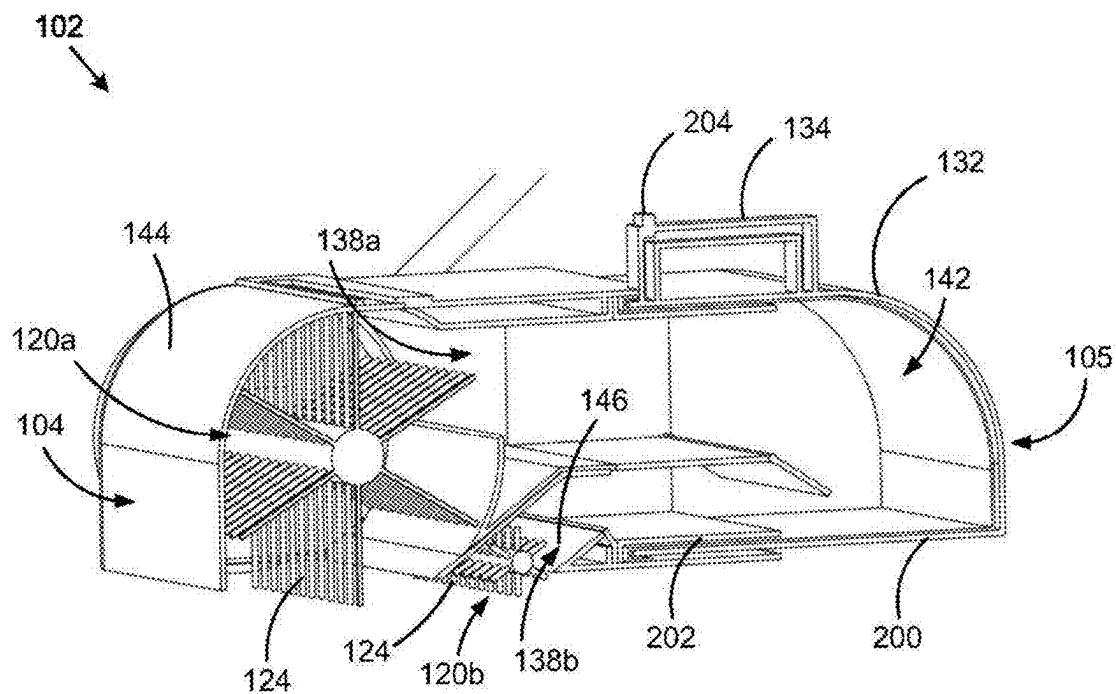


FIG. 45A

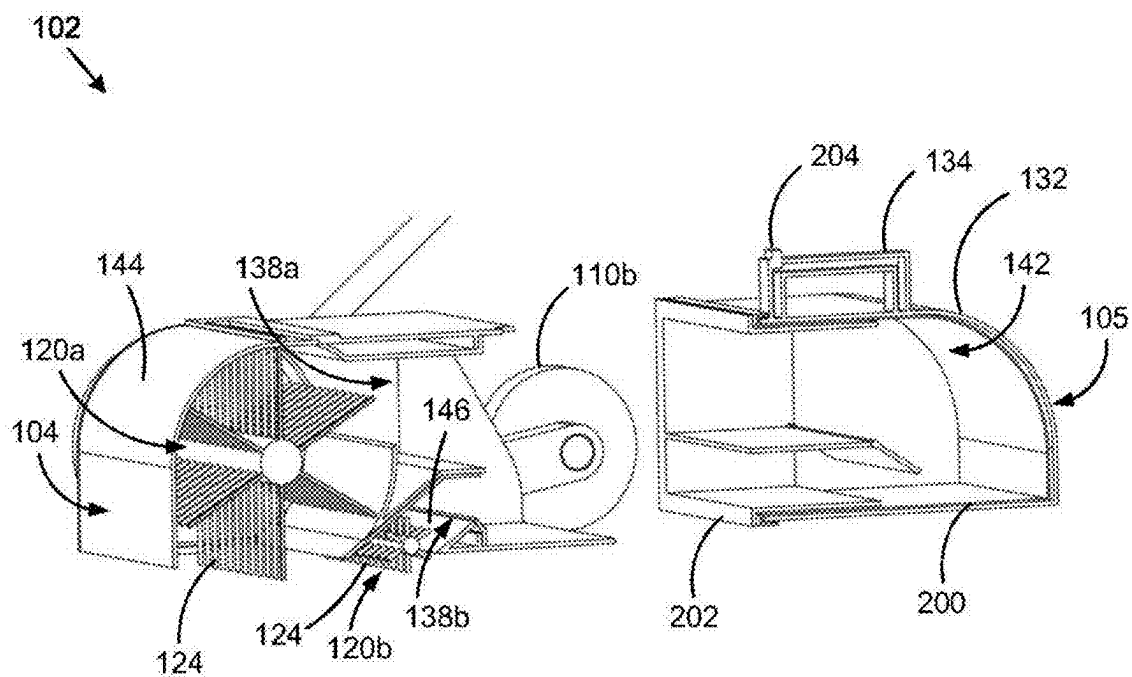


FIG. 45B

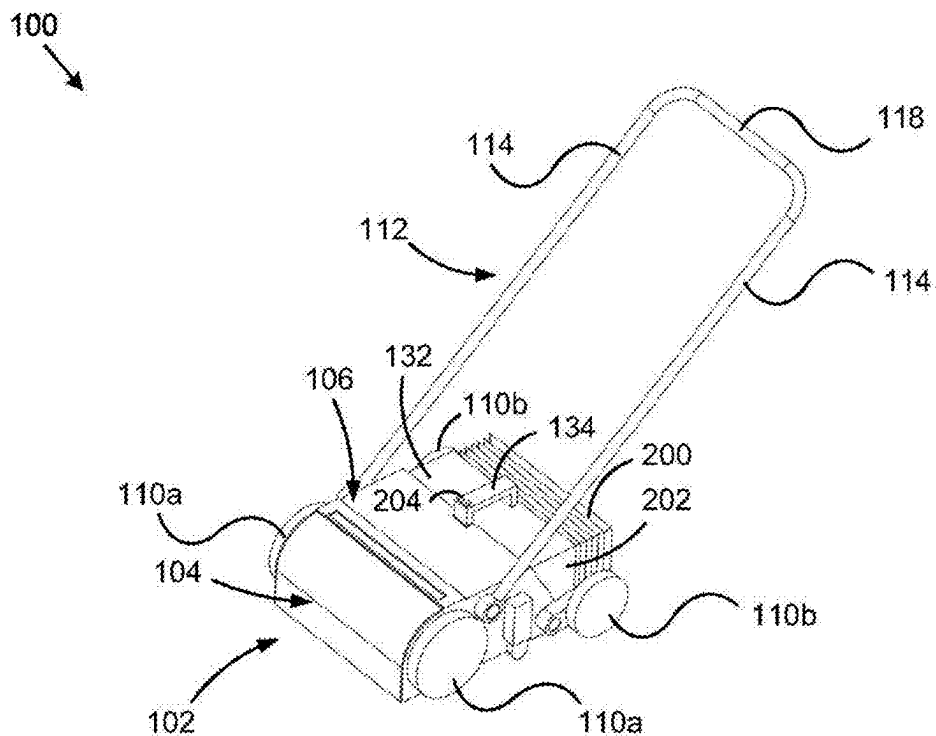


FIG. 46

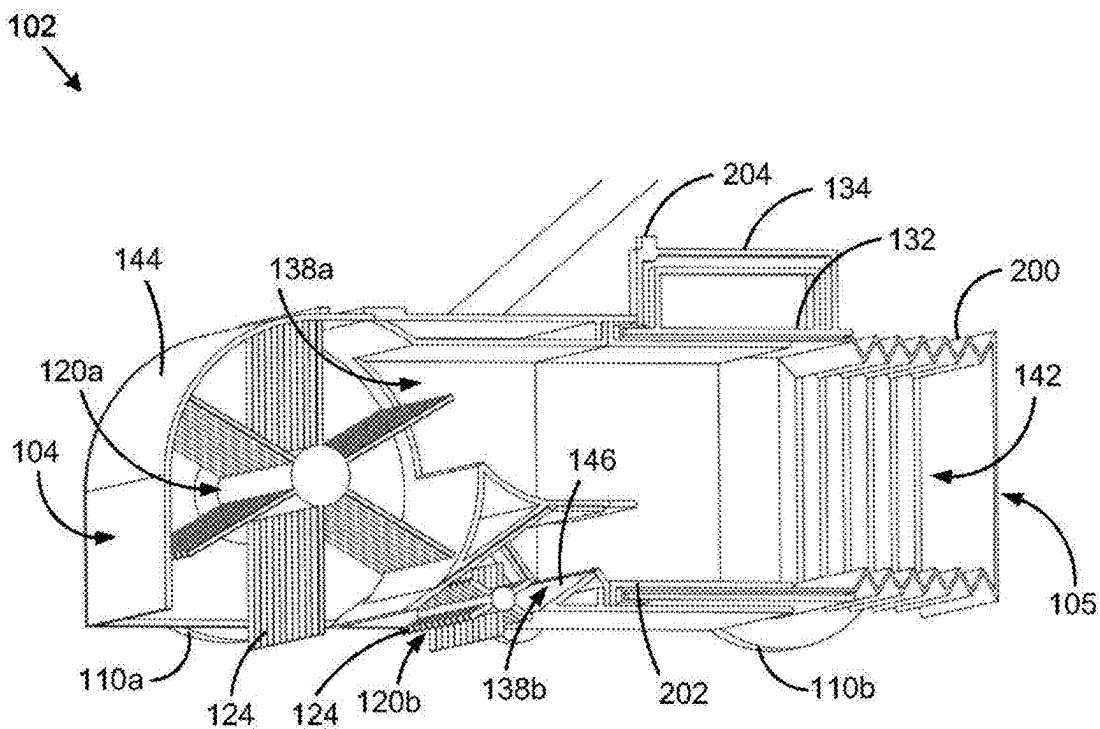


FIG. 47

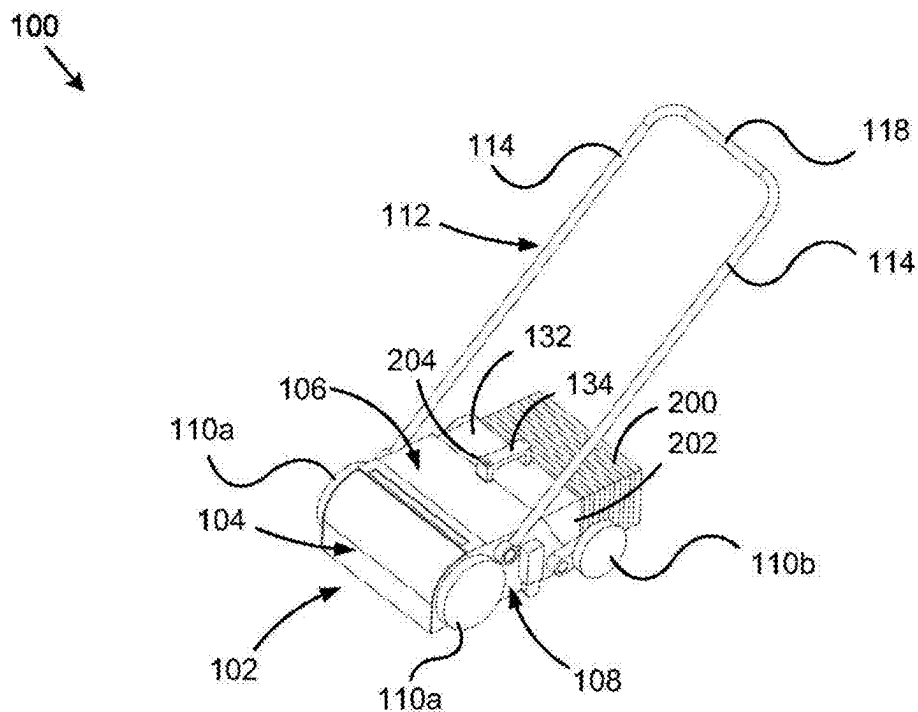


FIG. 48

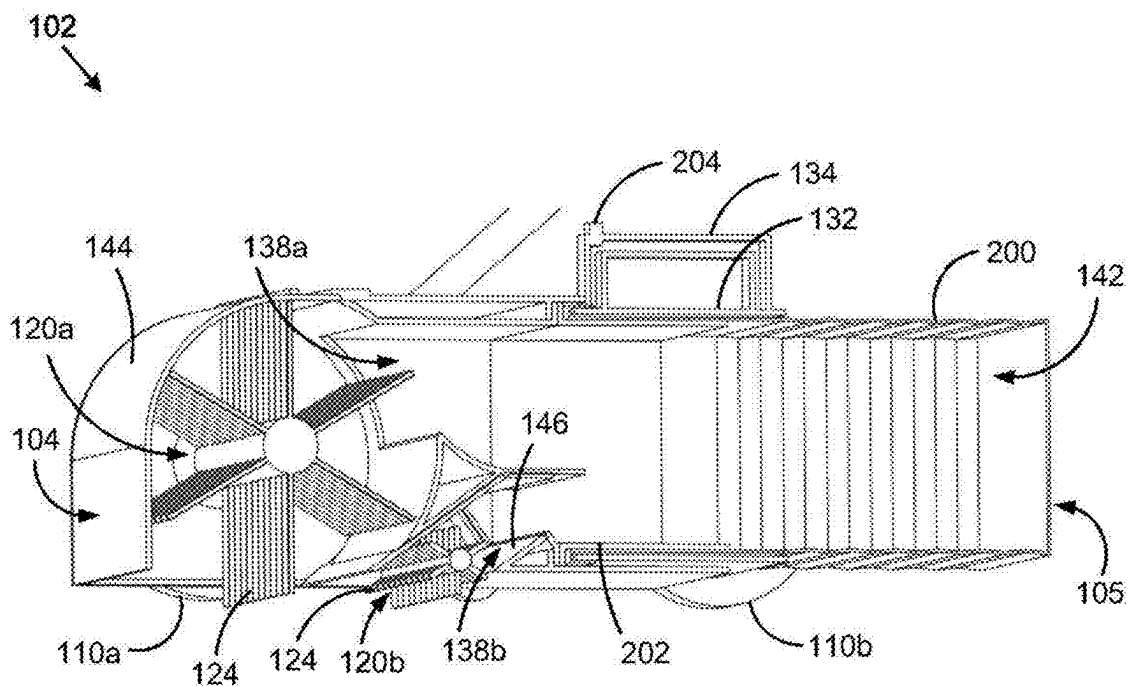


FIG. 49

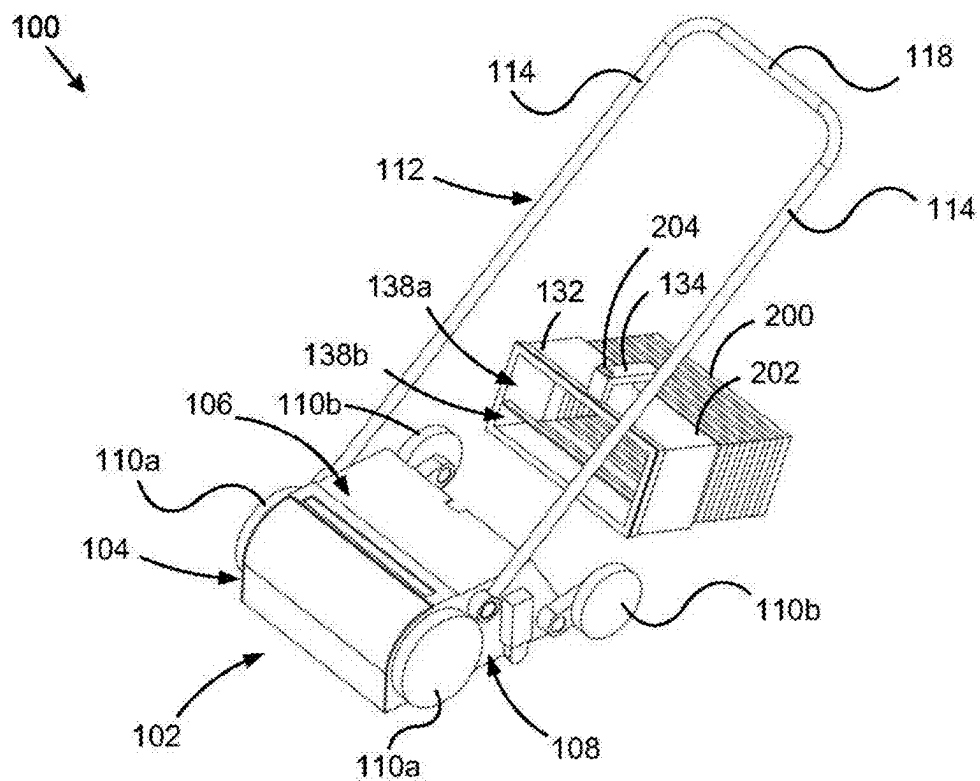


FIG. 50

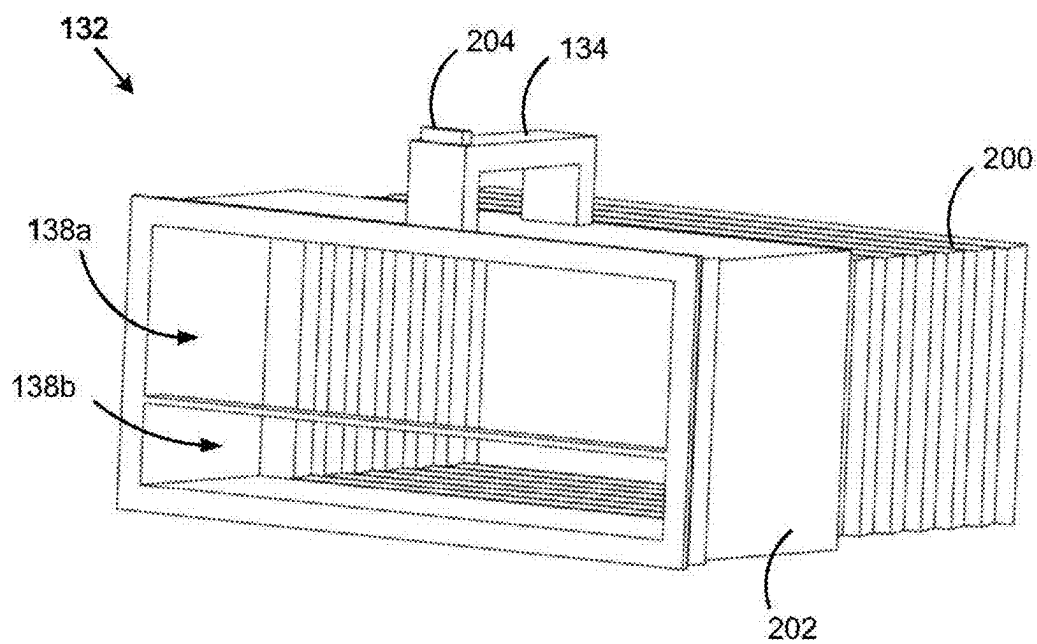


FIG. 51

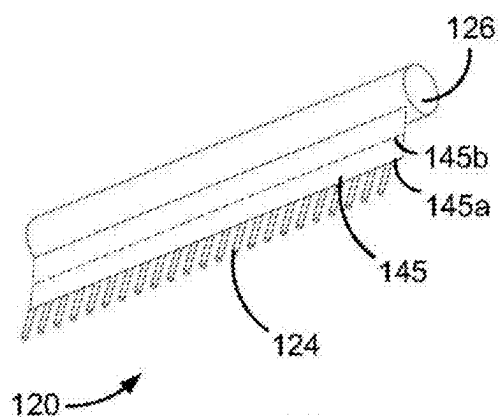


FIG. 52A

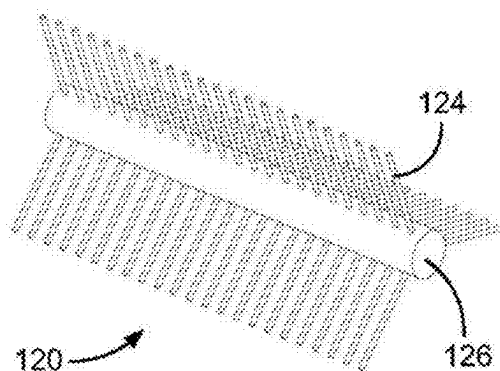


FIG. 52C

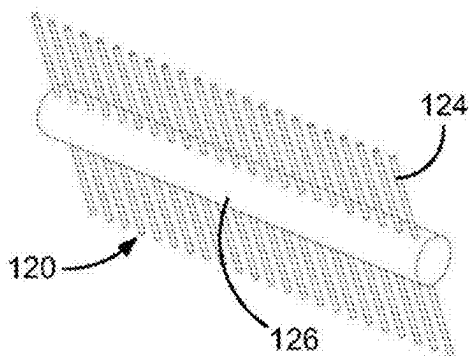


FIG. 52B

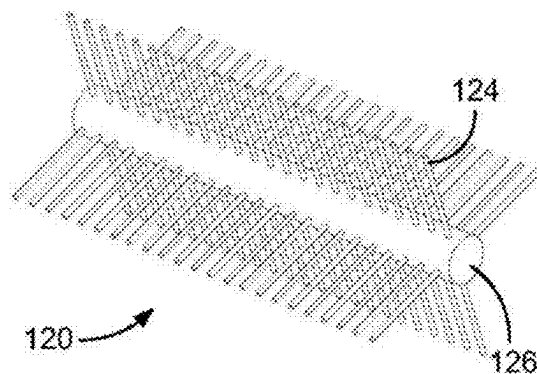


FIG. 52D

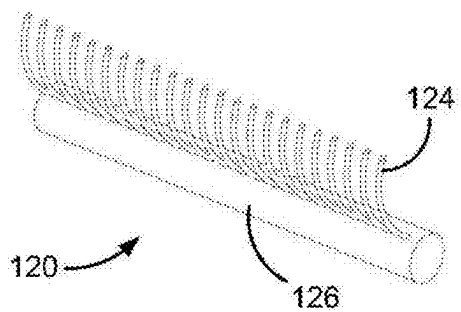


FIG. 53A

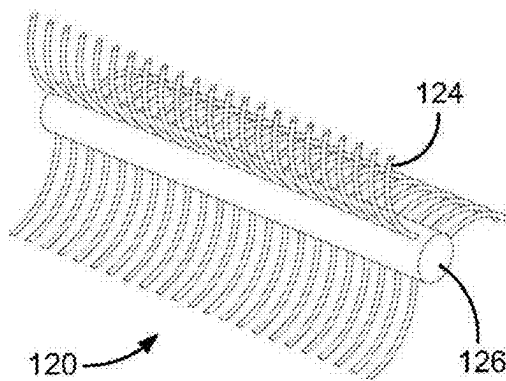


FIG. 53C

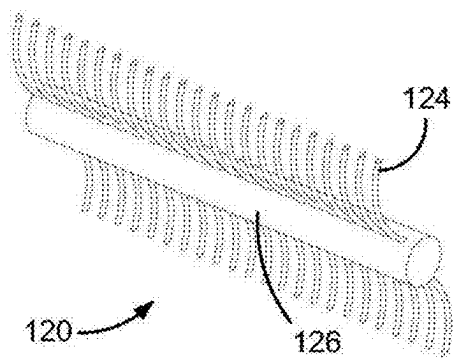


FIG. 53B

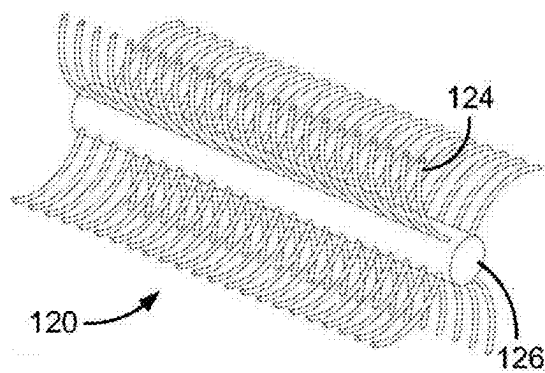


FIG. 53D

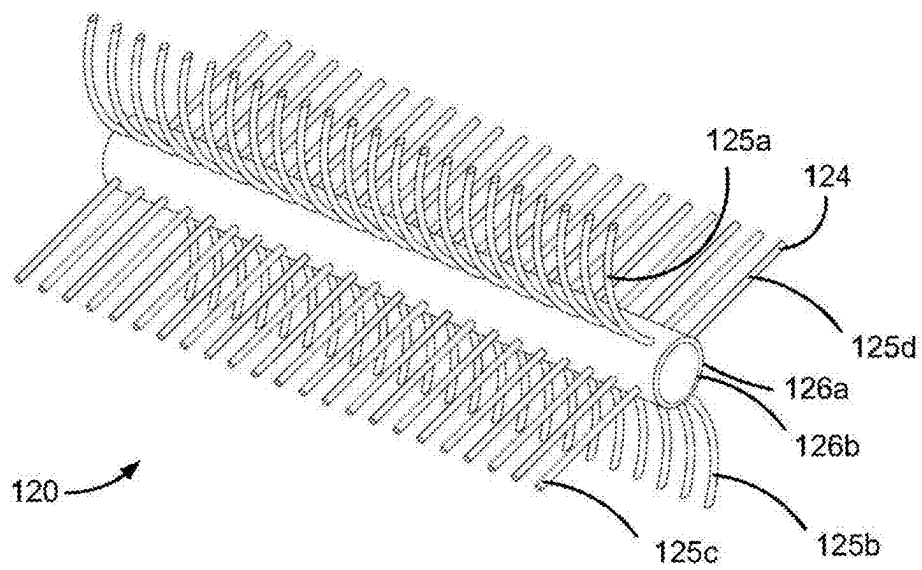


FIG. 54

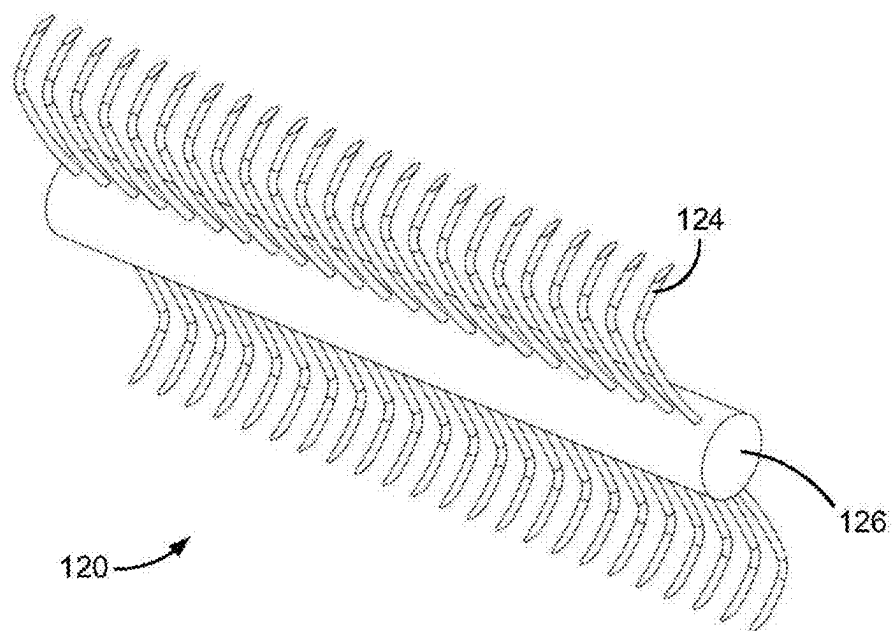


FIG. 55

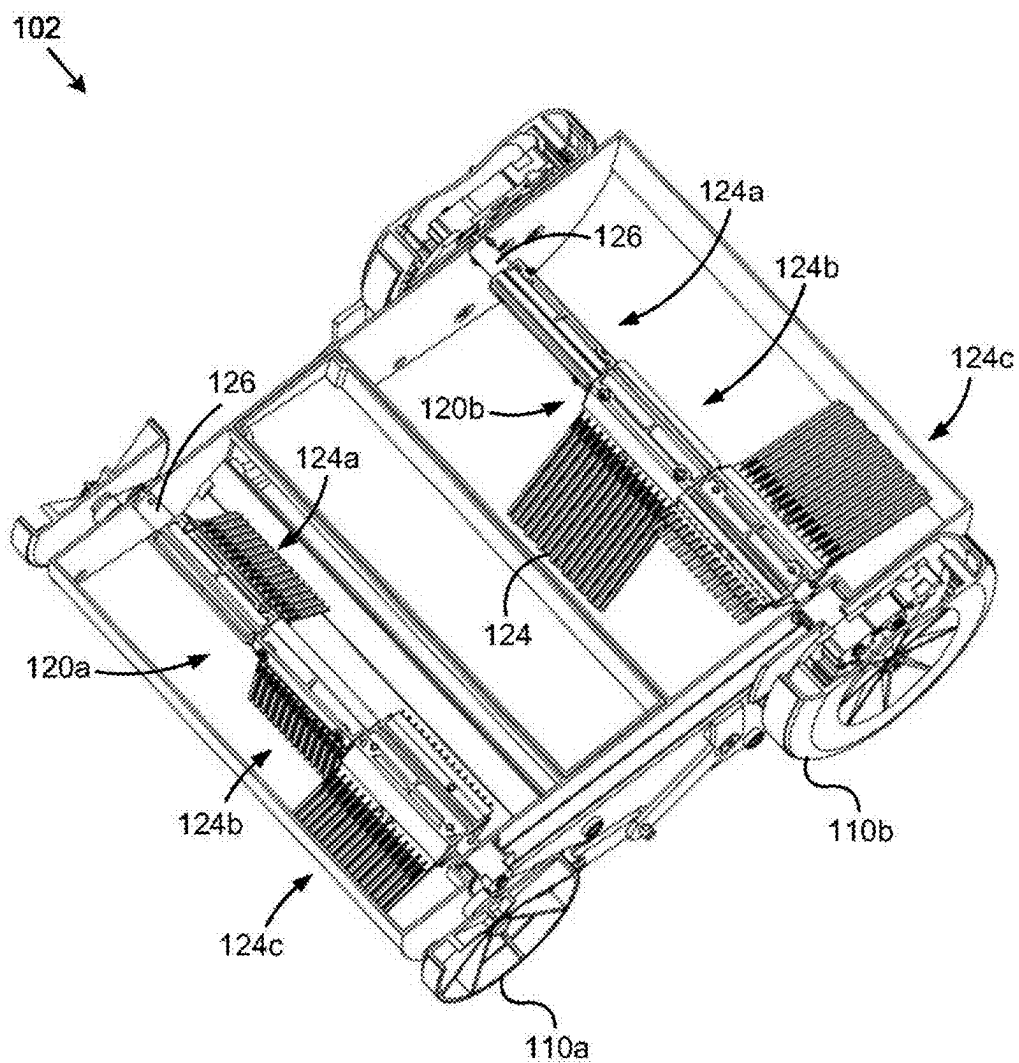


FIG. 56

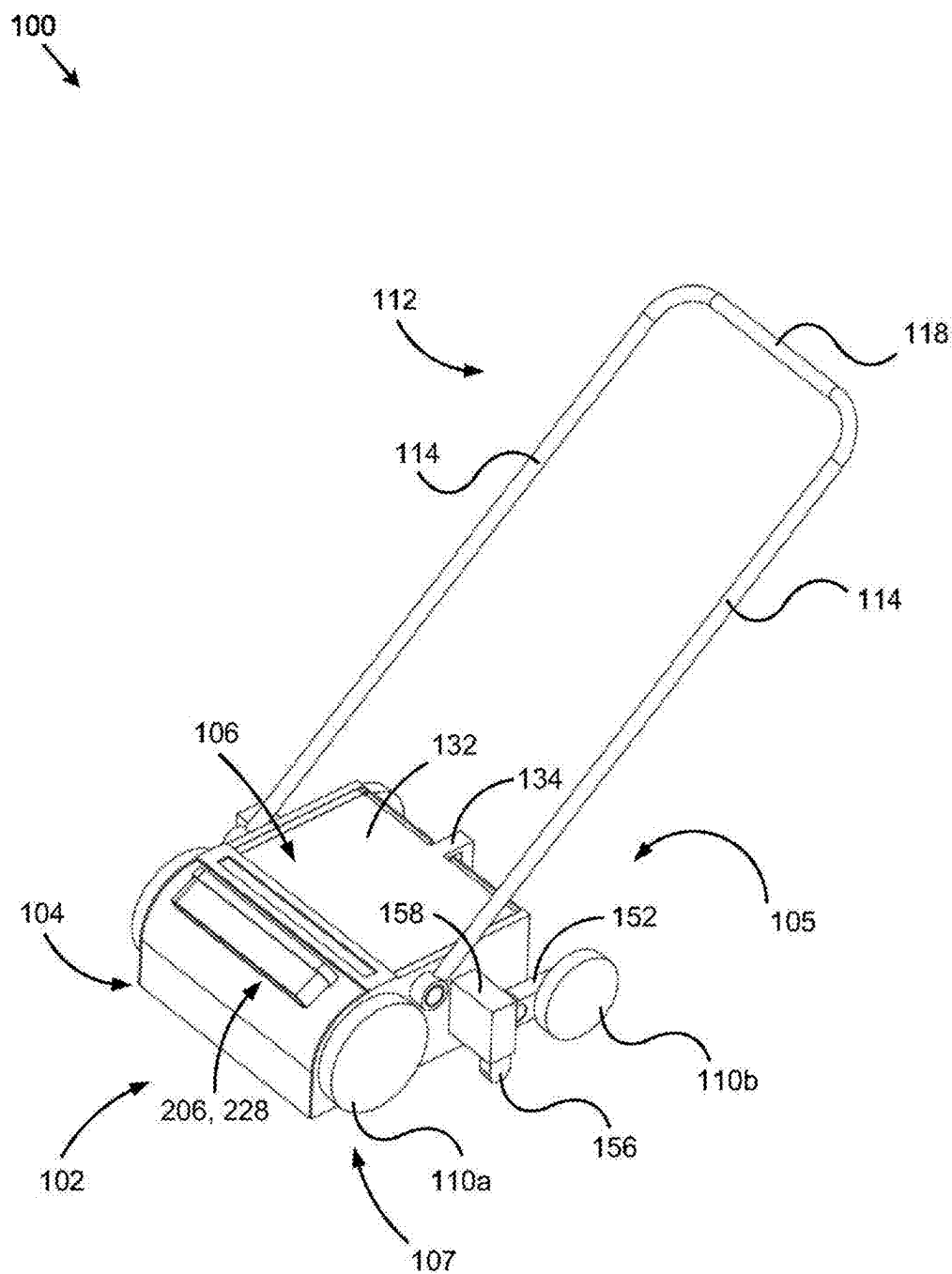


FIG. 57

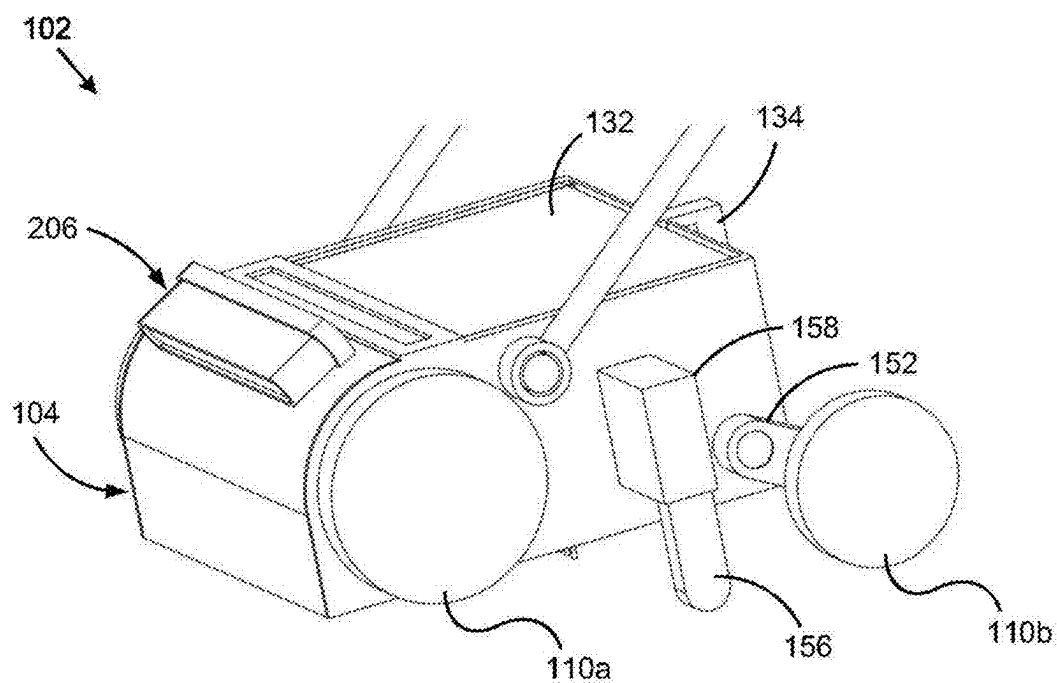


FIG. 58

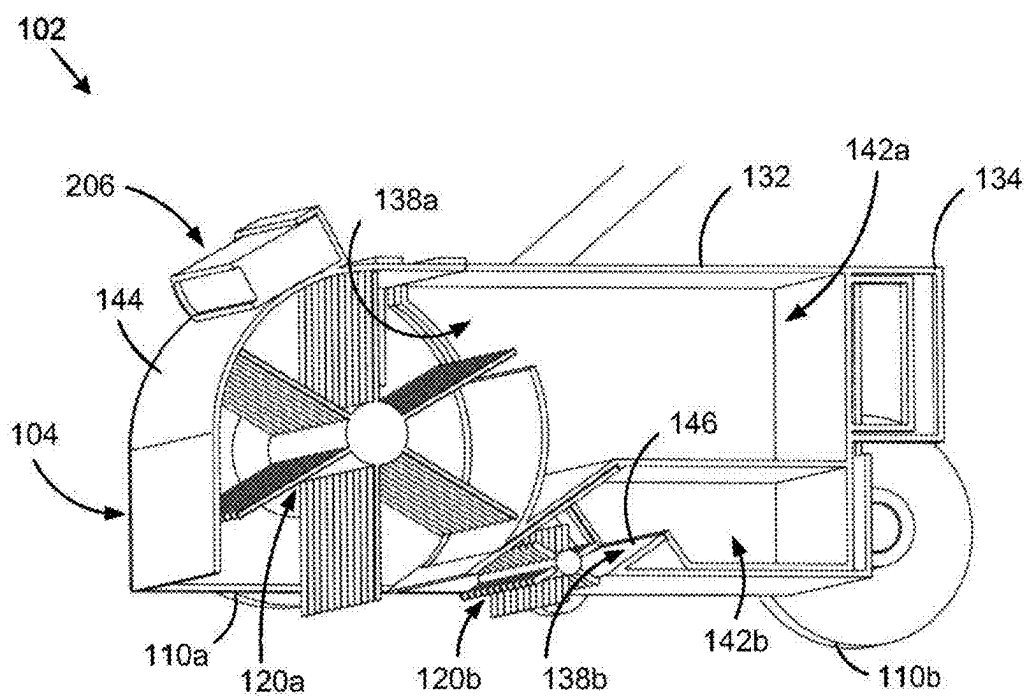


FIG. 59

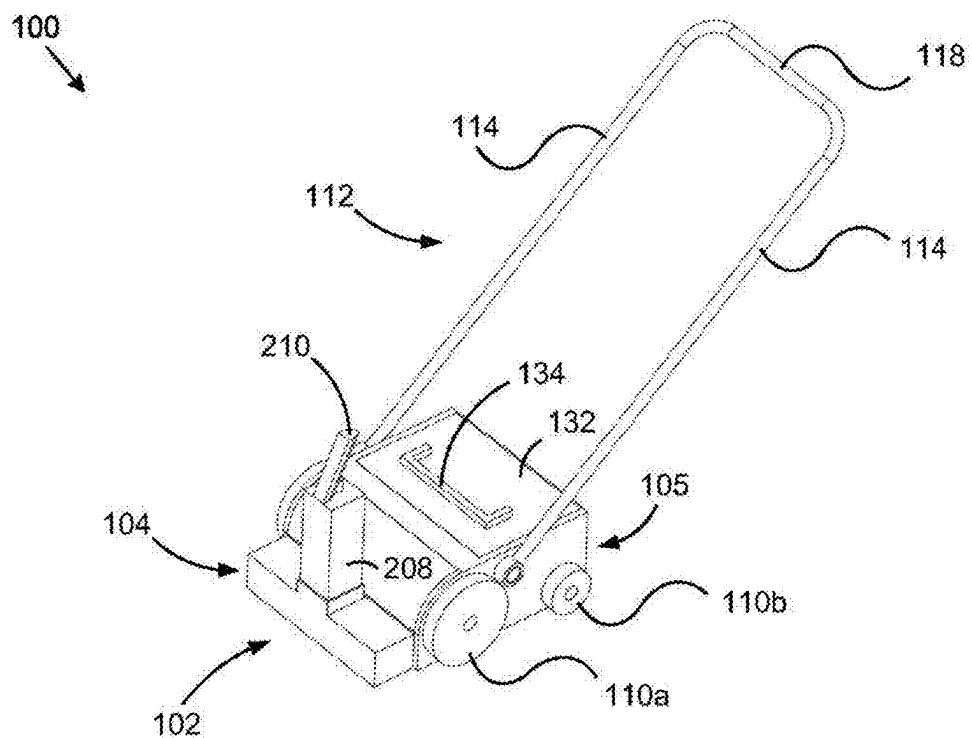


FIG. 60

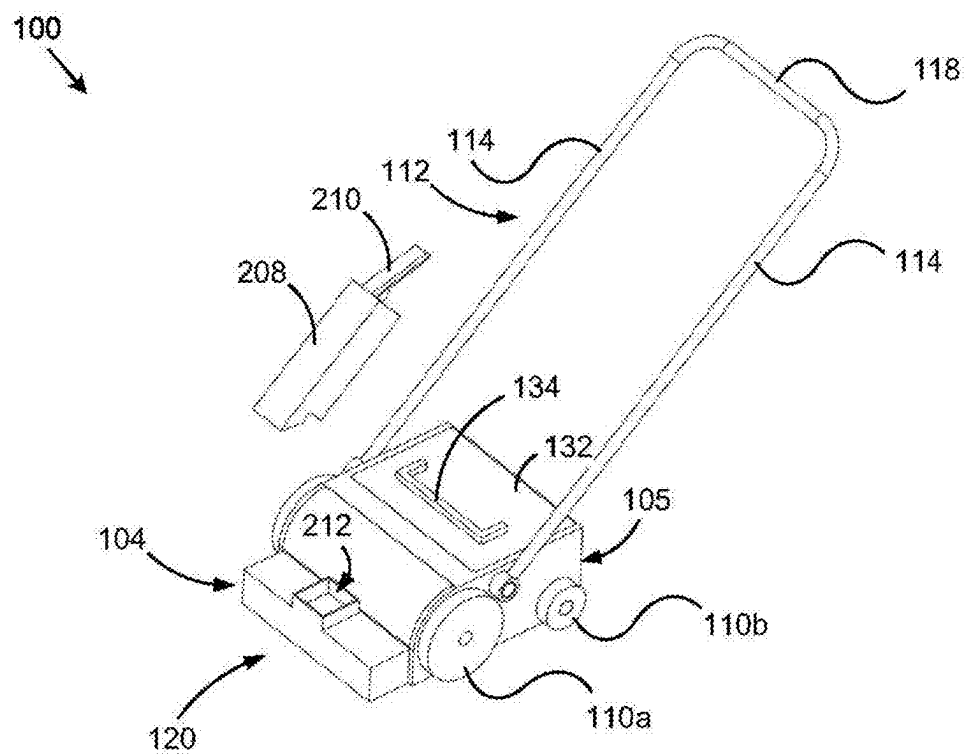
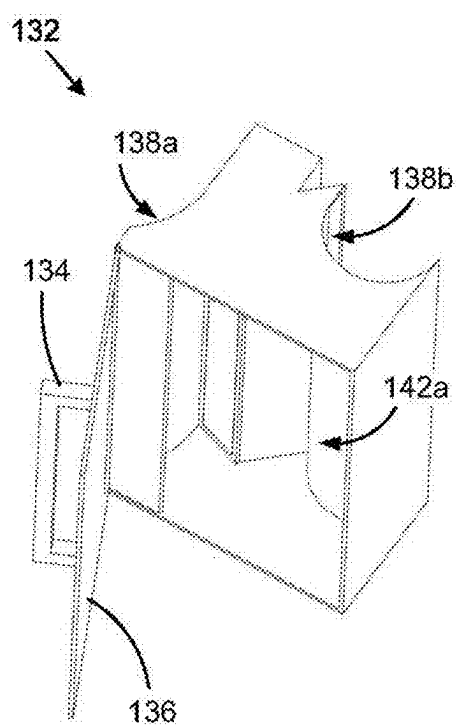
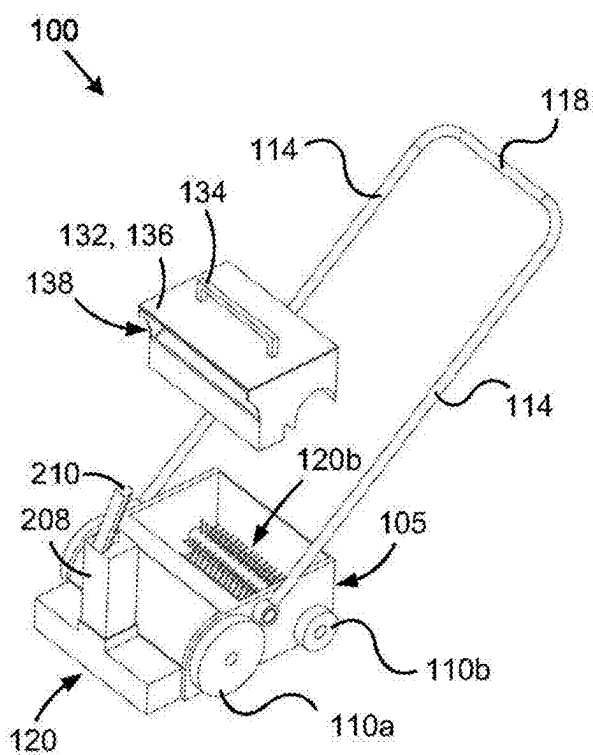
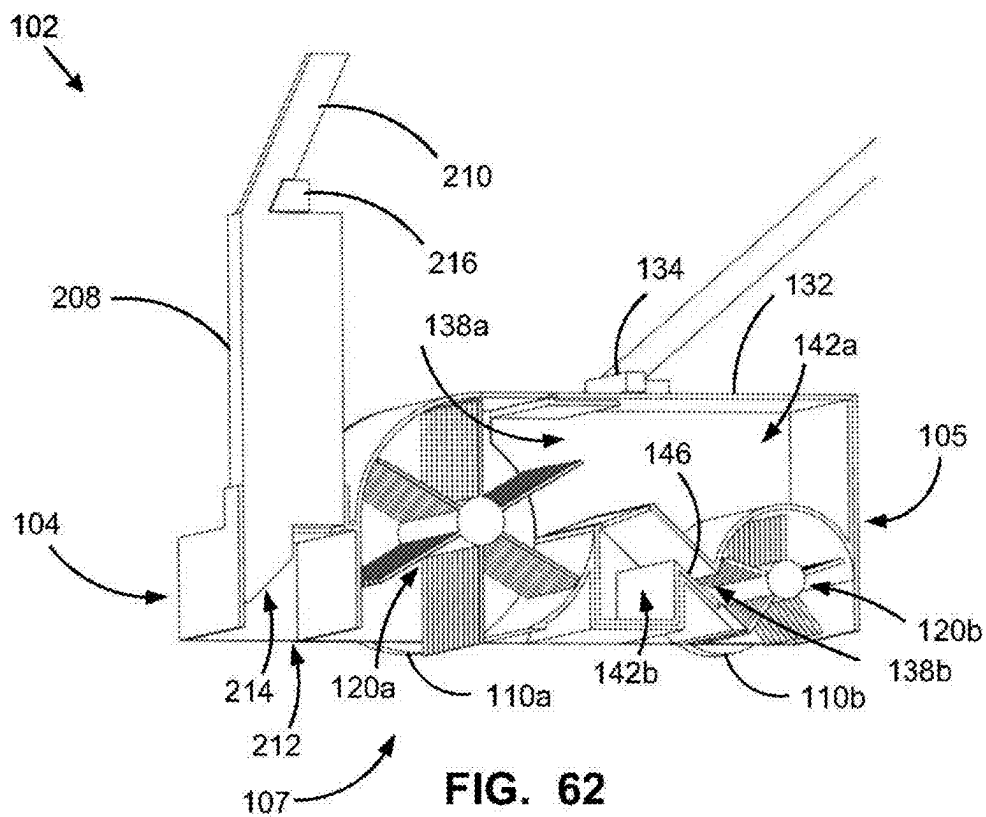


FIG. 61



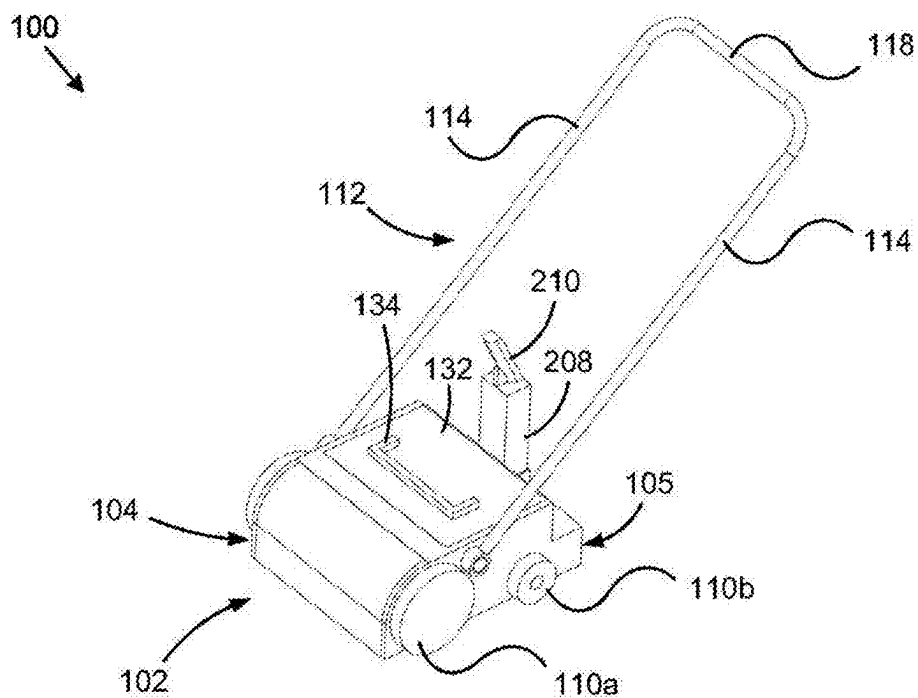


FIG. 65

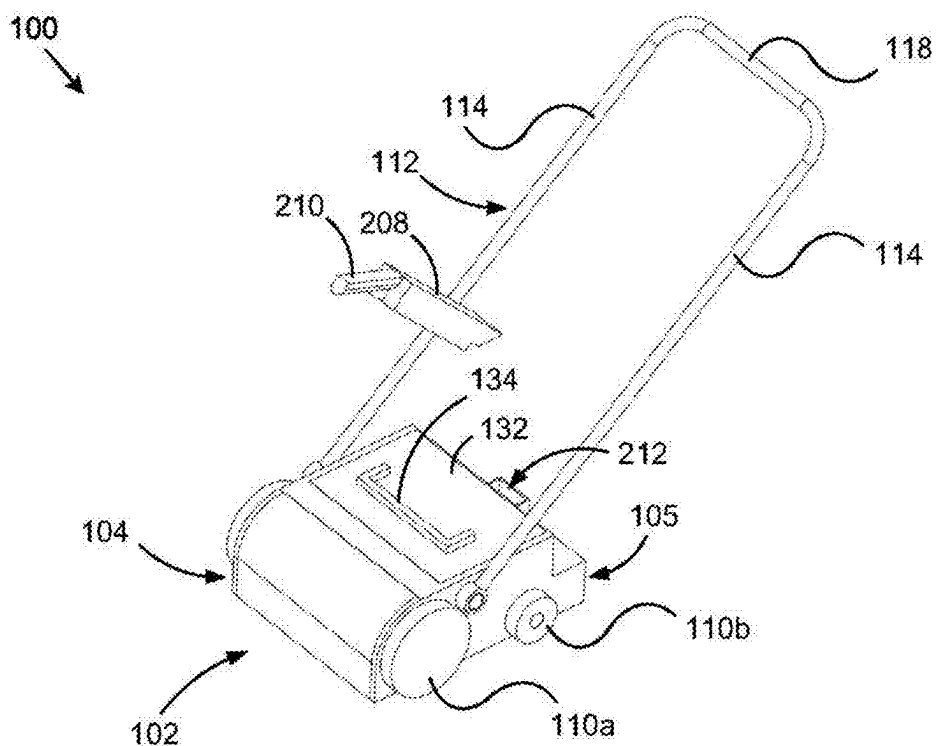


FIG. 66

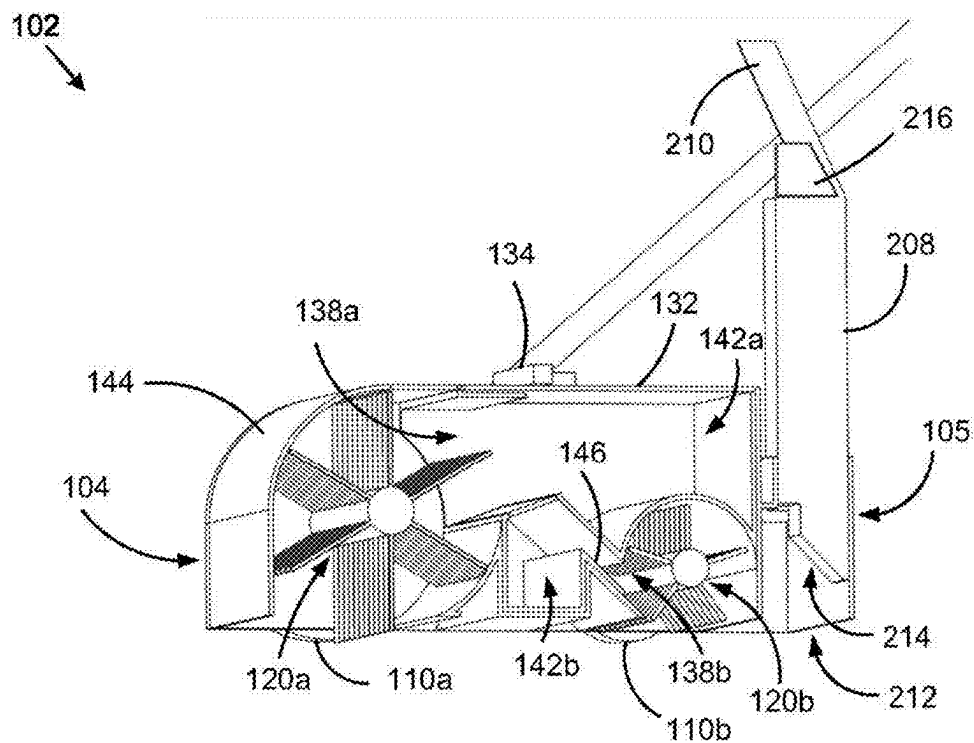


FIG. 67

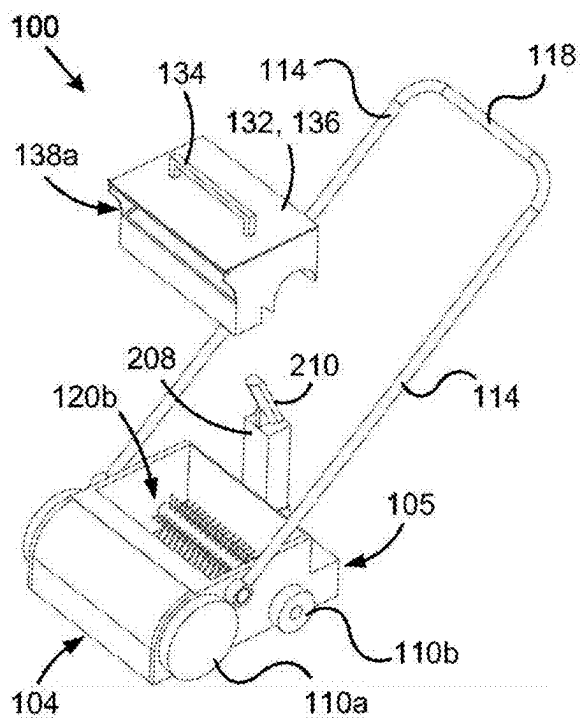


FIG. 68

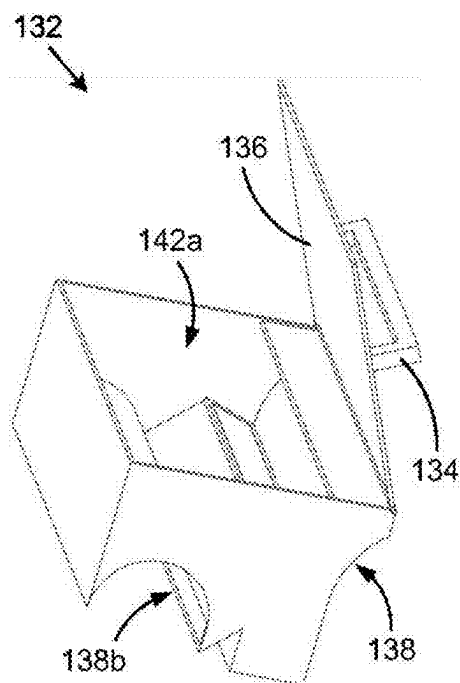


FIG. 69

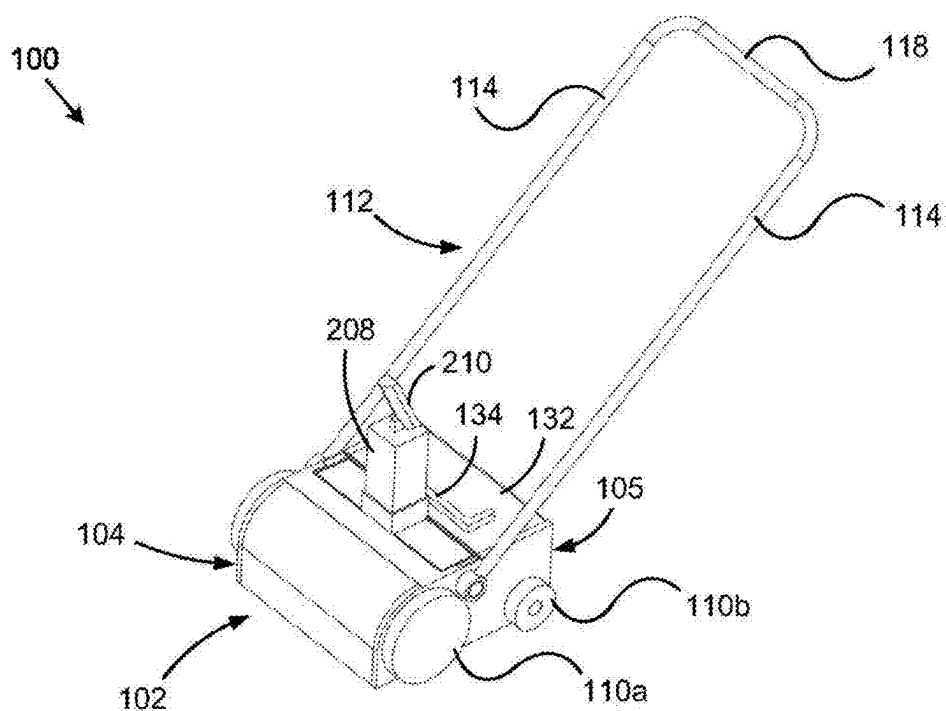


FIG. 70

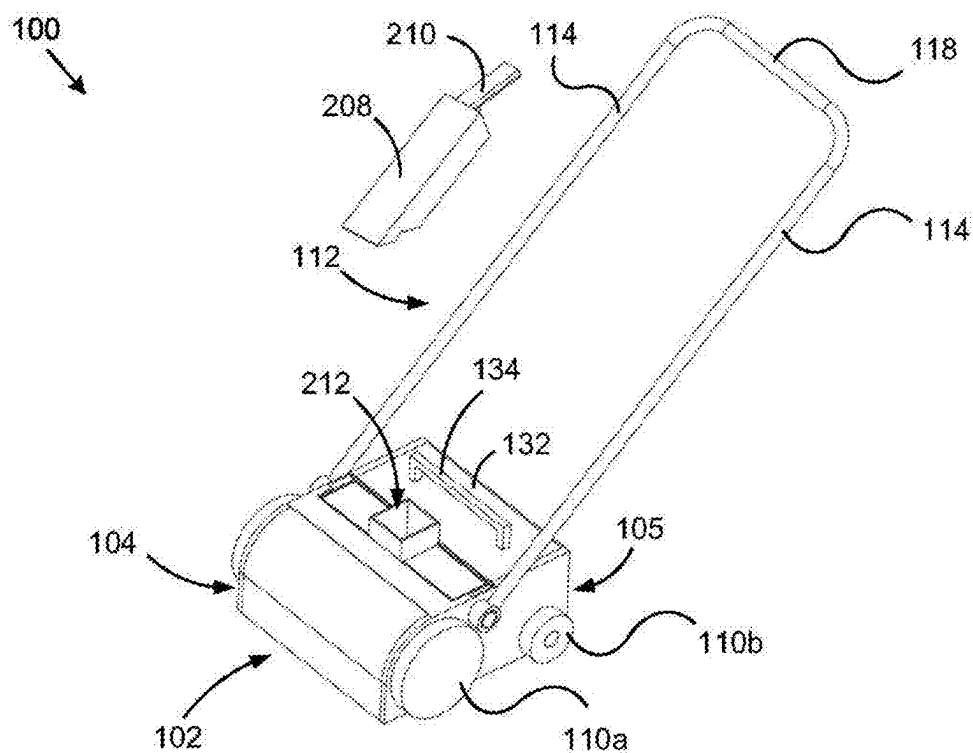


FIG. 71

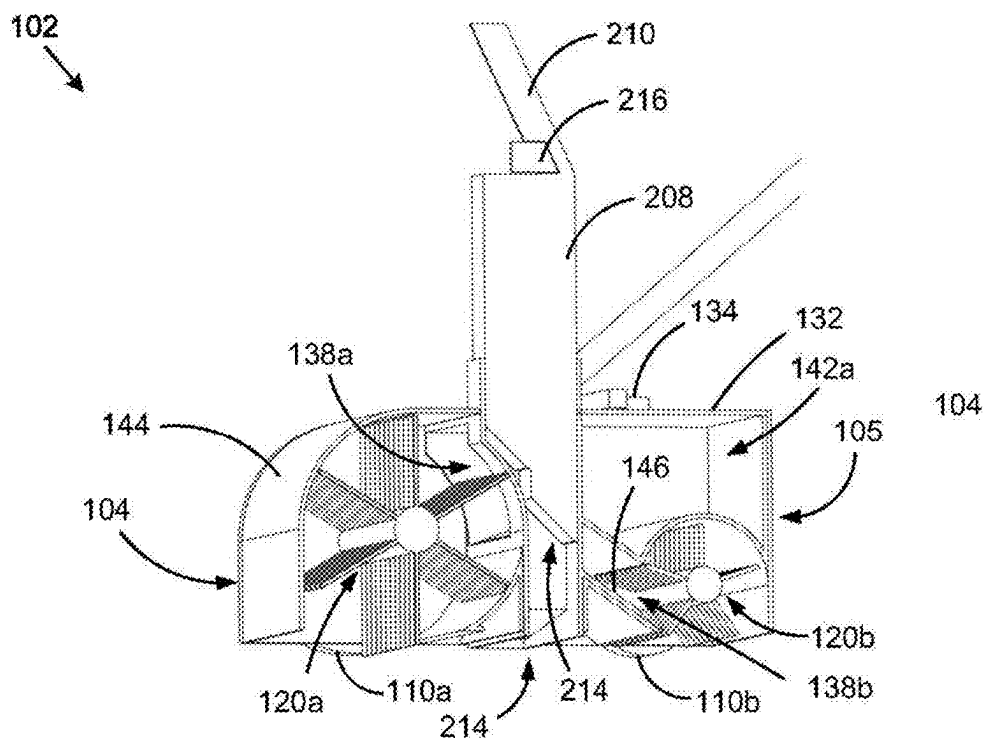


FIG. 72

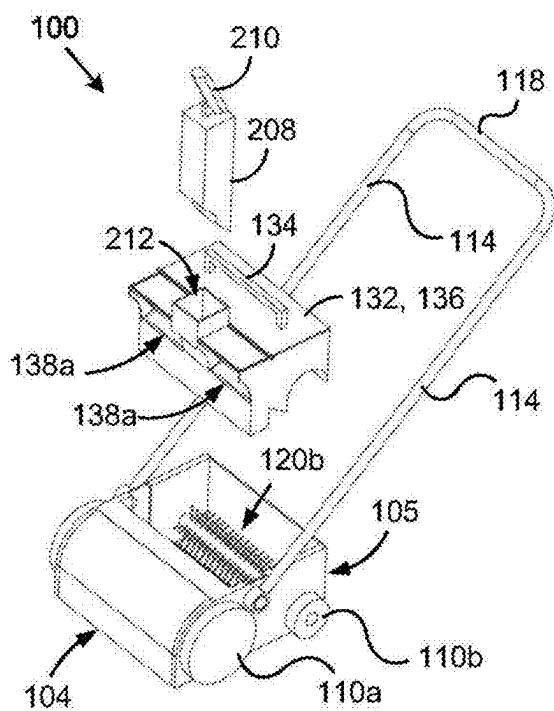


FIG. 73

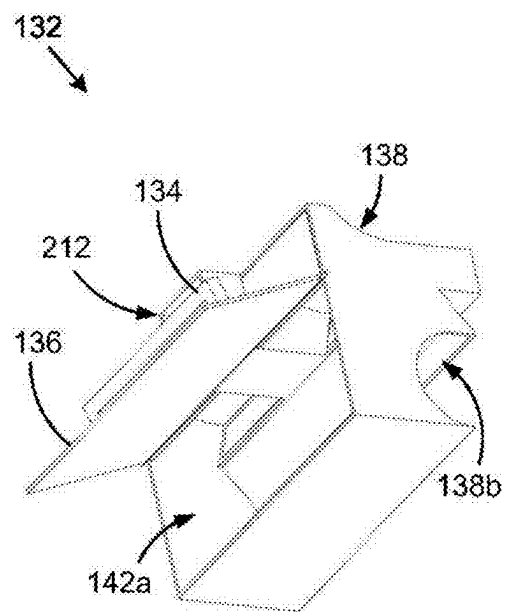


FIG. 74

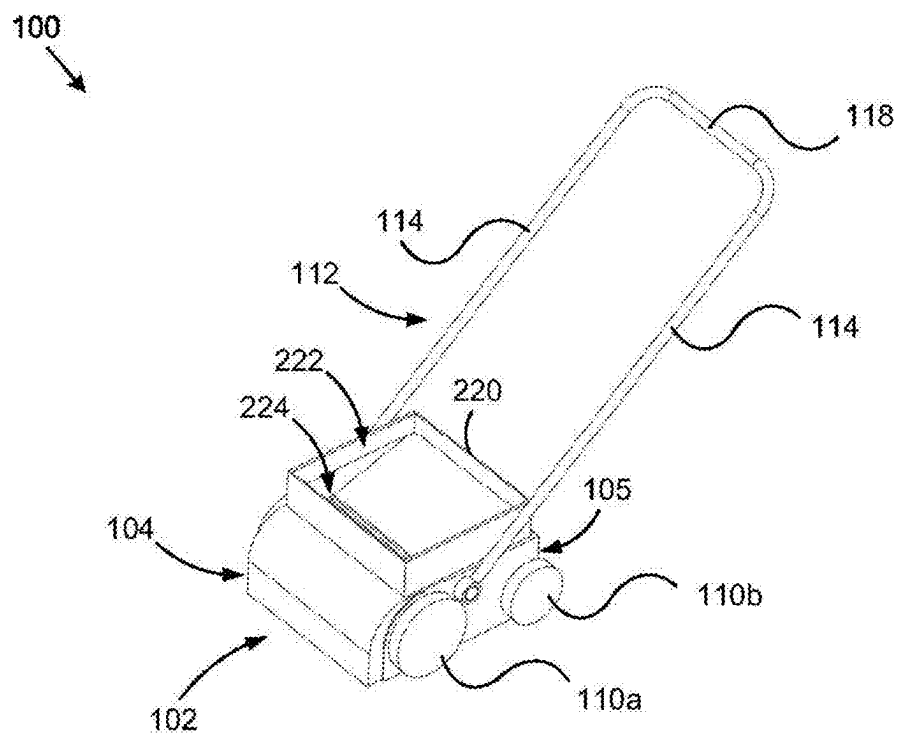


FIG. 75

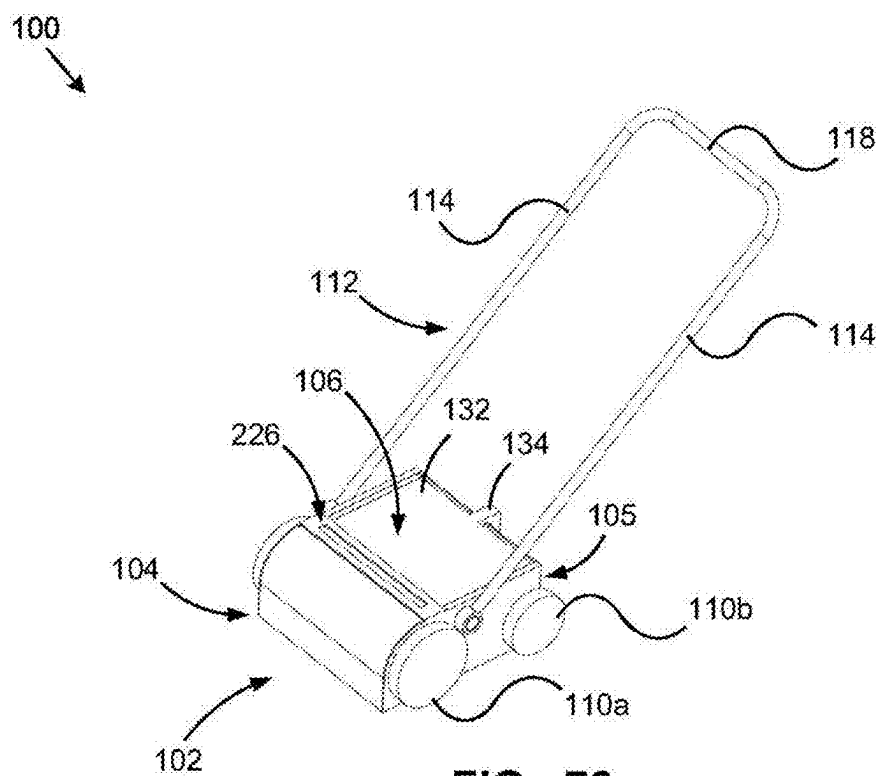


FIG. 76

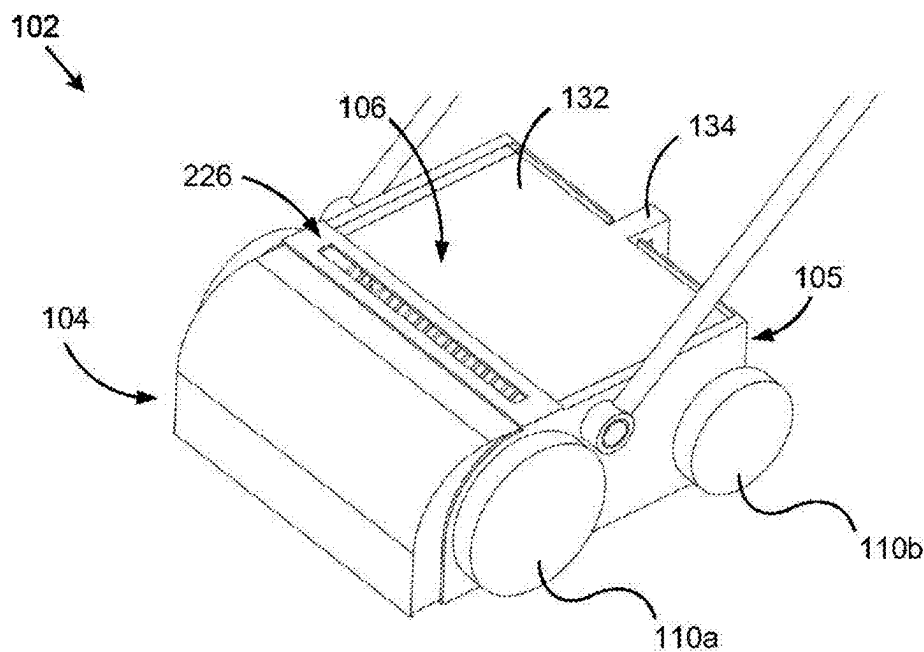


FIG. 77

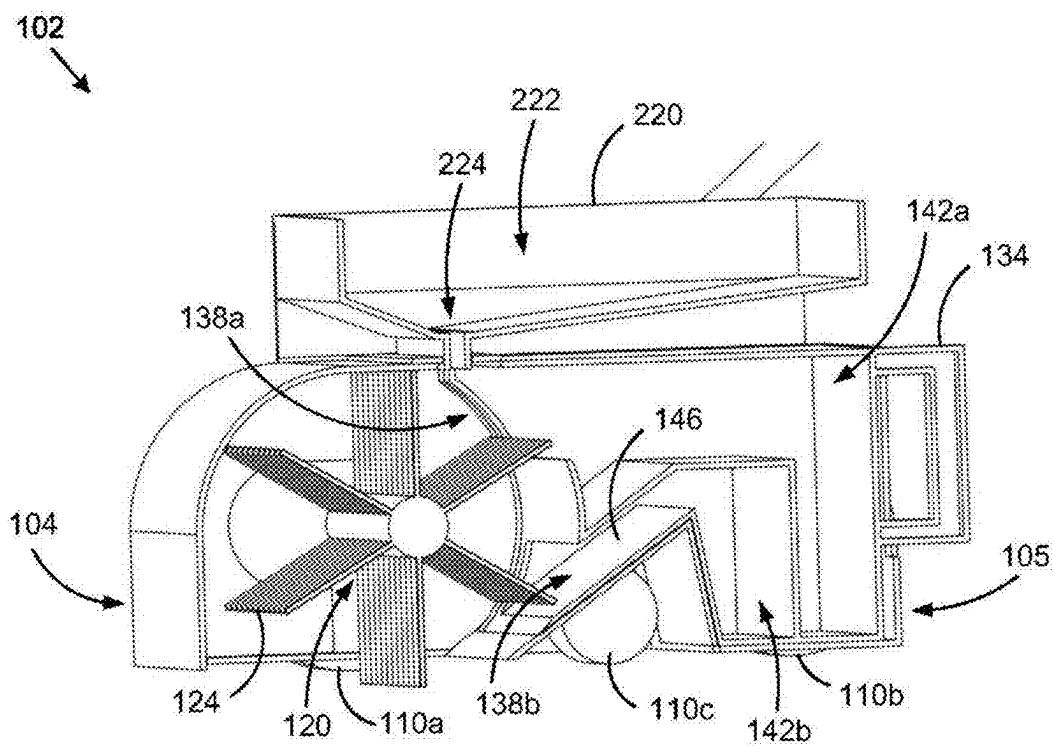


FIG. 78

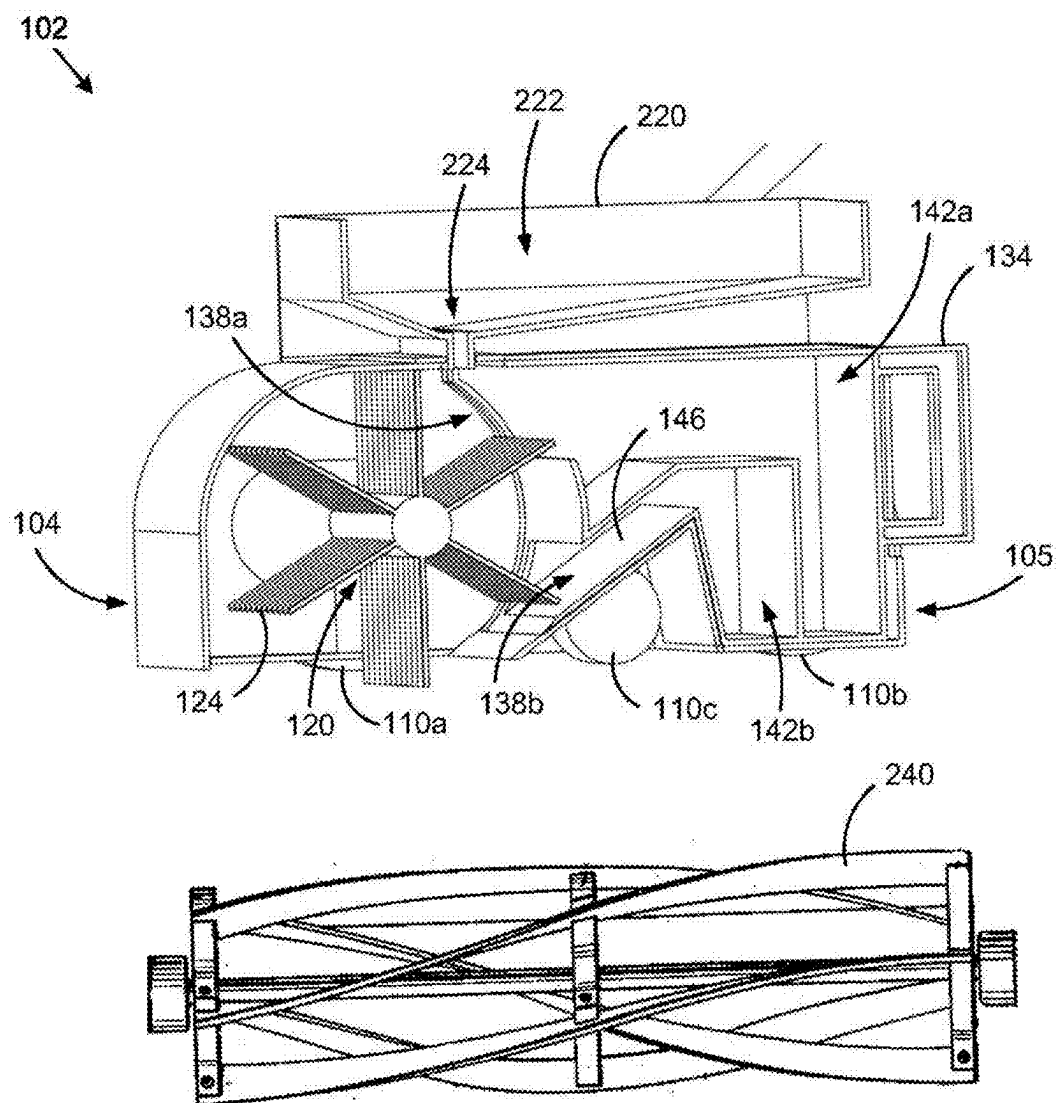


FIG. 79

SURFACE CLEANING APPARATUS

FIELD

[0001] This disclosure relates generally to surface cleaning apparatuses, and more particularly to lawn and patio sweepers.

INTRODUCTION

[0002] The following is not an admission that anything discussed below is part of the prior art or part of the common general knowledge of a person skilled in the art.

[0003] Various constructions for lawn and patio sweepers are known in the art. Known lawn and patio sweepers have a brush that rotates. As the brush rotates, debris such as grass trimmings and leaves may be picked up by the brush and then deposited in a collection bin of the lawn and patio sweeper. When the collection bin is full of debris, a user of the lawn and patio sweeper may empty the collection bin into, for example, a garbage bin or a paper yard waste bag.

SUMMARY

[0004] This summary is intended to introduce the reader to the more detailed description that follows and not to limit or define any claimed or as yet unclaimed invention. One or more inventions may reside in any combination or sub-combination of the elements or process steps disclosed in any part of this document including its claims and figures.

[0005] A surface cleaning apparatus has a surface cleaning head and a drive handle extending from the surface cleaning head. The drive handle includes a hand grip portion at an upper end thereof and the surface cleaning head includes one or more wheels, tractor treads or the like supporting the surface cleaning head on a target surface.

[0006] The surface cleaning apparatus may operate like a sweeper (or with a clean and/or dirty air motor and fan assembly to produce an air flow through the surface cleaning head) with one or more brushes. The brush(es) are rotatably mounted and positioned to engage the target surface on which the head is supported. In operation, the brush(es) rotate about a rotational axis that is generally horizontal and transverse to a direction of motion of the head over the target surface to collect debris that is on the target surface in one or more on board debris collection regions. As set out in more detail subsequently, the brush or brushes may, e.g., operate in different modes and/or be reconfigurable and/or be positioned in different locations and/or operate at different times to pick up the debris and direct it into the debris collection region(s).

[0007] In accordance with an aspect of this disclosure, which may be used alone or in combination with any other aspect, the surface cleaning apparatus includes a large debris path and a small debris path discrete from the large debris path. Different paths for large and small debris allows the paths to be adapted to the type of debris being handled. In use, a surface cleaning apparatus may pick up particulate matter with a wide range of particle sizes. The large debris path may be adapted to handle large debris, such as leaves, grass, long pine needles, and pine cones. The small debris path may be adapted to handle small debris, such as sand, short coniferous needles, dirt, and sawdust. The debris paths may be adapted by being sized differently, such as having different minimum transverse dimensions or different minimum transverse cross sectional areas. The debris paths may

be adapted by including different features, such as different elevation features to lift debris upwards, such as lifting small debris by kicking it up a ramp and lifting large debris by carrying it upward on bristles of the brush. The debris paths may be adapted by leading to different collection regions, such as to allow a user to dispose of different types of debris separately.

[0008] In accordance with this aspect, there is provided a surface cleaning apparatus comprising a surface cleaning head having a front end, a rear end and a brush that is rotatably mounted about a rotation axis wherein, when the surface cleaning head is placed on a horizontal surface and moved in a forward direction with the brush in position to clean the surface, the rotational axis extends generally horizontally and generally transverse to the forward direction, wherein the brush is rotatable in a clockwise direction and wherein the brush is also rotatable in a counter clockwise direction.

[0009] In accordance with an aspect of this disclosure, which may be used alone or in combination with any other aspect, the surface cleaning apparatus includes a large debris brush to pick up large debris, and a small debris brush to pick up small debris. Different brushes for different types of debris allow the brush to be adapted for use with the selected type of debris. The large debris brush is adapted to pick up large debris. The small debris brush is adapted to pick up small debris. In some examples, a large debris brush has a larger diameter and/or longer bristles than a small debris brush. A larger diameter and/or bristle length may be conducive to collecting large debris by providing sufficient spacing between bristles to scoop and transport the large debris. Also, or alternatively, a shorter free extent of the bristles may allow the bristles of the small debris brush to be sufficiently rigid such that the force generated by the bristles returning to their original state after deflecting while in contact with the target surface is sufficient to kick small debris along the flow path (e.g., up a ramp and into a small particulate collection area). The large and small debris brushes may direct debris to different collection regions. The large debris may be collected in a large debris collection region and the small debris may be collected in a discrete small debris collection region, such as to allow a user to selectively dispose of the large or small debris.

[0010] In accordance with this aspect, there is provided a surface cleaning apparatus comprising:

[0011] (a) a surface cleaning head having a first brush which is rotatable about a first rotational axis in a clockwise direction and a second brush which is rotatable about a second rotational axis in a counterclockwise direction;

[0012] (b) a first path from the first brush to a large particulate collection area; and,

[0013] (c) a second path from the first brush to a small particulate collection area

[0014] wherein, when the surface cleaning head is placed on a horizontal surface and moved in a forward direction with the brush in position to clean the surface, the rotational axis extends generally horizontally and generally transverse to the forward direction.

[0015] In accordance with an aspect of this disclosure, which may be used alone or in combination with any other aspect, the surface cleaning apparatus includes a large debris pick up mode and a small debris pick up mode. The apparatus may be selectively switched between the large debris pick up

mode and the small debris pick up mode. Switching between large debris and small debris modes allows a user to direct large and small debris to different collection regions and/or adapt one or more aspects of the apparatus for the selected debris size. In some examples, one or more settings of a brush (e.g., rotational speed and/or direction) are changed between large debris and small debris modes. In some examples, a configuration of a brush is changed between large debris and small debris modes (e.g., the bristles are extended to be longer or retracted to be shorter). In some examples, a dirt flow path is changed (e.g., expanded or contracted) between large debris and small debris modes. In some examples, a brush is switched between operative and inoperative conditions, such as by being moved between raised and lowered positions and/or by toggling a rotational drive system of the brush between on and off conditions between large debris and small debris modes. Alternately, a large debris brush and a small debris brush may be spaced apart (e.g., at opposed ends of a surface cleaning head) and, optionally, the leading brush (in a direction of motion of the surface cleaning head) may be operable and the other may be inoperable.

[0016] In accordance with this aspect, there is provided a surface cleaning apparatus comprising a surface cleaning head, the surface cleaning head comprising:

[0017] (a) a first brush which is located at a front end of the surface cleaning head and wherein the first brush is rotatable in one of a clockwise and counterclockwise direction; and,

[0018] (b) a second brush which is located at a rear end of the surface cleaning head and wherein the second brush is rotatable in the other of the clockwise and counterclockwise direction.

[0019] In accordance with this aspect, there is also provided a surface cleaning apparatus comprising a surface cleaning head having a first end and a second opposed end, the surface cleaning head comprising:

[0020] (a) a first brush which is located at a first end of the surface cleaning head;

[0021] (b) a second brush which is located at a second opposed end of the surface cleaning head; and,

[0022] (c) a drive handle that is rotatable from a first position to a second position wherein, when the handle is in the first position, the handle is positioned outward of the first end of the surface cleaning head whereby the handle is operable by a user to move the surface cleaning head in a first direction, and when the handle is in the second position, the handle is positioned outward of the second end of the surface cleaning head whereby the handle is operable by a user to move the surface cleaning head a second opposed direction,

[0023] wherein the first brush has a larger diameter than the second brush, and

[0024] wherein, when the surface cleaning head is moved in the first direction on a surface, the first brush rotates and the second brush is disengaged from the surface and, when the surface cleaning head is moved in the second direction, the second brush rotates, and

[0025] wherein movement of the handle from the second position to the first position disengages the second brush from the surface

[0026] In accordance with another aspect of this disclosure, which may be used alone or in combination with any other aspect, adjusting a drive handle between a first position

and a second position switches the surface cleaning apparatus between the small debris mode of operation and the large debris mode of operation. The drive handle extends upward from the head to an accessible height, allowing a user to switch between modes without reaching down. In some examples, the handle also extends out from the coupling location a significant distance, giving a mechanical advantage to a user when the user uses the handle as a lever to mechanically adjust a system at the head. Moving the handle may shift, e.g., a clutch system to change how a rotational drive system is engaged with a brush.

[0027] In accordance with this aspect, there is provided a surface cleaning apparatus comprising a surface cleaning head having a first end and a second opposed end, the surface cleaning head comprising:

[0028] (a) a first brush which is located at a first end of the surface cleaning head and which has a first diameter;

[0029] (b) a second brush which is located at a second opposed end of the surface cleaning head and has a second diameter, wherein the diameter of the first brush is larger than the diameter of the second brush; and,

[0030] (c) a drive handle that is rotatable from a first position to a second position wherein, when the handle is in the first position, the handle is positioned outward of the first end of the surface cleaning head whereby the handle is operable by a user to move the surface cleaning head in a first direction, and when the handle is in the second position, the handle is positioned outward of the second end of the surface cleaning head whereby the handle is operable by a user to move the surface cleaning head a second opposed direction,

[0031] wherein, when the surface cleaning head is moved in the first direction on a surface, the first brush rotates and the second brush is inoperable to clean the surface and, when the surface cleaning head is moved in the second direction, the second brush rotates, and

[0032] wherein movement of the handle from the second position to the first position renders the second brush inoperable.

[0033] In accordance with this aspect, there is also provided a surface cleaning apparatus comprising a surface cleaning head having a first end and a second opposed end, the surface cleaning head comprising:

[0034] (a) a first brush which is located at a first end of the surface cleaning head and which has a first diameter;

[0035] (b) a second brush which is located at a second opposed end of the surface cleaning head and has a second diameter, wherein the diameter of the first brush is larger than the diameter of the second brush; and,

[0036] (c) a drive handle that is rotatable from a first position to a second position wherein, when the handle is in the first position, the handle is positioned outward of the first end of the surface cleaning head whereby the handle is operable by a user to move the surface cleaning head in a first direction, and when the handle is in the second position, the handle is positioned outward of the second end of the surface cleaning head whereby the handle is operable by a user to move the surface cleaning head a second opposed direction,

[0037] wherein, when the surface cleaning head is moved in the first direction on a surface, the first brush rotates and the second brush is inoperable to clean the

surface and, when the surface cleaning head is moved in the second direction, the second brush rotates, and

[0038] wherein movement of the handle from the second position to the first position renders the second brush inoperable.

[0039] In accordance with another aspect of this disclosure, which may be used alone or in combination with any other aspect, the collection chamber is removeable in a closed configuration in which the collection region is enclosed. A closed removal configuration inhibits debris (e.g., dust) from exiting and disturbing a user, such as while the user carries the collection chamber to a disposal location.

[0040] In accordance with another aspect of this disclosure, which may be used alone or in combination with any other aspect, the dirt collection chamber includes an openable dumping door. The door may selective close an opening that provides direct access to the collection region within. A direct access dumping door allows the user to more easily remove the debris than if a user was required to dump through the dirt flow path openings which lead into the dirt collection region. Alternatively, the door may selectively close, e.g., the dirt flow path that opens into the collection region. The dumping door is moveable between an open position in which an opening into the collection region is open and a closed position in which the opening into the collection region is closed. The openable dumping door is a wall or part of a wall of the collection chamber that is selectively openable. The door may be a lid, covering an opening through an upper end of the collection chamber.

[0041] In accordance with another aspect of this disclosure, which may be used alone or in combination with any other aspect, the surface cleaning apparatus includes a collection region that is emptiable through another collection region. For example, the small debris collection region may empty through the large debris collection region. In some examples, a dumping opening is an opening in a wall of the large debris collection area, and the large and small debris collection areas are separated by a screen to prevent the large debris from entering the small debris collection areas but allow the small debris to move into the large debris collection area to be dumped through the dumping opening in the large debris collection area.

[0042] In accordance with another aspect of this disclosure, which may be used alone or in combination with any other aspect, the surface cleaning apparatus includes separate large and small debris collection regions and the large and small debris collection regions may be independently removeable. The user may selectively dump the large or small debris collection regions. For example, if a user is primarily collecting large debris, the user may dump the large debris more frequently than the small debris. Also, or alternatively, a user may dump the large debris in one location (e.g., a compost bin) and the small debris in a different location (e.g., a garden).

[0043] In accordance with this aspect, there is provided a surface cleaning apparatus comprising a surface cleaning head having a first end, a second end and a brush assembly wherein, in a first mode of operation, the brush assembly rotates in one of a clockwise and counterclockwise direction and directs large particulate matter to a large particulate collection area and wherein, in a second mode of operate, the brush assembly rotates in the other of the clockwise and the counterclockwise direction and directs small particulate matter to a small particulate collection area that is discrete

from the large particle particulate collection area wherein, when the surface cleaning head is placed on a horizontal surface and moved in a first direction with the brush assembly in position to clean the surface, a rotational axis of the brush assembly extends generally horizontally and generally transverse to the first direction.

[0044] In accordance with another aspect of this disclosure, which may be used alone or in combination with any other aspect, the debris collection region is expandable. The volume of the region is expandable to contain more debris when needed and to take up less space when the collection volume is not needed. In some examples, the collection container includes or consists of a flexible portion, such as plastic or fabric bag. Alternatively, the collection container may be an expandable ridged or semi-rigid container, such as an accordion-style container expandable in sections. A collection container may expand under the weight of the collected debris and thus may provide additional debris collection volume. The collection container may be supported by an expandable frame or other support. A flexible container can form a common collection area or two or more discrete collection areas.

[0045] In accordance with this aspect, there is provided a surface cleaning apparatus comprising a surface cleaning head having a first end, a second end and a brush assembly wherein, in a first mode of operation, the brush assembly rotates in a clockwise direction and in a second mode of operate, the brush assembly rotates in a counterclockwise direction wherein, when the surface cleaning head is placed on a horizontal surface and moved in a first direction with the brush assembly in position to clean the surface, a rotational axis of the brush assembly extends generally horizontally and generally transverse to the first direction and wherein the collection assembly has a volume that is expandable and receives particulate matter in at least one of the first and second modes of operation.

[0046] In accordance with another aspect of this disclosure, which may be used alone or in combination with any other aspect, the surface cleaning apparatus includes a brush with curved bristles. The bristles may be curved in a direction of expected rotation. Curved bristles may form a shape that is conducive to scooping up debris. Also, or alternatively, the curved bristles may be flexed against the curve when encountering the target surface, and resultant force generated by the bristles returning to their original state after disengaging from the target surface may be increased by the curve thus assisting in kicking debris forward.

[0047] In accordance with this aspect, there is provided a surface cleaning apparatus comprising a surface cleaning head, the surface cleaning head comprising:

[0048] (a) a first brush which is rotatable about a first rotational axis in a clockwise direction, wherein the first brush has a first core and a plurality of first bristles which extend outwardly from the first core and at least some of the bristles are curved in the clockwise direction; and,

[0049] (b) a second brush which is rotatable about a second rotational axis in a counterclockwise direction, wherein the second brush has a second core and a plurality of second bristles which extend outwardly from the second core and at least some of the bristles are curved in the counterclockwise direction,

- [0050] wherein, when the surface cleaning head is placed on a horizontal surface and moved in a forward direction with the brush in position to clean the surface, the rotational axis extends generally horizontally and generally transverse to the forward direction.
- [0051] In accordance with this aspect, there is also provided a surface cleaning apparatus comprising a surface cleaning head, the surface cleaning head comprising:
- [0052] (a) a first brush which is rotatable about a first rotational axis in a clockwise direction, wherein the first brush has a first core and a plurality of first bristles which extend outwardly from the first core and at least some of the bristles are curved in the clockwise direction; and,
- [0053] (b) a second brush which is rotatable about a second rotational axis in a counterclockwise direction, wherein the second brush has a second core and a plurality of second bristles which extend outwardly from the second core,
- [0054] wherein, when the surface cleaning head is placed on a horizontal surface and moved in a forward direction with the brush in position to clean the surface, the rotational axis extends generally horizontally and generally transverse to the forward direction
- [0055] In accordance with this aspect, there is also provided a surface cleaning apparatus comprising a surface cleaning head, the surface cleaning head comprising:
- [0056] (a) a first brush which is rotatable about a first rotational axis in a clockwise direction, wherein the first brush has a first core and a plurality of first bristles which extend outwardly from the first core; and,
- [0057] (b) a second brush which is rotatable about a second rotational axis in a counterclockwise direction, wherein the second brush has a second core and a plurality of second bristles which extend outwardly from the core and at least some of the bristles are curved in the counterclockwise direction,
- [0058] wherein, when the surface cleaning head is placed on a horizontal surface and moved in a forward direction with the brush in position to clean the surface, the rotational axis extends generally horizontally and generally transverse to the forward direction.
- [0059] In accordance with another aspect of this disclosure, which may be used by itself or in combination with one or more other aspects of this disclosure, the surface cleaning apparatus can further include a suction assembly. The suction assembly includes an air flow path extending between an air inlet and an air outlet, with an air moving member (e.g., a clean or dirty air motor and fan assembly) in the air flow path. The suction assembly further includes an air treatment member (e.g., a filter bag, cyclone, or momentum separator) in the air flow path to remove dirt from an air flow in the air flow path. The suction assembly may cooperate with the brush to pick up debris, such as with the suction air inlet opening into a dirt flow path downstream of a dirt inlet through which the brush engages the target surface to draw debris toward the brush. Alternative, the suction assembly may work independently of the brush, such as with the suction assembly air inlet rearward of the dirt inlet through which the brush may engage the target surface to pick up fine debris that is not picked up by the brush. The debris picked up by the suction assembly may be collected in a common collection region with debris picked up by a brush or brushes or may be collected separately as discussed herein. The

suction assembly may be a removable and independently operable vacuum, such as a hand vacuum that is selectively dockable with the head to cooperate with the head in cleaning the target surface.

[0060] In accordance with another aspect of this disclosure, which may be used by itself or in combination with one or more other aspects of this disclosure, the surface cleaning apparatus includes a fence projecting down from the bottom end of the apparatus adjacent a dirt inlet, optionally at a rearward end of the dirty air inlet. The fence may extend along the dirt inlet, optionally the fence and the dirt inlet are generally coextensive. The fence improves collection efficiency of the surface cleaning apparatus by blocking debris that the dirt inlet would otherwise pass over. In some examples, the downwardly extending fence is a passive brush (i.e., a non-rotatable brush). Optionally, the fence includes a row of bristles extending downward from the head. The fence may be selectively lowered (e.g., raised when the surface cleaning head is reversing).

[0061] In accordance with another aspect of this disclosure, which may be used by itself or in combination with one or more other aspects of this disclosure, the surface cleaning apparatus includes a dispensing reservoir, which may be removably mountable, e.g., with a dispensing pump so as to provide a removably mountable reservoir module, to the surface cleaning apparatus. A dispensing reservoir allows the apparatus to be used in additional ways. The apparatus may dispense a material across a surface, using the brush to distribute the material (e.g., grass seed or fertilizer). The apparatus may dispense cleaning fluids (e.g., water and/or soap) onto the target surface, using the brush to scrub the surface.

[0062] In accordance with another aspect of this disclosure, which may be used alone or in combination with any other aspect, the surface cleaning apparatus is switchable between a debris collection mode and a scrubbing mode. In a scrubbing mode, the speed of rotation of a brush may be faster than in a debris collection mode. In a scrubbing mode, the brush may be brought closer to the target surface than in a debris collection mode. In a scrubbing mode, the apparatus may dispense a cleaning fluid such as water and/or soap onto the target surface and/or onto the brush.

[0063] In accordance with another aspect of this disclosure, which may be used by itself or in combination with one or more other aspects of this disclosure, the surface cleaning apparatus includes one or more lights to illuminate the target surface. Lights may aid the user in identifying debris on the surface to be cleaned, such as under low light conditions or debris that may otherwise blend with the surface. Lights may be low-power lights, such as light emitting diodes. Lights may be directed forward and/or rearward of the head.

[0064] In accordance with another aspect of this disclosure, which may be used by itself or in combination with one or more other aspects of this disclosure, the surface cleaning apparatus includes one or more magnets to pick up metallic components, such as when the surface cleaning apparatus is used to clean a shop floor or to clean a lawn following a construction project wherein nails or other metallic components may be hidden in the grass. The magnet may be provided (e.g., removably mounted as a magnet module) to an underside of the head to pass over the target surface in close proximity to the target surface. The magnet may be forward of a dirt inlet in the direction of travel to pick up the metallic component before it is encountered by a brush.

Alternatively, the magnet may be rearward of the dirt inlet in the direction of travel to pick up metallic components that are missed by the brush. It will be appreciated that the magnet may be used without operation of the brush in some examples, such as to pick up metallic components without sweeping.

[0065] In accordance with another aspect of this disclosure, which may be used by itself or in combination with one or more other aspects of this disclosure, the surface cleaning apparatus is selectively reconfigurable as a mower. A brush may be replaced with a rotatable mower blade, such as a reel blade. The rotatable mower blade may be driven by the brush drive system in a rotational movement, such as to cut grass or weeds. In some examples, the surface cleaning apparatus may be provided with a brush and with an interchangeable rotatable mower blade (e.g., the surface cleaning head may selectively receive a brush module and a mower module). The brush and/or mower blade may be secured to the head and/or drive system by an easy-release fastener for easy interchangeability.

[0066] It will be appreciated by a person skilled in the art that an apparatus or method disclosed herein may embody any one or more of the features contained herein and that the features may be used in any particular combination or sub-combination.

[0067] These and other aspects and features of various embodiments will be described in greater detail below.

BRIEF DESCRIPTION OF THE DRAWINGS

[0068] For a better understanding of the described embodiments and to show more clearly how they may be carried into effect, reference will now be made, by way of example, to the accompanying drawings in which:

[0069] FIG. 1 is a perspective view of an example surface cleaning apparatus;

[0070] FIG. 2 is a rear perspective view of the surface cleaning apparatus of FIG. 1;

[0071] FIG. 3A is a cross sectional view of an embodiment of the surface cleaning apparatus of FIG. 1 having a common collection area;

[0072] FIG. 3B is a cross sectional view of an embodiment of the surface cleaning apparatus of FIG. 1 having discrete collection areas;

[0073] FIG. 4 is a perspective view of the surface cleaning apparatus of FIG. 1 with a collection container removed;

[0074] FIG. 5 is a perspective view of the collection container of FIG. 5 wherein the collection container is opened and inverted for emptying;

[0075] FIG. 6 is a perspective view of another example surface cleaning apparatus;

[0076] FIG. 7 is a rear perspective view of the surface cleaning apparatus of FIG. 6;

[0077] FIG. 8 is a cross sectional view of the surface cleaning apparatus of FIG. 6;

[0078] FIG. 9 is a perspective view of the surface cleaning apparatus of FIG. 6 with a collection container removed;

[0079] FIG. 10 is a perspective view of the collection container of the surface cleaning apparatus of FIG. 6 with a lid opened for emptying;

[0080] FIG. 11 is a perspective view of another example surface cleaning apparatus;

[0081] FIG. 12 is a rear perspective view of the surface cleaning apparatus of FIG. 11;

[0082] FIG. 13 is a cross sectional view of the surface cleaning apparatus of FIG. 11;

[0083] FIG. 14 is a perspective view of the surface cleaning apparatus of FIG. 11 with a collection container removed;

[0084] FIG. 15 is a perspective view of the collection container of the surface cleaning apparatus of FIG. 11 with a lid opened for emptying;

[0085] FIG. 16 is a perspective view of another example surface cleaning apparatus;

[0086] FIG. 17 is a cross sectional view of the surface cleaning apparatus of FIG. 16;

[0087] FIG. 18A is a perspective view of the surface cleaning apparatus of FIG. 16 with a collection container removed;

[0088] FIG. 18B is a perspective view of the surface cleaning apparatus of FIG. 16 with a large particulate collection container removed;

[0089] FIG. 18C is a perspective view of the surface cleaning apparatus of FIG. 16 with a small particulate collection container removed;

[0090] FIG. 19 is a side cross-sectional view of another example surface cleaning apparatus;

[0091] FIG. 20A is an expanded view of a front wheel of the surface cleaning apparatus of FIG. 19;

[0092] FIG. 20B is an expanded view of a rear wheel of the surface cleaning apparatus of FIG. 19;

[0093] FIG. 21 is a perspective view of another example surface cleaning apparatus with a drive handle in a first position;

[0094] FIG. 22 is a perspective view of the surface cleaning apparatus of FIG. 21 with the drive handle in a second position;

[0095] FIG. 23 is a side view of the surface cleaning apparatus of FIG. 21 with the drive handle in the first position;

[0096] FIG. 24 is a side cross-sectional view of the surface cleaning apparatus of FIG. 21 with the drive handle in the first position;

[0097] FIG. 25 is a side view of the surface cleaning apparatus of FIG. 21 with the drive handle in the second position;

[0098] FIG. 26 is a side cross-sectional view of the surface cleaning apparatus of FIG. 21 with the drive handle in the second position;

[0099] FIG. 27 is a perspective cross-sectional view of the surface cleaning apparatus of FIG. 21 with the drive handle in the first position;

[0100] FIG. 28 is a side cross-sectional view of the surface cleaning apparatus of FIG. 21 with the drive handle in the first position;

[0101] FIG. 29 is a perspective cross-sectional view of the surface cleaning apparatus of FIG. 21 with the drive handle in the second position;

[0102] FIG. 30 is a side cross-sectional view of the surface cleaning apparatus of FIG. 21 with the drive handle in the second position;

[0103] FIG. 31 is a perspective view of another example surface cleaning apparatus;

[0104] FIG. 32 is an expanded partial cross-sectional view of a surface cleaning head of the surface cleaning apparatus of FIG. 31 showing a clutch assembly in a second position engaged with a first tier of a front wheel gear;

[0105] FIG. 33 is an expanded partial cross-sectional view of the surface cleaning head of the surface cleaning apparatus of FIG. 31 showing the clutch assembly in a first position engaged with the first tier of the front wheel gear;

[0106] FIG. 34 is a top perspective cross-sectional view of the surface cleaning apparatus of FIG. 31 showing the clutch assembly in the first position engaged with the first tier of the front wheel gear;

[0107] FIG. 35 is an expanded view of the clutch assembly of the surface cleaning apparatus of FIG. 31 showing the clutch assembly in the second position engaged with the first tier of the front wheel gear;

[0108] FIG. 36 is an expanded view of the clutch assembly of the surface cleaning apparatus of FIG. 31 showing the clutch assembly in the second position engaged with the second tier of the front wheel gear;

[0109] FIG. 37 is an expanded partial cross-sectional view of the surface cleaning head of the surface cleaning apparatus of FIG. 31 showing the clutch assembly disengaged from the front wheel gear;

[0110] FIG. 38 is a top perspective cross-sectional view of the surface cleaning apparatus of FIG. 31 showing the clutch assembly disengaged from the front wheel gear;

[0111] FIG. 39 is a top perspective cross-sectional view of the surface cleaning apparatus of FIG. 31 showing the clutch assembly in the first position engaged with the second tier of the front wheel gear;

[0112] FIG. 40A is an expanded side view of the clutch assembly of the surface cleaning apparatus of FIG. 31 showing the clutch assembly in the second position engaged with the first tier of the front wheel gear;

[0113] FIG. 40B is an expanded side view of the clutch assembly of the surface cleaning apparatus of FIG. 31 showing the clutch assembly in the first position engaged with the first tier of the front wheel gear;

[0114] FIG. 40C is an expanded side view of the clutch assembly of the surface cleaning apparatus of FIG. 31 showing the clutch assembly in the second position engaged with the second tier of the front wheel gear;

[0115] FIG. 40D is an expanded side view of the clutch assembly of the surface cleaning apparatus of FIG. 31 showing the clutch assembly in the first position engaged with the second tier of the front wheel gear;

[0116] FIG. 41A is a perspective view of another example surface cleaning apparatus having brush drive motors;

[0117] FIG. 41B is a front perspective cross-sectional view of the surface cleaning apparatus of FIG. 41A;

[0118] FIG. 41C is a rear perspective cross-sectional view of the surface cleaning apparatus of FIG. 41A;

[0119] FIG. 42A is a perspective view of another example motor for use with the surface cleaning apparatus of FIG. 41A;

[0120] FIG. 42B is a front perspective cross-sectional view of the surface cleaning apparatus of FIG. 41A;

[0121] FIG. 42C is a rear perspective cross-sectional view of the surface cleaning apparatus of FIG. 41A;

[0122] FIG. 43 is a perspective view of another example surface cleaning apparatus;

[0123] FIG. 44 is a perspective view of the surface cleaning apparatus of FIG. 43 with a collection container separated from a surface cleaning head;

[0124] FIG. 45A is a perspective cross-sectional view of the surface cleaning head of the surface cleaning apparatus of FIG. 43;

[0125] FIG. 45B is a perspective view of the surface cleaning head of the surface cleaning apparatus of FIG. 43 with the collection container separated from the surface cleaning head;

[0126] FIG. 46 is a perspective view of another example surface cleaning apparatus with an expandable collection container;

[0127] FIG. 47 is a perspective cross-sectional view of the surface cleaning apparatus of FIG. 46;

[0128] FIG. 48 is a perspective view of the surface cleaning apparatus of FIG. 46 with the collection container expanded;

[0129] FIG. 49 is a perspective cross-sectional view of the surface cleaning apparatus of FIG. 46 with the collection container expanded;

[0130] FIG. 50 is a perspective view of the surface cleaning apparatus of FIG. 46 with the collection container removed;

[0131] FIG. 51 is a front perspective view of the collection container of the surface cleaning apparatus of FIG. 46;

[0132] FIG. 52A is a perspective view of an example brush having one row of straight bristles;

[0133] FIG. 52B is a perspective view of an example brush having two rows of straight bristles;

[0134] FIG. 52C is a perspective view of an example brush having three rows of straight bristles;

[0135] FIG. 52D is a perspective view of an example brush having four rows of straight bristles;

[0136] FIG. 53A is a perspective view of an example brush having one row of curved bristles;

[0137] FIG. 53B is a perspective view of an example brush having two rows of curved bristles;

[0138] FIG. 53C is a perspective view of an example brush having three rows of curved bristles;

[0139] FIG. 53D is a perspective view of an example brush having four rows of curved bristles;

[0140] FIG. 54 is a perspective view of an example brush having alternating rows of straight and curved bristles;

[0141] FIG. 55 is a perspective view of an example brush having two rows of rigid bristles;

[0142] FIG. 56 is a top cross-sectional perspective view of an example surface cleaning apparatus having brushes with adjustable bristles;

[0143] FIG. 57 is a perspective view of another example surface cleaning apparatus;

[0144] FIG. 58 is an enlarged perspective view of the surface cleaning apparatus of FIG. 57 with a rear wheel in a lowered position;

[0145] FIG. 59 is a perspective cross sectional view of the surface cleaning apparatus of FIG. 57;

[0146] FIG. 60 is a perspective view of another example surface cleaning apparatus having a hand vacuum at a front end;

[0147] FIG. 61 is a perspective view of the surface cleaning apparatus of FIG. 60 with the hand vacuum removed;

[0148] FIG. 62 is an enlarged cross-sectional view of the surface cleaning apparatus of FIG. 60;

[0149] FIG. 63 is a perspective view of the surface cleaning apparatus of FIG. 60 with a collection container removed;

[0150] FIG. 64 is a perspective view of the collection container of the surface cleaning apparatus of FIG. 60 with the lid opened;

[0151] FIG. 65 is a perspective view of another example surface cleaning apparatus having a hand vacuum at a rear end;

[0152] FIG. 66 is a perspective view of the surface cleaning apparatus of FIG. 65 with the hand vacuum removed;

[0153] FIG. 67 is an enlarged cross-sectional view of the surface cleaning apparatus of FIG. 65;

[0154] FIG. 68 is a perspective view of the surface cleaning apparatus of FIG. 65 with a collection container removed;

[0155] FIG. 69 is a perspective view of the collection container of the surface cleaning apparatus of FIG. 65 and the lid opened;

[0156] FIG. 70 is a perspective view of another example surface cleaning apparatus having a hand vacuum intermediate a front and rear end;

[0157] FIG. 71 is a perspective view of the surface cleaning apparatus of FIG. 70 with the hand vacuum removed;

[0158] FIG. 72 is an enlarged cross-sectional view of the surface cleaning apparatus of FIG. 70;

[0159] FIG. 73 is a perspective view of the surface cleaning apparatus of FIG. 70 with a collection container removed;

[0160] FIG. 74 is a perspective view of the collection container of the surface cleaning apparatus of FIG. 70 with the lid opened;

[0161] FIG. 75 is a perspective view of another example surface cleaning apparatus having a module mounted to a top side of a surface cleaning head;

[0162] FIG. 76 is a perspective view of the surface cleaning apparatus of FIG. 75 having the module removed;

[0163] FIG. 77 is an enlarged perspective view of the surface cleaning apparatus of FIG. 75 having the module removed;

[0164] FIG. 78 is a perspective cross-sectional view of the surface cleaning apparatus of FIG. 75; and,

[0165] FIG. 79 is a perspective cross-sectional view of the surface cleaning apparatus of FIG. 75 with a perspective view of an interchangeable reel blade.

[0166] The drawings included herewith are for illustrating various examples of articles, methods, and apparatuses of the teaching of the present specification and are not intended to limit the scope of what is taught in any way.

DESCRIPTION OF VARIOUS EMBODIMENTS

[0167] Various apparatuses will be described below to provide an example of an embodiment of each claimed invention. No embodiment described below limits any claimed invention and any claimed invention may cover apparatuses that differ from those described below. The claimed inventions are not limited to apparatuses having all of the features of any one apparatus described below or to features common to multiple or all of the apparatuses described below. It is possible that an apparatus described below is not an embodiment of any claimed invention. Any invention disclosed in an apparatus described below that is not claimed in this document may be the subject matter of another protective instrument, for example, a continuing patent application, and the applicants, inventors or owners do not intend to abandon, disclaim or dedicate to the public any such invention by its disclosure in this document.

[0168] The terms “an embodiment”, “embodiment”, “embodiments”, “the embodiment”, “the embodiments”, “one or more embodiments”, “some embodiments”, and

“one embodiment” mean “one or more (but not all) embodiments of the present invention(s), “unless expressly specified otherwise.

[0169] The terms “including”, “comprising”, and variations thereof mean “including but not limited to”, unless expressly specified otherwise. A listing of items does not imply that any or all of the items are mutually exclusive, unless expressly specified otherwise. The terms “a”, “an”, and “the” mean “one or more”, unless expressly specified otherwise.

[0170] As used herein and in the claims, two or more parts are said to be “coupled”, “connected”, “attached”, or “fastened” where the parts are joined or operate together either directly or indirectly (i.e., through one or more intermediate parts), so long as a link occurs. As used herein and in the claims, two or more parts are said to be “directly coupled”, “directly connected”, “directly attached”, or “directly fastened” where the parts are connected in physical contact with each other. As used herein, two or more parts are said to be “rigidly coupled”, “rigidly connected”, “rigidly attached”, or “rigidly fastened” where the parts are coupled so as to move as one while maintaining a constant orientation relative to each other. None of the terms “coupled”, “connected”, “attached”, and “fastened” distinguish the manner in which two or more parts are joined together.

[0171] It should be noted that terms of degree such as “substantially”, “about”, and “approximately” as used herein mean a reasonable amount of deviation of the modified term such that the end result is not significantly changed. These terms of degree may also be construed as including a deviation of the modified term, such as by 1%, 2%, 5% or 10%, for example, if this deviation does not negate the meaning of the term it modifies.

[0172] Furthermore, the recitation of numerical ranges by endpoints herein includes all numbers and fractions subsumed within that range (e.g., 1 to 5 includes 1, 1.5, 2, 2.75, 3, 3.90, 4, and 5). It is also to be understood that all numbers and fractions thereof are presumed to be modified by the term “about” which means a variation of up to a certain amount of the number to which reference is being made if the end result is not significantly changed, such as 1%, 2%, 5%, or 10%, for example.

[0173] Some elements herein may be identified by a part number, which is composed of a base number followed by an alphabetical or subscript-numerical suffix (e.g., 112a, or 112₁). Multiple elements herein may be identified by part numbers that share a base number in common and that differ by their suffixes (e.g., 112₁, 112₂, and 112₃). All elements with a common base number may be referred to collectively or generically using the base number without a suffix (e.g., 112).

[0174] Furthermore, it will be appreciated that for simplicity and clarity of illustration, where considered appropriate, reference numerals may be repeated among the figures to indicate corresponding or analogous elements. In addition, numerous specific details are set forth in order to provide a thorough understanding of the example embodiments described herein. However, it will be understood by those of ordinary skill in the art that the example embodiments described herein may be practiced without these specific details. In other instances, well-known methods, procedures, and components have not been described in detail so as not to obscure the example embodiments

described herein. Also, the description is not to be considered as limiting the scope of the example embodiments described herein.

General Description of a Surface Cleaning Apparatus

[0175] Referring to the FIGS. 1-2, an exemplary embodiment of a surface cleaning apparatus is shown generally as 100. In use, the surface cleaning apparatus 100 is applied to an interior or exterior target surface to be cleaned, such as a lawn, deck, turf, driveway, or sidewalk, and a brush contacts the target surface to displace debris. The exemplary surface cleaning apparatus 100 of FIGS. 1 and 2 includes a dirt collection region, and the brush sweeps up debris from the target surface and directs the debris into the dirt collection region, as described further elsewhere herein.

[0176] The exemplary surface cleaning apparatus 100 of FIGS. 1 and 2 includes a surface cleaning head 102. The surface cleaning head 102 includes a main body 101. The main body 101 has a longitudinal axis 103a extending between a front end 104 and a rear end 105. The main body 101 has a vertical axis 103b extending between a top end 106 and a bottom end 107 (see FIG. 23). The main body 101 also has a transverse axis 103c extending between first and second lateral sides 108. The axes 103 are mutually perpendicular.

[0177] The main body 101 may be any suitable shape, such as a prism or hemispherical shape. In the example illustrated, the main body 101 is generally shaped as a rectangular prism with a rounded edge between the top end 106 and the front end 104. In some embodiments, the head 102 has a generally flat bottom surface at the bottom end 107 to pass over the target surface.

[0178] The surface cleaning apparatus 100 has one or more wheels 110. The wheels 110 are arranged to contact the target surface to support the head above the target surface. The surface cleaning apparatus 100 may include any positional arrangement of wheels 110 suitable to support the head 102. The surface cleaning apparatus 100 may have one or more wheels proximate the front end 104 (e.g., secured to the head at a position within the forward 10% or 25% of the length of the head), and/or one or more wheels proximate the rear end 105 (e.g., secured to the head at a position within the rearward 10% or 25% of the length of the head), and/or one or more wheels in a central portion of the head between the front and rear ends 104, 105 (e.g., secured to the head at a position more than 25% of the length of the head from each of the forward and rear ends). As exemplified, front and/or rear wheels support the front and/or rear edges of the head 102 above the target surface (e.g., to prevent the lower surface of the front and/or rear ends of the head 102 from contacting the target surface). A central wheel can be arranged adjacent a dirt inlet to limit how close the dirt inlet can come to the target surface (e.g., to inhibit obstruction of the dirt inlet). It will be appreciated that any other members known in the art may be used to enable the surface cleaning head to travel across a target surface such as tank treads, glides and the like.

[0179] In the example illustrated in FIGS. 1-2, the surface cleaning apparatus 100 includes a pair of front wheels 110a mounted proximate the front end 104, a pair of rear wheels 110b mounted proximate the rear end 105, and a central wheel 110c (see FIG. 3) mounted at a position in between the front and rear ends 104, 105. In another example, FIGS. 6-7

show a surface cleaning apparatus 100 with front wheels 110a and rear wheels 110b and without the central wheel 110c.

[0180] The wheels 110 can be any shape and size conducive for movement of the cleaning head 102 across the surface to be cleaned. In some examples, one or more wheel is generally cylindrical (e.g., shaped as a disc or as an elongated transversely extending roller) or spherical. In the illustrated example, the head 102 includes five wheels 110. The wheels 110 include a pair of disc-shaped front wheels 110a positioned on laterally opposed sides 108 proximate the front end 104 of the cleaning head 102, a pair of disc-shaped rear wheels 110b positioned on laterally opposed sides 108 proximate the rear end 105 of the cleaning head 102, and an elongated roller wheel 110c positioned generally centrally in the surface cleaning head 102 between the front and rear ends 104, 105 and extending between the laterally opposed sides 108.

[0181] The wheels 110 may be mounted to the surface cleaning apparatus 100 at any suitable location to support the head 102 above the target surface. In some examples, one or more wheels 110 are mounted within the main body 101 of the head 102 (i.e., an inboard wheel). The exemplary roller wheel 110c of FIGS. 1 and 2 is an inboard wheel mounted within the main body 101. In some examples, one or more wheels 110 are mounted to an outside of the main body 101 of the head 102 (i.e., an outboard wheel). The exemplary front wheels 110a and rear wheels 110b of FIGS. 1 and 2 are outboard wheels mounted to an outside of the main body 101 of the head 102.

[0182] The surface cleaning apparatus 100 may include any number of wheels suitable to support the head 102. The exemplary surface cleaning apparatus 100 of FIGS. 1 and 2 includes five wheels 110, however other surface cleaning apparatus 100 may include one, two, three, four, six, or more than six wheels.

[0183] Referring again to FIGS. 1-2, the exemplary surface cleaning apparatus 100 includes a drive handle 112. In use, the drive handle enables a user to push or pull the surface cleaning head 102 over the target surface. The drive handle 112 extends upwardly from the head 102 to be grasped by a user. The drive handle 112 may be for use by a standing user, and may extend from the head 102 to an elevation suitable for a standing user. In some examples, the handle 112 extends to an elevation of at least 2 feet, at least 2.5 feet, or at least 3 feet above the target surface on which the head 102 is supported by the wheels 110. The handle 112 may include a hand grip portion 118 to be grasped by a user. The exemplary hand grip portion 118 of FIGS. 1 and 2 is at an upper end of the handle 112. It will be appreciated that one or more actuators (e.g., main on/off switch) may be provided on the handle, such as on the hand grip portion 118.

[0184] The drive handle 112 is drivably coupled to the head 102. The drive handle 112 may be coupled to the surface cleaning head 102 at any suitable position or positions on the head 102, such as coupled to the rear end, the front end, the top end, and/or one or both of the sides of the head 102. The drive handle 112 may be any suitable shape extending from the head. The exemplary drive handle 112 of FIGS. 1 and 2 has a pair of arms 114 coupled to a central portion (in the direction of the longitudinal axis) of the laterally opposed sides 108 of the cleaning head 102. Each arm 114 extends from a coupling 116 on the respective sides 108 to a hand grip portion 118. The exemplary couplings 116

of FIGS. 1 and 2 are generally centrally located on the laterally opposed sides 108 between the front and rear ends 104, 105.

[0185] The drive handle 112 may be a fixed handle, fixedly coupled to the head 102. Alternatively, the drive handle 112 may be an adjustable handle moveable between a first position and a second position. The handle may be adjusted to accommodate different users (e.g., different heights) and/or different uses of the apparatus 100 (e.g., rotatably mounted so as to enable pushing the head 102 forward and pushing the head 102 rearward, as exemplified in FIGS. 21 and 22 and as discussed further elsewhere herein). The handle 112 may be selectively moveable between only the first and second positions (and optionally lockable in one or more of the positions), selectively moved between the first and second positions and one or more intervening positions (and optionally lockable in one or more of the positions), or continuously adjusted between the first and second positions (and optionally lockable in any or all locations between the first and second positions).

[0186] The exemplary handle 112 of FIGS. 21 and 22 is moveable between a first position (FIG. 21) extending generally rearwardly of the head 102 and a second position (FIG. 22) extending generally forwardly of the head 102. When the handle extends generally rearwardly of the head 102, the user can move the head 102 while the user is in a position rearward of the head (e.g., push the surface cleaning head forward with the front end leading or pull rearward). When the handle extends generally forwardly of the head 102, the user can move the head 102 while the user is in a position generally forward of the head (e.g., push the surface cleaning head forward with the rear end leading or pull rearward). In some example, the couplings 116 are rotatable (e.g., pivotal) couplings having a coupling axis of rotation 119, to allow the handle 112 to be adjusted between the first and second positions.

[0187] Referring again to FIGS. 1 and 2, the exemplary surface cleaning apparatus 100 is a sweeper. A sweeper includes one or more brushes 120 for application to a target surface. The brush or brushes are arranged to sweep against the target surface in use. In use, the surface cleaning apparatus 100 is applied to an interior or exterior target surface that is to be cleaned, such as a lawn, deck, turf, driveway, or sidewalk, and a brush 120 contacts the target surface to displace debris that is on the target surface.

[0188] The exemplary head 102 of FIGS. 1 and 2 includes a brush 120 extending between the laterally opposed sides 108. The exemplary brush 120 of FIGS. 1 and 2 is rotatably mounted such that it is rotatable in the clockwise and/or counter clockwise direction about a rotation axis 122. As exemplified, when the surface cleaning head 102 is placed on a horizontal surface, the rotational axis 122 extends generally horizontally and generally transverse to the forward/rearward directions.

[0189] A brush 120 may be positioned at any suitable location in the head 102. The exemplary brush 120 of FIGS. 1 and 2 is mounted proximate the front end 104. The exemplary head 102 of FIGS. 6 and 7 includes a brush 120 mounted proximate the rear end 105. The exemplary head 102 of FIGS. 11 and 12 includes a brush 120 mounted generally centrally in the surface cleaning head 102 between the front and rear ends 104, 105.

[0190] In embodiments in which the brush 120 is rotationally driven, the brush may be driven by any suitable

drive system. In some examples, the brush is rotationally driven by a motor, such as a dedicated motor or a motor that also drives one or more other components of the apparatus 100. Alternatively, or additionally, a brush may be driven rotationally by rotation of one or more wheels 110, such as coupled to the wheel directly or indirectly.

[0191] Referring still to FIGS. 1 and 2, the exemplary surface cleaning apparatus 100 includes a debris collection region 142 to hold debris displaced by the brush 120. The debris collection region 142 is also referred to as a dirt collection region herein. The debris collection region 142 may be provided at any suitable location on the surface cleaning apparatus 100, such as on the handle 112 or in the head 102. In the illustrated example of FIGS. 3A and 3B, the dirt collection region 142 is in the head 102. Providing the dirt collection region in the head 102 lowers the center of gravity of the apparatus 100 compared to locating the dirt collection region on the handle 112, which may improve the stability and/or hand-feel of the apparatus 100. In accordance with some aspects as discussed elsewhere herein, in some examples, an apparatus 100 includes more than one discrete dirt collection region, as described further elsewhere herein. In embodiment in which the apparatus 100 includes two or more brushes 120, two or more brushes may direct debris to a common collection region or to different collection regions.

[0192] The dirt collection region 142 is provided in a collection chamber 132. In some examples, the collection chamber 132 is removably mounted to the surface cleaning head 102. The collection chamber 132 may comprise or consist of a hard-sided container. Alternatively, or additionally, the collection chamber 132 may comprise or consist of a removable liner (e.g., a flexible bag). As shown in FIGS. 4-5 and FIGS. 9-10, the collection container 132 can include a handle 134 to facilitate removal of the collection container 132 from the surface cleaning head 102. The exemplary handle 134 is discrete from the drive handle 112. In embodiments wherein the head 102 includes multiple discrete collection regions 142, the collection regions 142 may be separate collection chambers which may be removable concurrently or independently. In some examples, a large debris collection region and a discrete small debris collection chamber are both in a single one-piece collection assembly.

[0193] Referring again to FIGS. 1-2, the surface cleaning head 102 includes a dirt inlet 128 in the bottom side 107 thereof. The brush 120 is adjacent the dirt inlet 128 to direct debris into the dirt inlet. In some examples, the brush is mounted within the dirt flow path. In some examples, part of the brush 120 protrudes through the dirt inlet 128. In the exemplary embodiment of FIGS. 1 and 2, the bristles 124 of the brush 120 protrude through the dirt inlet 128.

[0194] As exemplified in FIGS. 6-7, if a plurality of brushes are provided, then a dirt inlet 128 may be provided for each brush 120a, 120b. The head 102 may include a discrete dirt inlet 128 for each brush 120, however in some examples the head 102 includes a common dirt inlet 128 for more than one brush. In any example, the bristles 124 may extend at least as far below the bottom side 107 of the surface cleaning head 102 as the wheels 110 such that the bristles 124 of the brush 120 can sufficiently engage and sweep the surface to be cleaned. As discussed elsewhere herein, if one or more of the wheels 110 and/or brushes 120 are retractable, the degree of engagement of the bristles 124

of the brushes **120** with the surface to be cleaned can be adjusted by extending/retracting the wheels **110** and/or brushes **120**. For example, where two brushes are present, it may be desirable in some use cases (e.g., in at least one mode) to adjust the wheels **110** such that one or more brush **120** is raised away from the target surface.

[0195] In some examples, the dirt inlet **128** is sized to include a spacing gap **127** beyond a sweep **137** of the bristles **124** in one or both directions along the longitudinal axis **103a**. The exemplary dirt outlet **128** of FIGS. 3A and 3B includes a spacing gap **127** forward of the sweep **137** of the bristles **124** in one direction of rotation of the bristles (i.e., a clockwise rotation in FIGS. 3A and 3B). The spacing gap **127** is provided for the passage of debris, particularly large debris, in through the debris inlet **128** between the bristles **124** and the edge of the dirt inlet **128**. The spacing gap may prevent the debris from being kicked forward of the surface cleaning head **102** by the sweeping action of the bristles **124** of the brush **120**. The spacing gap **127** can be any suitable size. In some examples, the spacing gap **127** is at least 5% of the radius of the brush **120** in use, between 5% and 75% of the radius of the brush **120** in use, at least 10% of the radius of the brush **120** in use, between 10% and 75% of the radius of the brush **120** in use, or between 10% and 50% of the radius of the brush **120** in use. A spacing gap **127** may be constant along the length of the brush **120**, variable along the length, and/or provided along only one or more portions of the length of the brush **120**.

[0196] Referring to FIGS. 3A-3B, the surface cleaning head **102** includes a dirt flow path **138** between the dirt inlet **128** and a dirt collection region **142**. The dirt flow path **138** extends from the dirt inlet **128** to the dirt collection region **142**. The dirt flow path **138** may be any suitable length. For example, the path **138** may consist of an opening in a wall (i.e., the dirt inlet) which opening opens directly into the dirt collection chamber (e.g., if the brush **120** is mounted outside the dirt flow path), or the path **138** may include a passageway between the dirt inlet and the dirt collection chamber.

[0197] Referring still to FIGS. 3A and 3B, the exemplary brush **120** is mounted within the dirt flow path **138**. Dirt travelling from the exemplary dirt inlet **128** of FIGS. 3A and 3B to the dirt collection region **142** passes the brush **120**. The exemplary head **102** of FIGS. 3A and 3B includes a brush chamber **121** in the dirt flow path **138**, and the brush **120** is mounted in the brush chamber **121**.

Brush Drive

[0198] In accordance with an aspect of this disclosure, which may be used alone or in combination with any other aspect, the brush or brushes of the surface cleaning head may be electrically and/or mechanically driven. It will be appreciated that a plurality of brushes are provided, one or more, and optionally all, may be driven a common motor or mechanical linkage. Alternately, a motor or mechanical linkage may drive only one or only some of the brushes, and accordingly a plurality of motors or mechanical linkages may be provided or one or more motors and one or more mechanical linkages may be provided.

[0199] Referring to FIG. 14, the exemplary brush **120b** is driven by a motor **117** powered by a power supply **123**. The power supply **123** may be any suitable power supply, such as an on-board energy storage member (e.g., a battery or capacitor) or a power cord to an external power source (e.g., a power cord to a residential power outlet). As another

example, referring to FIGS. 41A-41C, the exemplary surface cleaning apparatus **100** includes a first brush motor **190a** operatively connected to the first brush **120a** and a second brush motor **190b** operatively connected to the second brush **190b**. When the surface cleaning head **102** is moved forwardly, the first brush motor **190a** can be energized to rotate the first brush **120a** in one of the clockwise direction and the counter clockwise direction, and the second brush motor **190b** can be deenergized to stop rotation of the second brush **120b** (e.g., to lock the brush in position or to leave the brush free to rotate but undriven by the apparatus **100**). When the surface cleaning head **102** is moved rearwardly, the second brush **190b** can be energized to rotate the second brush **120b** in one of the clockwise direction and the counter clockwise direction, and the first brush motor **190a** can be deenergized to stop rotation of the first brush **120a**.

[0200] The brush motor **190** can be any suitable motor known in the art. For example, the brush motor **190** can be any rotary electrically motor, such as a direct current (DC) motor. FIGS. 42A-42C illustrate an example brush motor **190** for use with the surface cleaning apparatus **100**. The surface cleaning apparatus **100** can be corded or can include a rechargeable battery to provide electrical power to the brush motor(s) **190**. As shown, the brush motor **190** includes a drive shaft **192** that is rotatably driven by the brush motor **190**. The drive shaft **192** can be a linear toothed gear, such as a worm gear. Referring again to FIGS. 41A-41C, the drive shaft **192** of the brush motors **190a**, **190b** is provided with a worm gear engaged with a toothed portion **194** of the central member or core **126** of the first and second brushes **120a**, **120b**. In the illustrated example, the toothed portion **194** of the central members **126** meshes with the drive shaft **192** at a shaft angle of about 90°. When the motors **190a**, **190b** cause the drive shaft **192** to rotate, the brushes **120a**, **120b** are thus correspondingly driven to rotate by the engagement of the teeth of the drive shaft **192** and the toothed portion **194** of the central member **126**.

[0201] It will be appreciated that in any embodiment according to any aspect disclosed herein, a user-controlled actuator can be operable to selectively control the brush(es). The surface cleaning apparatus **100** can include electrical actuator controls to energize or deenergize the brush motors **190a**, **190b** and/or to selectively control the direction of rotation of the brushes **120a**, **120b**. The electrical actuator controls dictating the motors **190a**, **190b** can be, for example, a switch on the surface cleaning head **102** or drive handle **112** (e.g., on the hand grip portion **118**), a sensor for detecting the direction of rotation of one or more wheels **110**, a sensor for detecting the position of the drive handle **112** (i.e., forwardly extending or rearwardly extending), or the drive handle **112** can act as a switch when moved between positions. It will be appreciated that, if more than one motor is provided, then each may be individually energized or deenergized or they may be concurrently energized or deenergized.

[0202] Alternatively, or additionally, in any embodiment according to any aspect disclosed herein, the brush is coupled to the wheels **110** such that rotation of the wheels drives rotation of the brush **120**. The brush may be coupled to the wheels **110** directly or indirectly (e.g., through a mechanical linkage and/or gearing). A rotational direction of the brush may be the same as a rotational direction of the wheels **110**, opposite to the rotational direction of the

wheels, or selectively switchable between being the same and opposite to the rotational direction of the wheels.

[0203] Referring to FIG. 19, shown therein is a cross-sectional view of the surface cleaning apparatus 100 showing an example gearing whereby the first and second brushes 120a, 120b are driven mechanically by rotation of the front and rear wheels 110a, 110b. The front wheel 110a includes a front wheel gear 160a, which is shown to be an internal spur gear, and which rotates with the front wheel 110a in the same direction of rotation as the front wheel 110a. A first brush drive gear 162a is coupled to the first brush 120a such that the first brush 120a rotates with the first brush drive gear 162a. The front wheel gear 160a engages the first brush drive gear 162a and drives the first brush drive gear 162a, and thereby the first brush 120a, to rotate in the same direction of rotation as the front wheel 110a. The rear wheel 110b includes a rear wheel gear 160b, which is shown to be an external spur gear, and which rotates with the rear wheel 110b in the same direction of rotation as the rear wheel 110b. A second brush drive gear 162b is coupled to the second brush 120b such that the second brush 120b rotates with the second brush drive gear 162b. The rear wheel gear 160b engages the second brush drive gear 162b and drives the second brush drive gear 162b, and thereby the second brush 120b, to rotate in the opposite direction of the rear wheel 110b.

[0204] FIGS. 20A and 20B shown expanded views of the front wheel 110a and rear wheel 110b, respectively, wherein the first and second brush drive gears 162a, 162b each include an internal ratchet 164 and pawl 166. The teeth of the ratchets 164 are etched into the first and second brush drive gears 162a, 162b such that the ratchets 164 rotate with the first and second brush drive gears 162a, 162b in the same direction of rotation. The pawls 166 are linear members extending across the diameter of the ratchets 164. When the ratchets 164 rotate in one of the clockwise and counter clockwise directions, the pawls 166 catch the teeth of the ratchets 164 and are thereby rotated with the ratchets 164 in the same direction of rotation. When the ratchets 164 rotate in the other of the clockwise and counter clockwise directions, the pawls 166 pass over the teeth of the ratchets 164 and thus remain stationary. The first and second brushes 120a, 120b are coupled to the pawls 166 such that the first and second brushes 120a, 120b rotate with the pawls 166 in the same direction of rotation.

[0205] In the illustrated example, the ratchet 164 of the first brush drive gear 162a is configured such that the pawl 166 catches the teeth of the ratchet 164 when the ratchet 164 rotates in the counter clockwise direction. Accordingly, when the front wheels 110a rotate in the counter clockwise direction (i.e., when the surface cleaning head 102 is moved in the forward direction), the first brush 120a also rotates in the counter clockwise direction, and when the front wheels 110a rotate in the clockwise direction (i.e., when the surface cleaning head 102 is moved in the rearward direction), the first brush 120a does not rotate. The ratchet 164 of the second brush drive gear 162b is configured such that the pawl 166 catches the teeth of the ratchet 164 when the ratchet 164 rotates in the clockwise direction. Accordingly, when the rear wheels 110b rotate in the counter clockwise direction, the second brush 120b rotates in the clockwise direction, and when the rear wheels 110b rotate in the clockwise direction, the second brush 120b does not rotate. It will be appreciated that any combination of internal/

external wheel gears and brush drive gears, with or without ratchets and pawls, can be used to mechanically control the direction of rotation of the brushes 120a, 120b based on the direction of rotation of the wheels 110a, 110b.

[0206] A head 102 may include only a single brush 120, two discrete brushes 120, or more than two discrete brushes 120. In embodiments wherein a head 102 includes more than one brush, two or more brushes may be driven concurrently or separately. Two or more brushes may be driven by a common drive system (e.g., a motor or linkage to another rotating component such as the wheels 110). Two or more brushes may be driven by discrete drive systems. Brushes driven by a common drive system may be driven concurrently, or selectively (e.g., via a clutch assembly). Using a clutch assembly to link a brush 120 to a wheel 110 enables a brush 120 to be operated mechanically (e.g., to reduce the need for one or more electric motors) and still adjustably (e.g., allowing switching between being driven and not driven, switching between drive directions, and/or switching between speeds of rotation).

Brush Location

[0207] In accordance with an aspect of this disclosure, which may be used alone or in combination with any other aspect, a head 102 may include one or more brushes, each of which may be located as different positions in the head 102, as discussed previously herein and also subsequently). It will be appreciated that the type of brush or brushes which are used may be selected based on the surface or surfaces that a brush is to be used on. For example, the target surface of one brush may be a lawn and the target surface for another brush may be patio tiles or indoor/outdoor carpet.

[0208] In embodiments wherein a head 102 includes more than one brush, the brushes 120 may be positionally arranged in any suitable arrangement. In some examples, a single brush 120 may extend from one side 108 to the other side 108, and in use the brush 120 may sweep the target surface across substantially all of the width of the head 102. In some examples, more than one brush 120 is used to sweep substantially all of the width of a target surface, such as two or more brushes, which may substantially abut one another, having coaxial axes of rotation or two or more brushes having axes of rotation spaced along the longitudinal axis 103a and extending across different portions of the width of the head 102.

[0209] In some examples, two or more brushes are spaced apart along the longitudinal axis behind one another along the longitudinal axis 103a. In use, brushes spaced apart along the longitudinal axis behind one another may be applied sequentially to a target surface and an order of application may be determined by a direction of motion of the head 102. The exemplary brushes 120 of FIG. 8 are spaced apart along the longitudinal axis 103a behind one another.

[0210] Referring again to FIGS. 1 and 2, the exemplary brush 120 includes a plurality of bristles 124 each extending from a central support 126. Bristles 124 may be spaced angularly about the central support 126. The exemplary brush 120 of FIGS. 1 and 2 includes a plurality of bristles 124 in each of a plurality of angularly spaced rows 125 on the central support 126. Alternatively, or additionally, a brush 120 may include bristles 124 that are not arranged in a row 125 with other bristles. The exemplary rows 125 of FIGS. 1 and 2 are linear and parallel to the axis of rotation 122

of the brush 120. Alternatively, or additionally, a brush 120 may include one or more rows 125 that are non-linear and/or not parallel to the axis of rotation 122, such as one or more helical rows of bristles. Any suitable number of rows may be used. For example, the brushes 120 can have one row of bristles 124 (see e.g., FIGS. 52A, 53A), two rows of bristles 124 (see e.g., FIGS. 52B, 53B), three rows of bristles 124 (see e.g., FIGS. 52C, 53C), four rows of bristles 124 (see e.g., FIGS. 52D, 53D), or six rows of bristles 124 (see e.g., FIG. 49).

[0211] Referring now to FIG. 55, the bristles 124 of the brushes 120 can be a rigid plastic and/or metal for some uses. For example, rigid plastic or metal bristles 124 may be suitable for collection of large debris, such as leaves, pinecones, and the like. In this way, the rigid plastic or metal bristles 124 can be used in a “rake” mode for collection of yard waste.

[0212] The brushes 120 and/or the bristles 124 thereof of any configuration described herein can be swappable or replaceable such that the surface cleaning apparatus 100 may be reconfigured by switching brush types for a wide range of uses and/or modes of operation.

Brush Operation

[0213] In accordance with an aspect of this disclosure, which may be used alone or in combination with any other aspect, a head 102 may have a mode of operation in which one or more brushes are not operated. In such a case, the brush may be configured to not impede the cleaning of a target surface with another brush.

[0214] A brush may be selectively applied to a target surface on which the apparatus 100 is supported. A brush that is not to be applied to the target surface may be rendered inoperable, by, e.g., being locked against rotational movement, having a rotational drive system disengaged but otherwise leaving the brush free to rotate (such as if rotated by debris or the target surface), and/or raising the brush up away from the target surface. A brush may be moved between a first position applied to the surface and a second position lifted away from the surface. For example, a brush for collecting smaller debris may be raised relative to the target surface when, e.g., the surface to be cleaned is a lawn or other soft surface (e.g., turf, carpet, and the like), and lowered when the surface to be cleaned is a patio or other hard surface (e.g., deck, driveway, tile, and the like). A brush meant to collect small debris can be disengaged from surfaces where small debris collection is not desired and/or suitable (i.e., large particulate matter collection only).

[0215] A brush may be rendered inoperable in response to a change in direction of rotation of wheels of the head 102, a change in position of a handle or a user manually operating a switch (e.g., deenergizing a brush motor).

[0216] In some examples, the brush is moved between the first and second positions by moving one or more wheels 110 between first a first position in which one or more brushes are positioned to clean a surface and a second position in which one or more brushes are spaced from the surface. In some examples, one or more wheels 110 is moveable and, in use, moving the one or more wheels may raise or lower some or all of the head 102 relative to a target surface on which the head 102 is resting. The brush or brushes may be moved by a motor. Alternately, a mechanical linkage may be provided whereby rotating the handle from one end (e.g., the

front end) to the other end (e.g., the rear end) drives a mechanical linkage which drives the lift-off mechanism to raise and/or lower wheels.

[0217] Referring to FIGS. 23 to 26, the exemplary wheels 110a are vertically moveable by a lift off mechanism between a first position (FIGS. 23 and 24) and a second position (FIGS. 25 and 26) with respect to the main body 101 of the head 102, the first position generally raised relative to the second position along the vertical axis 103b. Lowering the rear wheels 110a lowers a portion of the head 102 (as exemplified) and/or it may lower a component of the apparatus 100, such as a brush 120. Similarly, raising the rear wheels 110a raises a portion of the head 102 (as exemplified) and/or it may raise component of the apparatus 100, such as a brush 120. A wheel 110 may be selectively moveable between only the first and second positions, selectively moved between the first and second positions and one or more intervening positions, or continuously adjustable between the first and second positions. In the embodiment of FIGS. 23 to 26, it will be appreciated that the front wheels 110b and the rear wheels 110a may each be selectively raised and lowered.

[0218] Alternatively the lift off mechanism may use a rotationally mounted member to raise and/or lower a brush relative to a target surface. Referring to FIGS. 11-12, in the illustrated example, the surface cleaning apparatus 100 includes a lift off mechanism 150. The lift off mechanism 150 includes a pair of legs 152. Each leg 152 is rotatably coupled to a respective one of the opposed sidewalls 108 of the surface cleaning head 102 proximate the rear end 105. The rear wheels 110b are coupled to the distal end of each leg 152. The legs 152 can be moved between the lowered position as shown in FIGS. 11-12 and the raised position (see e.g., in FIG. 58). To transition from the lowered position to the raised position, the legs 152 can rotate about a coupling axis 154 of the rotatable coupling to drive the rear wheels 110b toward the surface to be cleaned, and thereby raise the rear end 105 of the surface cleaning head 102. As the rear end 105 is raised, the second brush 120b is separated from the surface. Conversely, to transition from the raised position to the lowered position, the legs 152 can rotate about the coupling axis 154 of the rotatable coupling to drive the rear wheels 110b away from the surface to be cleaned, and thereby lower the rear end 105 of the surface cleaning head 102. As the rear end 105 is lowered, the second brush 120b is reengaged with the surface. Referring still to FIGS. 11-12, in the illustrated example, the lift off mechanism 150 of the surface cleaning apparatus 100 includes a pair of extendable arms 156 mounted to the laterally opposed sides 108 of the surface cleaning head 102.

[0219] In this embodiment, the arms 156 can be moved between a retracted position as shown in FIGS. 11-12 and an extended position (see e.g., FIG. 58) driven by respective arm motors 158. To transition from the retracted position to the extended position, one or more arm motors 158 may drive the arms 156 toward the surface to be cleaned, whereby contact with the surface raises the rear end 105 of the surface cleaning head 102. The legs 152 can rotate about the coupling axis 154 of the rotatable coupling to maintain engagement of the rear wheels 110b with the surface. The legs 152 can be biased toward the surface such that the lifting force to raise the rear end 105 of the surface cleaning head 102 is shared between the arm motors 158 and the biasing force. The biasing force may be insufficient to

independently move the legs 152 from the lowered position to the raised position under the weight of the surface cleaning head 102. Alternatively, the arms 156 can be coupled to the legs 152 such that transitioning the arms 156 from the retracted position to the extended position drives the legs 152 from the lowered position to the raised position. This may advantageously avoid contact of the arms 156 with the surface to be cleaned, which may generate resistive friction against the user moving the surface cleaning head 102. Alternately, a mechanical linkage may be provided whereby rotating the handle from one end (e.g., the front end) to the other end (e.g., the rear end) drives a mechanical linkage which drives the lift-off mechanism to raise and/or lower wheels.

[0220] Alternatively, or additionally, a brush 120 itself may be raised and/or lowered relative to the main body 101 of the head 102 between first and second positions. For example, the brush may be adjusted within the main body 101 itself (e.g., retracted into the main body 101 or extended such that at least a portion of the brush protrudes below the main body 101). A brush may be selectively moveable between only the first and second positions, selectively moved between the first and second positions and one or more intervening positions, or continuously adjustable between the first and second positions.

[0221] A position of a wheel or brush may be manually adjusted by a user. Also, or alternatively, a position of a wheel or brush may be automatically adjusted. Automatic adjustment may be, e.g., in response to a sensed condition (e.g., a target surface over which the head travels), in response to a mode selection (e.g., on selection of a mode in a user interface that is communicatively coupled to the wheel or brush wherein the mode may be a type of target surface to be cleaned), and/or in response to an adjustment of another component of the apparatus 100 (e.g., movement of the handle 114). A position of a wheel or brush may be automatically adjusted by a motor, such as a linear motor extending or retracting an arm on which the wheel is mounted. A position of a wheel or brush may be automatically adjusted by a mechanical linkage to another element, such as a mechanical linkage to the handle such that moving the handle from one position to another moves the wheel or brush from one position to another.

Multiple Dirt Flow Paths and/or Dirt Containers

[0222] In accordance with an aspect of this disclosure, which may be used alone or in combination with any other aspect, if a head 102 has more than one brush, then the head 102 may have more than one dirt flow path. Alternately or in addition, the head 102 may have two or more dirt collection regions. Accordingly, in some examples, two or more dirt flow paths may share a common dirt inlet and/or share a common dirt collection region. Further, in some examples, two or more dirt flow paths each open from a discrete dirt inlet and/or open to a discrete dirt collection region. Two or more dirt flow paths may be used concurrently or alternatively.

[0223] It will be appreciated that, if first and second debris flow paths 138a, 138b are provided, then they may have a common path segment with each includes at least one portion that is discrete from the other. In some examples, substantially all of the first path 138a is discrete from the second path 138b.

[0224] The exemplary head 102 of FIGS. 3A and 3B includes a first debris flow path 138a that is substantially

fully separate from the second debris flow path 138b, with the choice of path along which debris is directed controlled by the direction or rotation of the brush 120. However, in some examples only a portion, e.g., an upstream end of the paths, is separate. As exemplified in FIG. 8, the first brush 120a can direct dirt along a first path 138a or a third path 138c, and the first and third paths 138a, 138c share a common outlet end 148b but have separate upstream ends 148a. As exemplified in FIG. 8 and as discussed elsewhere herein, clockwise rotation of the brush 120a directs dirt through the first path 138a extending forward of the brush 120a and over the brush 120a to an outlet from the brush chamber 121, while counterclockwise rotation of the brush 120a directs dirt in the third path 138c extending rearward of the brush 120a to the same outlet from the brush chamber 121 (e.g., up to a ramp 146).

[0225] The brush chamber 121 may be any suitable shape, including rectangular or cylindrical. The exemplary brush chamber 121 of FIGS. 3A and 3B is a generally cylindrical chamber. The exemplary brush chamber 121 of FIGS. 3A and 3B includes a first end wall 121a and an opposed second end wall 121b. The exemplary brush of FIGS. 3A and 3B is mounted between the first and second end walls 121a, 121b. A sidewall 121c extends between the exemplary first and second end walls. The brush chamber includes at least one dirt inlet port 129 and at least one dirt outlet port 133. The exemplary inlet port 129 of the apparatus 100 of FIGS. 3A and 3B forms the dirt inlet 128. The exemplary brush chamber 121 of FIGS. 3A and 3B includes two outlet ports 133a and 133b, one for each of the dirt flow paths 138a and 138b. In some embodiments in which a brush chamber 121 includes two or more outlet ports 133, two or more of the outlet ports 133 are angularly spaced from one another. In some embodiments in which a brush chamber 121 includes two or more outlet ports 133, in use two or more of the outlet ports 133 are sequential following the debris inlet 128 such that debris encounters one port 133 and then another (e.g., to primarily direct the debris through the first port 133 that is encountered). In some examples, an outlet port 133 of a brush chamber 121 is an inlet port to a collection area 142, as exemplified by the port 133 of the brush chamber of the first brush 120a of FIG. 13.

[0226] In some examples, the brush chamber 121 includes walls sized and shaped to remain in close proximity to, or in contact with, the brush 120 mounted within the chamber 121. Chamber walls sized and shaped to remain in close proximity to, or in contact with, the brush 120 to retain debris against and/or between bristles 124 to be carried by the bristles as the brush 120 rotates in use. In embodiments in which a spacing gap 127 is provided, the wall chambers extend away from the brush sweep to accommodate the spacing gap 127, as exemplified in FIGS. 3A and 3B.

[0227] The sidewall 121c includes one or more curved surfaces extending generally parallel to the brush axis of rotation to assist in guiding debris about the axis of rotation when the brush is in use. The exemplary apparatus 100 of FIGS. 3A and 3B includes a deflector 144 forming a curved surface in the surface cleaning head 102. The deflector 144 forms part of the sidewall 121c. The deflector 144 forms a wall of part of a debris path 138a. The curvature of the deflector 144 generally follows the radius of the brush 120. In use, during rotation of the brush 120, a radially outward end of the bristles 124 may pass in close proximity to, or in contact with, the deflector 144. The front end 104 extends

downwardly from the deflector **144** toward the target surface. The exemplary deflector **144** of FIGS. 3A and 3B also forms the rounded edge of the surface cleaning head **102** between the top side **106** and the front end **104**.

[0228] The exemplary dirt flow paths **138** of FIGS. 3A-3B each include a portion downstream of the brush chamber, extending from the brush chamber **121** to the dirt collection region **142**. The portion downstream of the brush chamber **121** may be any suitable length. The exemplary dirt flow path **138b** of FIG. 3A includes an extended passageway between the dirt collection chamber **142** and the dirt inlet **128**. The exemplary path **138a** of FIG. 3A includes an opening in a wall separating the brush chamber **121** from the dirt collection region **142**.

[0229] As shown in FIGS. 3A-3B and FIG. 8, the bristles **124**, through rotation of the brush(es) **120**, can direct dirt and debris through the dirt inlet **128** and to the collection region **142** via the path **138**. Each dirt inlet **128** opens to at least one dirt flow path **138**. In some examples, a dirt inlet **128** opens to more than one discrete dirt flow path **138**. Discrete dirt flow paths **138** may lead to different dirt collection regions **142** or to a common dirt collection region **142**. In the example illustrated in FIG. 3A, the dirt inlet **128** opens to discrete first and second dirt collection paths **138a**, **138b** each leading to the same dirt collection region **142**. In the example illustrated in FIG. 3B, the dirt inlet **128** opens to discrete first and second dirt collection paths **138a**, **138b** leading to separate dirt collection regions **142a**, **142b**, respectively. It will be appreciated that discrete dirt flow paths opening from discrete dirt inlets **128** may lead to a common dirt collection region **142** or to different dirt collection regions **142**.

[0230] In embodiments in which an apparatus **100** includes discrete collection areas **142a**, **142b**, the collection areas **142a**, **142b** can be positioned relative to one another in any configuration, such as spaced in the forward/rearward direction, vertically spaced (i.e., one above the other), laterally spaced (i.e., one beside the other in a direction transverse to the forward/rearward direction), or any combination thereof.

Change in Mode of Operation

[0231] In accordance with an aspect of this disclosure, which may be used alone or in combination with any other aspect, the surface cleaning apparatus includes two or more different modes of operation, including any of the modes of operation described herein. In any embodiment in which the surface cleaning apparatus **100** includes two or more modes of operation, the surface cleaning apparatus **100** may be switched between modes in any suitable way including any of the ways described herein.

[0232] In some examples, the surface cleaning apparatus **100** is manually switchable between modes. A manual switching system includes a manual toggle or actuator, such as a lever, mechanically coupled to an adjustable system to effect a switching of the system (e.g., a lever coupled to the rear wheels **110b** to lower the wheels and thereby lift a rear brush, or coupled to drive system to reverse the drive system and thereby switch the rotational direction that the drive system drives a brush). For example, referring to FIGS. 11 and 12, the legs **152** can be fixedly coupled to the opposed sidewalls **108**, and the user can loosen the couplings, adjust the angle at which the legs **152** extend from the surface

cleaning head **102**, and retighten the couplings to fix the legs **152** at a new raised or lower position.

[0233] In some examples, the surface cleaning apparatus **100** is automatically switchable between modes. Automatic switching may be controlled in any suitable way, such as electronically and/or mechanically. An electronic automatic system may include an electronic control system having a user interface (e.g., an actuator such as a button or touch-screen) to receive a user mode selection input, the electronic control system communicatively coupled to an adjustable system (e.g., a brush, wheel, and/or drive system) to control a condition thereof (e.g., communicatively coupled to a motor to extend an arm to lower the rear wheels, or communicatively coupled to a drive system to reverse the drive system to switch the rotational direction that the drive system drives a brush). An electronic automatic system may use a motor, solenoid or the like. A mechanical automatic system includes a mechanical linkage (one or more links) to another component of the apparatus **100** such that the mode is changed when the other component is adjusted (e.g., in response to moving a handle, such as described elsewhere herein).

[0234] A surface cleaning apparatus **100** may include more than one switching system operable to switch between modes of operation, and a user can select between the switching systems.

[0235] A surface cleaning apparatus may be switched between modes in response to a sensed input. Referring to FIGS. 57-59, in the example shown, the surface cleaning apparatus **100** includes a sensor **206** mounted to the front end **104** of the surface cleaning head **102**. The sensor **206** may be configured to detect whether the target surface to be cleaned is grass or another soft surface (e.g., carpet, turf) or a hard surface (e.g., concrete, asphalt, stone, wood). For example, the sensor **206** may be configured to detect the height of the surface, where a surface that raises higher may be a soft surface and a lower surface may be a hard surface. The sensor **206** may be configured to detect the density of the surface, where a lower density surface may be a soft surface and a higher density surface may be a hard surface. The detection means of the sensor may be optical, ultrasonic, sonic, or any other detection means suitable to determining the type of surface. In response to the sensor **206** determining the type of surface, the surface cleaning apparatus **100** may automatically convert to a mode of operation suitable for the detected surface type.

Differential Conveyance of Large and Small Debris

[0236] In accordance with an aspect of this disclosure, which may be used alone or in combination with any other aspect, the surface cleaning apparatus includes a large debris path and a small debris path discrete from the large debris path. Different paths for large and small debris allows the paths to be adapted to the type of debris being handled.

[0237] As discussed herein, a single brush may convey debris to the large debris path (e.g., when the brush rotates in one direction) and to the small debris path (e.g., when the brush rotates in the opposite direction). Alternately, or in addition, one brush (a large debris brush) may direct debris to the large debris path and a different brush (a small debris brush) may direct debris to the small debris path. In the latter case, the large debris brush and the small debris brush may operate at different times depending, e.g., on the target surface to be cleaned and/or the expected debris to be

collected. In the latter case, both the large and small debris brushes may rotate in the same direction (e.g., they are located at the same end of the head 102) or they may rotate in different directions (e.g., they are at opposite ends of the head 102). In any case, the brush rotates to direct debris to the inlet to a path 138a, 138b.

[0238] In use, a surface cleaning apparatus may pick up particulate matter with a wide range of particle sizes. The large debris path is adapted to handle large debris, such as leaves, grass, long pine needles, and pine cones. The small debris path is adapted to handle small debris, such as sand, short coniferous needles, dirt, and sawdust. The debris paths may be adapted by, e.g., being sized differently, by including different features, and/or by leading to different collection regions, as described further elsewhere herein.

[0239] In accordance with this aspect, as exemplified in FIGS. 3A and 3B, the exemplary large debris path 138a may be larger than the exemplary small debris path 138b. The large debris path 138a is generally larger in order to accommodate the larger debris. In some examples, the large and small debris paths have a different minimum transverse cross sectional areas 139 in a plane transverse to a direction of flow therethrough. The minimum transverse cross sectional area 139 of the large debris path 138a may be at least 10%, at least 25%, or at least 50% larger than the minimum transverse cross sectional area 139 of the small debris path 138b. Optionally, the small debris path 138b may be more slot-shaped than the large debris path.

[0240] Optionally, the debris travels (e.g., is swept up, e.g., a ramp) to an elevation at an upper end of the dirt collection region for the debris. Accordingly, the debris enters the dirt collection region at an elevation above a floor of the dirt collection region. In such a case, each debris path 138 may open into a collection region at an elevated location above a floor of the collection region to allow the debris to fall into the chamber and collect below the entry location, inhibiting the return of the debris back out the debris flow path. The exemplary collection region 142 of FIG. 3A is a common collection region for a small debris flow path 138b and a large debris flow path 138a, while the exemplary head 102 of FIG. 3B includes discrete or separate large and small collection regions 142a, 142b. As exemplified in FIG. 3A, the inlet of a dirt collection region may be at a higher elevation for a large debris path 138a than a small debris flow path 138b, even if separate collection regions are used.

[0241] To accommodate both the large and small debris paths in a compact head 102, one of the large and small debris paths may be generally above the other of the large and small debris and, optionally the large debris path may be generally above the small debris path. The large debris path 138a and/or the portion thereof immediately downstream of the brush chamber may be generally above the small debris path 138b and/or the portion thereof immediately downstream of the brush chamber, as exemplified in FIG. 3B.

[0242] The debris path may include a ramp such that the brush sweeps debris up the ramp to or towards the dirt collection region. If there is no air flow produced by a motor and fan assembly, then the brush should impart sufficient momentum to convey debris to the dirt collection region. As exemplified in FIGS. 3A and 3B, the small debris path 138b includes a ramp 146 extending upwardly towards an outlet end 148b of the small debris path 138b. As exemplified in FIG. 3B, the small debris path opens from a lower end of the brush chamber 121, and extends upward to direct debris into

the collection region 142b at an elevated location. The ramp 146 may support the small debris in the path 138b as the debris is raised to the elevated location. A ramp may extend along all or a portion of a debris path. The large debris path 138a may also include a ramp, however the exemplary large debris path 138a of FIG. 3A does not include a ramp (e.g., the brush 120 may carry large debris to an elevated location and then throw the debris to the entry of the collection region).

[0243] As exemplified by the ramp 146 of FIG. 3B, in some examples a lower end of the ramp 146 is adjacent the dirt inlet 128. In some examples, a lower end of the ramp 146 is separated from the dirt inlet 138 by less than 5%, 10%, or 20% of the radius of the brush 120. A ramp close to the dirt inlet reduces the distance that the brush needs to move the debris before the debris reaches the ramp.

[0244] In some examples, a lower end of a ramp may be at or close to a bottom end of the apparatus 100. In use as the surface cleaning apparatus 100 passes over the target surface, the ramp may be maintained close to (e.g., within 1 mm, 5 mm, or 10 mm) or against the target surface to inhibit the brush 120 from kicking debris rearward past the ramp. A wheel (e.g., wheel 110c of FIG. 2) may be adjacent (e.g., rearward of) the ramp 146 to support the ramp 146 at an appropriate height above the target surface.

[0245] In any embodiment, a ramp 146 and/or a portion of the debris flow path upstream of the ramp can include a flexible portion 147 (e.g., resilient material, such as rubber) at the lower end thereof adjacent the dirt inlet 128. The flexible portion 147 (e.g., exemplary flexible portion 147 of FIG. 3A) may deform during impact with irregularities in the surface to be cleaned (e.g., large particulate matter, embedded stones, cracks in the surface, roots, etc.), to ease movement of the apparatus 100 over the target surface with the ramp kept close to or against the surface.

[0246] The larger debris may be lighter and easier for the bristles 124 to carry (i.e., lift) than the small debris. As exemplified in FIGS. 3A and 3B, the large debris path may open from an upper end of the brush chamber 121. The exemplary inlet end 148a of the large debris path 138a of FIGS. 3A and 3B opens from the upper end of the brush chamber 121. In some examples, the entire inlet end 148a of the large debris path 138a opens into the top 75%, the top 60%, or the top 50% of the height of the brush chamber 121. As exemplified in FIGS. 3A and 3B, the small debris path 138b may open from a lower end of the brush chamber 121. The exemplary inlet end 148a of the small debris path 138b of FIG. 3B opens from the lower end of the brush chamber 121. In some examples, at least a portion of the inlet end 148a of the small debris path 138b opens into the bottom 5%, 10%, or 15% of the height of the brush chamber 121.

[0247] As exemplified in FIG. 3B, a portion of each of the large and small debris paths immediately downstream of the brush chamber may extend generally transversely the same length as the brush to receive debris from along the entire transverse length of the brush.

[0248] One or both of the debris paths 138a, 138b may extend over a brush. Such a design may be used if a single brush is used for large and small debris or if a large debris brush and a small debris brush are provided.

[0249] As exemplified in FIG. 3B, the large debris path 138a may extend over the brush 120 relative to the vertical axis 103b. This design is useable if a debris brush (large or small debris) rotates clockwise and the inlet of the debris

path is forward of the brush (e.g., the brush is at the front end of the head **102**) of if the debris brush rotates counterclockwise and the inlet of the debris path is rearward of the brush (e.g., the brush is at the rear end of the head **102**).

[0250] Alternately, one or both of the debris paths **138a**, **138b** may not extend over a brush. Such a design may be used if a single brush is used for large and small debris or if a large debris brush and a small debris brush are provided.

[0251] As exemplified in FIG. 3B, the small debris path **138b** may extend upwardly and rearwardly from the brush **120**. This design is useable if a debris brush (large or small debris) rotates counterclockwise and the inlet of the debris path is rearward of the brush (e.g., the brush is at the front end of the head **102**) of if the debris brush rotates clockwise and the inlet of the debris path is forward of the brush (e.g., the brush is at the rear end of the head **102**).

[0252] Alternatively or in addition, in some examples as exemplified in FIG. 3B, a longitudinal length of a debris path (e.g., the large debris path **138a** as exemplified) that is within a brush chamber **121** may be longer than a longitudinal length of a small debris path **138b** that is within the brush chamber **121**, such as at least 50%, at least 75% or at least 100% longer. Such a design may be used if a debris path extends over the brush.

[0253] In any embodiment, the large debris path **138a** may include a spacing gap **127** and the small debris path **138b** may not include a spacing gap **127** or may include a smaller spacing gap than the large debris path (e.g., at least 50% smaller or at least 75% smaller) and which may be a minimal gap spacing.

[0254] Whether a single brush is used to selectively direct debris to the large debris path **138a** and the small debris path **138b**, or a large debris brush is used to direct debris to the large debris path **138a** and a small debris brush is used to direct debris to the small debris path **138b**, the large and small debris paths may lead to different dirt collection regions **142**. For example, the exemplary large debris path **138a** of FIG. 3B opens into a large debris collection region **142a** and the exemplary small debris path **138b** opens into a discrete small debris collection area **142b**. Alternately, as exemplified in FIG. 3A that may lead to a common collection region.

[0255] If large and small collection regions are provided, they may be separately emptiable. For example, large debris such as leaves may be emptied into a compost area and small debris such as sand may be emptied into a garden or disposal bin. For example, they may be individually removable or individually openable.

[0256] It will be appreciated that in any embodiment, the different collection regions may be adapted for the different debris. For example, the larger debris may be sent to a larger collection region so that the region is able to hold a reasonable volume of the larger debris. The large debris collection region **142a** may be, e.g., at least 25% larger, at least 50% larger, or at least 75% larger than the small debris collection area **142b**.

[0257] Whether a single brush is used to selectively collect large and small debris, or a large debris brush is used to collect large debris and a small debris brush is used to collect small path, a large debris collection region **142a** and/or a large debris flow path **142a** may be separated from a small debris collection area by a porous member such that smaller debris may pass through the porous member by gravity to thereby separate small and large debris. For example, a large

debris collection region **142a** and/or a large debris flow path **142a** may extend over a small debris collection area and/or flow path that is separated from the small debris collection area by a porous member. Referring to FIGS. 8-10, the exemplary head **102** includes a porous member **130** between the large debris flow path **148a** and the small debris collection area **142b**. The exemplary porous member **130** is a screen with apertures sized to allow small debris through but to block large debris. For example, an aperture size (e.g., diameter) of the porous member **130** may be less than 10 mm, less than 5 mm, or less than 1 mm.

Discrete Large and Small Debris Brushes

[0258] In accordance with an aspect of this disclosure, which may be used alone or in combination with any other aspect, the surface cleaning apparatus **100** includes a large debris brush to pick up large debris, and a small debris brush to pick up small debris. The large debris brush is adapted to pick up large debris such as leaves, grass, long pine needles, and pine cones. The small debris brush is adapted to pick up small debris such as sand, short coniferous needles, dirt, and sawdust. The large and small debris brushes may direct debris to different collection regions or a common collection region. As discussed herein, a single debris path may lead to a common collection region or a large debris path **138a** may lead to a large debris collection region **142a** and a small debris path **138b** may lead to a small debris collection region **142b**.

[0259] Large and small debris brushes may differ in any suitable way. In some examples, a large debris brush has a larger diameter and/or longer bristles than a small debris brush. A larger diameter and/or bristle length may be conducive to collecting large debris by providing sufficient angular spacing between adjacent rows of bristles to scoop up and transport the large debris. Also, or alternatively, a longer free diameter of the bristles (i.e., the distance from the core to the radial outer end of the bristles) may allow the bristles of the large debris brush to be more flexible, and the bristles of the small debris brush to be sufficiently rigid such that the force generated by the bristles **124** returning to their original state after deflecting while in contact with the target surface is sufficient to kick small debris along the flow path (e.g., up the ramp **146** and into the small particulate collection area **142b**). The diameter of the large debris brush may be at least 20%, 25%, or 50% larger than the diameter of the small debris brush. The free length of the bristles **124** of the large debris brush may be at least 20%, 25%, or 50% longer than the bristles **24** of the small debris brush.

[0260] The large debris brush **120a** can have a diameter in the range of 2-15 inches, 3-12 inches, or 4-8 inches. The small debris brush **120b** can have a diameter in the range of 1-6 inches, 2-5 inches, or 3-4 inches. Referring to FIG. 8, the exemplary large debris brush **120a** has a diameter of about 5 inches and the exemplary small debris brush **120b** has a diameter of about 2 inches. Referring to FIG. 13, the exemplary large debris brush **120a** has a diameter of about 6 inches and the exemplary small debris brush **120b** has a diameter of about 1 inch.

[0261] Also, or alternatively, in some examples, large and small debris brushes are driven at different speeds of rotation. In some examples, the large debris brush **120a** is operable at a first speed of rotation and the small debris brush **120b** is operable at a second speed of rotation. A small debris brush may be driven at a speed of rotation that is

faster than that of a large debris brush, such as at least 25% faster, at least 40% faster, or at least 50% faster. A faster rotational speed assists the small debris brush **120b** in generating sufficient force to kick the small particulate matter along the debris flow path (e.g., up the ramp **146** along the second path **138b**), while a slower speed of the large debris brush **120a** inhibits the brush from kicking larger debris away.

[0262] As discussed previously, the large debris brush **120a** and the small debris brush **120b** may rotate in the same direction or in opposite directions.

Small and Large Debris Pick Up Modes

[0263] In accordance with an aspect of this disclosure, which may be used alone or in combination with any other aspect, the surface cleaning apparatus picks up small debris differently from large debris. The apparatus **100** may be selectively switched between a first mode and a second mode. The first mode is a large debris pick up mode to pick up large debris, such as leaves, grass, long pine needles, and pine cones. The second mode is a small debris pick up mode to pick up small debris, such as sand, short coniferous needles, dirt, and sawdust.

[0264] A surface cleaning apparatus according to this disclosure may be useable on a variety of surfaces. Accordingly, different modes of operation may be used for different surfaces. For example, a large debris mode may be used when using the surface cleaning apparatus on grass or a patio wherein large debris is present (e.g., after a storm has deposited pine cones, leaves, etc. on a patio). A small debris mode may be used when the surface cleaning apparatus is used to clean a surface having small debris. Switching between large debris and small debris modes may comprise one or more of:

- [0265] 1. Altering the operation of the surface cleaning apparatus to direct large and small debris to different collection regions.
- [0266] 2. Adjusting one or more settings of a brush **120** (e.g., rotational speed and/or direction of rotation) between large debris and small debris modes.
- [0267] 3. Changing the configuration of a brush **120** between large debris and small debris modes (e.g., longer or shorter bristles).
- [0268] 4. Changing a dirt flow path **138** (e.g., expanded or contracted) between large debris and small debris modes.
- [0269] 5. Rendering a brush inoperable (e.g., moving a brush between raised and lowered positions and/or a rotational drive system of a brush is toggled between on and off conditions wherein the brush will not rotate or is freely rotatable but disengaged from a drive mechanism) between large debris and small debris modes.

[0270] It will be appreciated that any one or more of the foregoing may be used in any surface cleaning apparatus using any one or more of the aspects set out herein.

[0271] In accordance with this aspect, a brush **120** may selectively rotatable clockwise in one of the small debris mode and large debris mode and counterclockwise in the other of the small debris mode and the large debris mode. Different rotation directions may be used to direct debris to different debris flow paths. As exemplified in FIG. 3B, a clockwise rotation setting of brush **120** directs debris along a first debris flow path **138a** which extends through a forward portion of the brush chamber **121**, and a counter-

clockwise rotation setting of brush **120** directs debris in a second debris flow path **138b** which extends through a rearward portion of the brush chamber **121** and up ramp **146**. The different debris flow paths may have different features adapted for the different debris sizes (e.g., referring to FIG. 8 the debris is carried up by the exemplary brush **120a** in the first path **138a** and swept up the ramp **146** in the third path **138c**) and/or lead to different collection regions (e.g., referring to FIG. 3B, the exemplary first path **138a** leads to a first region **142a** and the second path leads to a second region **142b**).

[0272] Switching rotational direction may be accomplished in any suitable way. For example, a manual clutch may be used. Referring to FIGS. 31-33, the exemplary surface cleaning apparatus **100** includes a hand lever **182** coupled to a frame **174** of a clutch assembly **170** provided for the front wheels **110a**. The clutch assembly **170** includes a brush drive gear **162** and first, second, and third clutch gears **172a-c**. While the surface cleaning apparatus shown includes only one brush, it will be appreciated that the clutch assembly **170** and hand lever **182** described herein could be used in a surface cleaning apparatus having more than one brush. For example, a single clutch assembly **170** and hand lever **182** could be used to change the rotational directions of a first brush and a second brush, or each brush could have its own clutch assembly **170**, which could be controlled by a single common lever or each brush and clutch assembly could have its own lever **182**.

[0273] Referring to FIGS. 32-33, in the illustrated example, the front wheel gear **160a** is shown as an external spur gear that rotates with the front wheel **110a** in the same direction of rotation as the front wheel **110a**. The clutch assembly **170** is shown in the second position in FIG. 32, wherein the third clutch gear **172c** is engaged with the front wheel gear **160a**. The user may grasp the hand lever **182** and pull the hand lever **182** in the generally rearward direction to move the clutch assembly **170** from the second position to the first position as shown in FIG. 33, wherein the first clutch gear **172a** is engaged with the front wheel gear **160a**. The user may conduct the reverse operation to move the clutch assembly **170** from the first position to the second position. The user may, for example, place the clutch assembly **170** in the second position to collect small particulate matter when the surface cleaning head **102** is moved in the forward direction. The user may, for example, place the clutch assembly **170** in the first position to collect large particulate matter when the surface cleaning head **102** is moved in the forward direction. The user may, for example, alternate the clutch between the first and second position when alternating between moving the surface cleaning head **102** in the forward and rearward directions in order to collect large particulate matter or collect small particulate matter in both the forward and rearward directions. It will be appreciated that instead of a manually operable lever, the clutch may be adjusted by electronic control (e.g., a motor or solenoid which may be manually actuated by a user, a sensor which detects the type of target surface or the type of debris to be collected or a direction or movement of the head **102**).

[0274] Referring now to FIG. 34, shown therein is a perspective cross-sectional view of the surface cleaning apparatus **100** through the clutch assembly **170**, front wheel gear **160a**, and brush drive gear **162**. The clutch assembly **170** is engaged with the front wheel gear **160a** to thereby rotate the brush drive gear **162** with the front wheel **110a**. As

shown, the brush drive gear **162** is coupled to the central member **126** of the brush **120** such that rotation of the brush drive gear **162** drives a corresponding rotation of the brush **120**.

[0275] Alternately or in addition to changing a direction of rotation of one or more brushes, the rotational speed of one or more brushes may be changes. The rotational speed may be adjusted by the clutch assembly **170**, adjusting the power to a motor that drives a brush, etc. Any actuator discussed herein may be used. For example, a brush of the surface cleaning apparatus may rotate at a first speed in a large debris mode and may rotate at a second speed in a small debris mode. The second speed may be greater than the first speed. A slower speed allows the brush to pick up larger debris. A faster speed helps propel smaller debris that is swept up by the brush **120** up a path to a collection region. In some examples, the user can choose between only the first speed or the second speed. In some examples, the user can choose between only the first speed, the second speed, and one or more setpoints in between. In some examples, the speed is continually adjustable between the first speed and the second speed.

[0276] For example, a clutch assembly **170** can further be operable in conjunction with, e.g., the front wheel gear **160a** to adjust the speed of rotation of the brush **120**. For example, referring now to FIG. **35** showing an expanded view of a portion of the cross-section of FIG. **34**, the front wheel gear **160a** has a first tier **184a** and a second tier **184b**, and the clutch assembly **170** can be moved between the first tier **184a** and second tier **184b**. The first tier **184a** has a greater diameter than the second tier **184b** and a greater number of teeth than the second tier **184b**. Accordingly, when the clutch assembly **170** is engaged with the first tier **184a** as shown in FIG. **35**, the rotational speed of the brush **120** resulting from rotation of the front wheel gear **160a** will be greater than when the clutch assembly **170** is engaged with the second tier **184b**. A user may wish to switch to a slower rotational speed of the brush **120** for collection of larger debris and may switch to a faster rotational speed of the brush **120** for collection of smaller debris or scrubbing applications.

[0277] The front wheels **110a** include a wheel casing **186** movable between a retracted position (shown in FIG. **35**) and an extended position (shown in FIG. **36**). The front wheel gear **160a** is coupled to the wheel casing **186**. The user may grip the wheel casing **186** and pull outwardly from side wall **108** of the surface cleaning head **102** to move the wheel casing **186**, and thereby the front wheel gear **160a**, from the retracted position to the extended position. The user may push the wheel casing **186** inwardly toward the side wall **108** of the surface cleaning head **102** to move the wheel casing **186**, and thereby the front wheel gear **160a**, from the extended position to the retracted position. When the wheel casing **186** is in the retracted position, the clutch assembly **170** can engage the first tier **184a** of the front wheel gear **160a**, and when the wheel casing **186** is in the extended position, the clutch assembly **170** can engage the second tier **184b** of the front wheel gear **160a**. It will be appreciated that the wheel casing may be adjusted by electronic control (e.g., a motor or solenoid) or that a mechanical linkage actuatable (e.g., manually by a user or by a position of the handle).

[0278] Referring to FIG. **37**, the clutch assembly **170** may further include a biasing means **188**, exemplified as a spring, biasing the clutch assembly **170** toward the front wheel gear **160a**. The biasing means **188** may help maintain engage-

ment of the clutch assembly **170** with the first and second tiers **184a**, **184b** of the front wheel gear **160a**. Accordingly, to switch from the first tier **184a** to the second tier **184b** of the front wheel gear **160a**, and thereby slow the rotational speed of the brush **120**, the user may grasp the hand lever **182** and pull the hand lever **182** in the generally rearward direction. As shown in FIG. **38**, the user may move the clutch assembly **170** against the biasing means **188** away from the first tier **184a** of the front wheel gear **160a** and hold the clutch assembly **170** in a third position intermediate the first and second positions, wherein none of the gears of the clutch assembly **170** engage the front wheel gear **160a**. The user may then pull the wheel casing **186** to the extended position as shown in FIG. **38** and return the clutch assembly **170** to the first or second position in engagement with the second tier **184b** of the front wheel gear **160a** such as shown in FIG. **36**. The user may conduct the reverse operation to move the clutch assembly **170** from the second tier **184b** to the first tier **184a** of the front wheel gear **160a**, and thereby increase the rotational speed of the brush **120**. It will be appreciated that such an adjustment may be electronically controlled (e.g., a motor or solenoid) or mechanically adjusted by a mechanical linkage actuatable (e.g., manually by a user or b a position of the handle).

[0279] Accordingly, following the operations described above, the user may shift the clutch assembly **170** to any of the second position on the first tier **184a** (see e.g., FIGS. **40A** and **32**), the first position on the first tier **184a** (see e.g., FIGS. **40B** and **33**), the second position on the second tier **184b** (see e.g., FIGS. **40C** and **36**), and the first position on the second tier **184b** (see e.g., FIGS. **40D** and **39**). In this way, the user may achieve any combination of rotational speeds and rotational directions of the brush **120** when moving the surface cleaning head **102**.

[0280] It will be appreciated that additional tiers can be used and/or differently sizes tiers. It will also be appreciated that differently sized gears can be used in the gear train of the clutch assembly **170**, such as the first or third clutch gears **172a**, **172c**, to produce a different rotational speed and rotational direction of the brush **120** when the clutch assembly **170** is in the first position than in the second position.

[0281] As discussed previously, a brush may be reconfigurable between a first configuration and a second configuration. The brush may be reconfigurable in any suitable way, such as by lengthening and shortening bristles, stiffening or unstiffening bristles, increasing or decreasing the number of discrete bristles and/or rows.

[0282] In some examples, at least some of the bristles **124** of the brushes **120** are retractable. For example, all of the bristles **124** of the brush **120** may be retractable to change the brush **120** from a first diameter for collecting large particulate matter to a second diameter for collecting small particulate matter. In such examples, operations such as those described herein may be performed to lower the surface cleaning head **102** when the bristles **124** of the brush **120** are retracted to maintain contact of the bristles **124** with the surface to be cleaned. In some examples, two or more patterns of the bristles **124** can be achieved by adjusting the length of the rows of bristles **124**. The bristles **124** may be adjustable to any number of lengths between fully extended and fully retracted, thereby enabling adjustment of the number of bristles of a given length and of the stiffness of the bristles **124**, such as depending on the size of the debris being collected.

[0283] Additionally, or in the alternative, the brushes 120 can include a backing member that is extendable and retractable behind the bristles 124 to adjust the stiffness of the bristles 124. The backing member may extend about the bristles 124 along a portion of the length of the bristles to support the bristles. For example, to increase the stiffness of the bristles 124, the backing member may be extended behind a substantial portion of the length of the bristles 124. To decrease the stiffness of the bristles 124, the backing member may thus be retracted to extend behind a lesser portion of the length of the bristles 124. Referring to FIG. 52A, a backing member 145 extends from the central member 126 along part of the length of the bristles 124 to support the bristles 124, and is adjustable between a first extent 145a and a second extent 145b. The backing member 145 lies against the bristles 124 to delineate a free end of the bristles that is able to flex freely (i.e., free of the backing member 145).

[0284] In any example, the number of rows of bristles 124 can be adjustable. As exemplified in FIG. 56, each brush 120a, 120b of the surface cleaning apparatus 100 includes three bristle segments 124a-124c. Each bristle segment 124a-124c includes two rows of bristles 124 extending across about one third of the length of the central member 126. The bristles 124 are staggered about the central member 126 such that each brush 120a, 120b includes six rows of bristles 124. The bristles 124 may be rotatable about the central member 126 to change the six rows of bristles 124. For example, the bristles 124 of each bristles segment 124a-124c can be brought into alignment. In this way, the brushes 120a, 120b are changed from six rows of bristles 124 each extending across about one third of the length of the central member 126, to three rows of bristles 124 each extending substantially across the length of the central member 126.

[0285] As another example, two or more rows of bristles 124 extending across the central member 126 or a segment thereof can be brought together. This may decrease the number of rows of bristles 124 and increase the stiffness of each row of bristles 124. For example, the brushes 120a, 120b may be changed from 8 rows of bristles 124 to four rows of bristles 124 having an increased stiffness. Any number of bristle segments may be used having any number of rows of bristles 124 extending therefrom. Any number of rows of bristles 124 can therefore be achieved with any combination of lengths and stiffnesses. Referring to FIG. 54, the exemplary brush 120 includes a first central member 126a to which are mounted first and second rows 125a, 125b and a second central member 126b to which are mounted second and third rows 125c and 125d. Rotation of first central member 126a and second central members 126b relative to one another can adjust the distance between the first set of rows and the second set of rows. In use, first central member 126a can be rotated such that the first row 125a lies against or within the third row 125c and the second row 125b lies against or within the fourth row 125d, thus effectively forming a brush 120 with two rows rather than four.

[0286] As discussed previously, the debris flow path 138 may be reconfigurable between small and large configurations. The debris flow path 138 may be reconfigurable by moving at least one wall along part or all of a debris flow path. Moving the wall changes at least one transverse dimension of the debris flow path, changes a spacing between the wall and a brush that is mounted in the debris

flow path, and/or changes a height above the target surface of a leading edge of the dirt inlet 128 opening into the dirt flow path.

[0287] Referring to FIGS. 3A and 3B, the exemplary front wall 149 is adjustable between a first position 149a and a second position 149b. The front wall 149 forms a wall of the dirt flow path 138. Shifting the wall between the first position 149a and the second position 149b changes the spacing between the wall 149 and the brush to allow for larger debris to pass or to pass more easily in the second position 149b. Shifting the wall between the first position 149a and the second position 149b changes the spacing gap 127 to allow for larger debris in the second position 149b. As exemplified, shifting the wall between the first position 149a and the second position 149b changes the width of the path 138 along the vertical 103b to allow for larger debris in the second position 149b. As exemplified in FIG. 3A, the front wall is pivoted upwardly changes the clearance 141 between the leading edge 128a of the dirt inlet 128 and the target surface 143 to allow for larger debris in the second position 149b. It will be appreciated that the width of the path 138 is increased more at the lower end of the wall 149 and the amount of the increase in width decreases in the upstream direction. It will be appreciated that instead of pivoting the wall 149 at a rear end thereof, the wall 149 could be translated forwardly and/or vertically. If the wall 149 is translated forwardly and vertically, then the increase in the width of the path may be generally uniform along the length of the path or at least the portion of the path that extends around the brush 120.

[0288] For example, part or all of the front wall 149 may be raised to pick up, e.g., pinecones, and lowered to pick up, e.g., acorns. Lowering the front wall 149 may prevent the smaller acorns from being kicked forward by the brush 120 when the brush is rotated in the clockwise direction.

[0289] In accordance with this aspect, a surface cleaning apparatus 100 may include more than one discrete brush 120 and switching between large and small debris modes may change which brush or brushes is driven on a target surface and/or in which order brushes sweep the target surface.

[0290] As exemplified in FIG. 8, the large debris brush 120a is mounted in the first debris flow path 138a leading to a first debris collection region 142a, while the small debris brush 120b of FIG. 8 is mounted in the second debris flow path 138b leading to a second debris collection region 142b discrete from the first debris collection region 142a.

[0291] Optionally, in use, the large and small debris brushes can be concurrently rotationally driven (at the same or different speeds of rotation and in the same or different directions) and applied to a target surface. For example, moving the head 102 of FIG. 8 forward applies different brushes in a first order (e.g., a large debris mode) and moving the head 102 rearward applies the different brushes in a second, different order (e.g., a small debris mode). Accordingly, the large debris brush 120a is applied ahead of the small debris brush 120b as the head 102 is moved forward across a target surface, and when the head 102 is moved rearward the large debris brush 120a follows the small debris brush 120b across the target surface. Accordingly, a user selects an order of application of large and small brushes by selecting a direction of motion of the head 102. If separate large and small debris containers are used, the apparatus 100 may be operated to sweep up large debris first to avoid filling the small debris collection area, which may

have a smaller collection volume than the large debris collection area, with large debris.

[0292] Instead of driving both brushes on a surface, a surface cleaning apparatus **100** having a large debris brush **120a** and a small debris brush **120b** may be switched between large and small debris modes by moving one or more brushes between raised and lowered positions (as discussed elsewhere herein) and/or toggling a rotational drive system of one or more brush between on and off conditions (as discussed elsewhere herein) and/or disengaging a drive system such that a brush or brushes are freely rotatable but not driven. For example, the apparatus **100** may use only one or more large debris brushes in a large debris mode and use only one or more small debris brushes in a small debris mode (i.e., only use the brush or brushes of the corresponding type when in a particular mode).

[0293] It will be appreciated that the apparatus **100** may use both large and small debris brushes in one of the large and small debris modes, but in the other of the large and small debris modes, the apparatus may use only brushes of the corresponding type. For example, the apparatus **100** may use both large and small debris brushes in the large debris mode, but use only small debris brushes in the small debris mode (e.g., to keep a large debris brush from kicking small debris away from the apparatus **100**).

[0294] A drive system may be switched between on and off conditions by any suitable system, including any of the systems described elsewhere herein. Referring to FIG. **8**, the exemplary brushes **120a**, **120b** are driven by the wheels **110a**, **110b**, respectively, and the drive system is linked to the brushes such that one or both of the brushes **120a**, **120b** is only driven to rotate when the surface cleaning head **102** is moved in one of the forward and rearward directions and the drive system is inactive when the surface cleaning head **102** is moved in the other of the forward and rearward directions.

[0295] A brush may be raised and lowered by any suitable system, including any of the systems described elsewhere herein (e.g., raising and lowering a set of wheels **110**). In some examples, the brush is raised even when the drive system is toggled off, to keep the brush from disturbing debris on the target surface and/or to keep the brush from wearing or snagging on the target surface. In some examples, a brush may be raised even while driven by the drive system, such as if the drive system is powered by wheel rotation and a simple coupling (e.g., a direct coupling) between the brush and the wheel is used.

Switching Modes of Operation by Adjusting Handle Position

[0296] In accordance with another aspect of this disclosure, which may be used alone or in combination with any other aspect, the surface cleaning apparatus **100** is switchable between a small debris mode of operation and a large debris mode of operation by adjusting the drive handle **112** between a first position and a second position.

[0297] As exemplified in FIGS. **1** and **2**, the drive handle **112** may extend upward from the head **102** to an accessible height, allowing a user to switch between modes without reaching down. The handle **112** may extend outwardly from the coupling (e.g., pivot) location a significant distance, giving a mechanical advantage to a user when the user uses the handle as a lever to mechanically adjust a system at the head **102**.

[0298] The exemplary drive handle **112** of FIGS. **21** and **22** is rotatable about the coupling axis **119** of the couplings **116** between a first position (FIG. **21**) and a second position (FIG. **22**). In the first position, the drive handle **112** is operable by the user to move the surface cleaning head **102** in a first direction (e.g., pushing the head **102** forward), and in the second position the drive handle **112** is operable by the user to move the surface cleaning head **102** in a second direction (e.g., pushing the head **102** rearward). When the exemplary drive handle **112** of FIGS. **21** and **22** is in the first position, the drive handle **112** extends rearward of the rear end **105** of the surface cleaning head **102** whereby the drive handle **112** is operable by the user to push the surface cleaning head **102** in the forward direction. When the exemplary drive handle **112** of FIGS. **21** and **22** is in the second position, the drive handle **112** extends forward of the front end **104** of the surface cleaning head **102** whereby the drive handle **112** is operable by the user to push the surface cleaning head **102** in the rearward direction.

[0299] Accordingly, switching the handle between first and second positions enables the user to select which of the first and second brushes **120a**, **120b** leads by switching which direction the head **102** will move when the handle **112** is pushed. Accordingly, moving the handle **112** between first and second positions may switch a mode of operation even if the handle **112** is not mechanically or electrically coupled to another system of the head **102**.

[0300] In embodiments in which a brush direction of rotation is governed by the wheel direction of rotation or the position of a clutch which is part of the drive system, switching the handle between first and second positions and driving the apparatus **100** from the new handle position reverses the direction of rotation of the brush.

[0301] Optionally, as discussed elsewhere herein, the drive handle **112** may be a lever arm for a mechanical linkage. Accordingly, when the drive handle **112** is rotated about the coupling axis **119** of the couplings **116** between the first and second positions, the drive handle **112** actuates the mechanical linkage to change the mode of operation of the surface cleaning apparatus **100**. For example, the surface cleaning apparatus **100** may be configured such that rotating the drive handle **112** effects one or more of retracts one of the first and second brushes **120a**, **120b**, changes a brush direction of rotation, changes a brush speed of rotation, and raises one of the front and rear ends **104**, **105** of the surface cleaning head **102**.

[0302] In the example illustrated in FIGS. **21-26**, the surface cleaning apparatus **100** is configured such that rotating the drive handle **112** about the coupling axis **119** from the first position to the second position moves the front wheels **110a** from the raised position to the lowered position. Rotating the drive handle **112** from the first position to the second position thus raises the front end **104** of the surface cleaning head **102**, thereby disengaging the brush **120b** from the surface to be cleaned. Alternately, or in addition, the surface cleaning apparatus **100** can be configured such that rotating the drive handle **112** about the coupling axis **119** from the first position to the second position (or vice versa) raises the rear end **105** of the surface cleaning head **102**, thereby disengaging the brush **120a** from the surface to be cleaned.

[0303] Reference is now made to FIGS. **27-30** showing cross-sectional views of a surface cleaning apparatus **100** with an example mechanical linkage assembly **151**. The

exemplary drive handle 112 of FIGS. 27-30 acts as a lever to move the front wheels 110a from the raised position to the lowered position. FIGS. 27 and 28 show the surface cleaning apparatus 100 with the drive handle 112 in the first position and the front wheels 110a in the raised position. FIGS. 29 and 30 show the surface cleaning apparatus 100 with the drive handle 112 in the second position and the front wheels 110a in the lowered position. In the illustrated example, the linkage assembly 151 includes a pair of legs 152, where each leg 152 is rotatably coupled to a respective one of the opposed sidewalls 108 of the surface cleaning head 102 proximate the front end 104. The front wheels 110a are coupled to the distal end of each leg 152.

[0304] The linkage assembly 151 further includes a pair of angled link members 168. Each link member 168 is coupled at one end to a respective leg 152 (at the end opposite the front wheels 110a) and coupled at the other end to a respective arm 114 of the drive handle 112 (at the end opposite the hand grip portion 118). Each link member 168 is further rotatably coupled to a respective one of the opposed sidewalls 108 of the surface cleaning head 102 at the vertex 153 of the angled link member 168. When the user rotates the drive handle 112 about the coupling axis 119 of the rotatable couplings 116 from the first position to the second position, the link members 168 are driven to rotate about their vertices, the coupling point between the link members 168 and the legs 152 is driven upward, and the front wheels 110a at the distal end of the legs 152 are driven toward the surface to be cleaned. In this way, rotating the drive handle 112 about the coupling axis 119 from the first position to the second position raises the front end 104 of the surface cleaning head 102 and disengages the first brush 120a from the surface. Conversely, when the user rotates the drive handle 112 about the coupling axis 119 of the rotatable couplings 116 from the second position to the first position, the link members 168 are driven to rotate about their vertices 153 in the opposite direction, the coupling point between the link members 168 and the legs 152 is drawn downward, and the front wheels 110a at the distal end of the legs 152 are driven toward the surface cleaning head 102. In this way, rotating the drive handle 112 about the coupling axis 119 from the second position to the first position lowers the front end 104 of the surface cleaning head 102 and re-engages the first brush 120a with the surface to be cleaned.

[0305] Additionally, or in the alternative, the rotatable drive handle 112 may also act as a switch for one or more electrical motors. When the drive handle 112 is rotated about the coupling axis 119 of the couplings 116 between the first and second positions, the drive handle 112 may switch the electrical motor(s) to change the mode of operation of the surface cleaning apparatus 100. For example, the surface cleaning apparatus 100 may be configured such that rotating the drive handle 112 activates one or more motors to retract one of the first and second brushes 120a, 120b and/or raise one of the front and rear ends 104, 105 of the surface cleaning head 102 as described previously. As another example, if rotation of the first brush 120a and/or second brush 120b is driven by one or more motors, rotating the drive handle 112 may change the direction of rotation of one or both brushes 120a, 120b or stop rotation of one of the brushes 120a, 120b (e.g., disengage or deenergize the drive motor). Stopping a motor reduces power consumption and/or extends the service life of a motor by eliminating unnecessary use.

[0306] In some examples, only one of the first and second brush 120a, 120b is lowered and/or rotationally driven by the apparatus 100 in the forward direction and only the other of the first and second brush 120a, 120b is lowered and/or rotationally driven by the apparatus 100 in the rearward direction. Accordingly, in each of a small and large debris mode of operation, only the respective small or large brush or brushes are used. It will be appreciated that, optionally, only the leading brush in the direction of motion of the surface cleaning head 102 may be lowered and/or rotationally driven by the apparatus 100, and the trailing brush in the direction of motion may be raised and/or not driven by the apparatus 100.

[0307] As exemplified in FIGS. 27-30, the exemplary drive handle 112 may alternately or in addition be coupled to a clutch assembly 170. The exemplary drive handle 112 of FIGS. 27-30 moves the clutch assembly to change the mode of operation of the brushes 120a, 120b when the drive handle 112 is moved between the first and second positions. Referring to FIGS. 28 and 30, the surface cleaning apparatus 100 includes a clutch assembly 170 provided in the rear wheels 110b and operatively coupled to the drive handle 112. The clutch assembly 170 includes a gear train having a first clutch gear 172a, second clutch gear 172b, and third clutch gear 172c, and the second brush drive gear 162b. The second brush drive gear 162b is coupled to the second brush 120b such that the second brush 120b rotates with the second brush drive gear 162b. The gear train is mounted on a frame 174 such that the teeth of the gears mesh for transmission of rotation from each gear to an adjacent gear. As shown, the first clutch gear 172a is positioned at one end of the gear train and meshes with the second clutch gear 172b, which also meshes with the second brush drive gear 162b. The third clutch gear 172c is positioned at the other end of the gear train and meshes with the second brush drive gear 162b.

[0308] The rear wheel gear 160b of the rear wheel 110b is shown as an internal spur gear that rotates with the rear wheel 110b in the same direction of rotation as the rear wheel 110b. The frame 174 of the gear train can be rotatable about the second brush drive gear 162b such that the first clutch gear 172a and the third clutch gear 172c can be alternately engaged with the rear wheel gear 160b. When the first clutch gear 172a is engaged with the rear wheel gear 160b, rotation of the rear wheel gear 160b is transmitted through the first clutch gear 172a and second clutch gear 172b to the second brush drive gear 162b, and thereby to the second brush 120b. In this way, when the first clutch gear 172a is engaged with the rear wheel gear 160b, the second brush 120b is driven to rotate in the same rotational direction as the rear wheel gear 160b. When the third clutch gear 172c is engaged with the rear wheel gear 160b, rotation of the rear wheel gear 160b is transmitted through the third clutch gear 172c to the second brush drive gear 162b, and thereby to the second brush 120b. In this way, when the third clutch gear 172c is engaged with the rear wheel gear 160b, the second brush 120b is driven to rotate in the opposite rotational direction as the rear wheel gear 160b.

[0309] The drive handle 112 can be configured to move the clutch assembly 170 between a first position, wherein the first clutch gear 172a is engaged with the rear gear wheel 160b, and a second position, wherein the third clutch gear 172c is engaged with the rear gear wheel 160b. For example, as shown in FIG. 28, when the drive handle 112 extends rearwardly from the surface cleaning head 102, the clutch

assembly 170 is in the first position. As shown in FIG. 30, when the drive handle 112 extends forwardly from the surface cleaning head 102, the clutch assembly 170 is in the second position. The frame 174 is biased to the first position. In the example shown in FIGS. 28 and 30, the surface cleaning apparatus 100 includes a wire 176 coupled at one end to the frame 174 of a respective clutch assembly 170 and coupled at the other end to a respective arm 114 of the drive handle 112 (at the end opposite the hand grip portion 118). The wire 176 extends around a guide 178 between the arm 114 and the frame 174. When the drive handle 112 is moved from the position shown in FIG. 28 to the position shown in FIG. 30, the wire pulls the frame 174 generally in the forward direction. As a result, the clutch assembly 170 is rotated about the second brush drive gear 162b from the first position to the second position.

[0310] In the example shown, when the drive handle 112 extends rearwardly from the surface cleaning head 102, the second brush 120b will rotate in the counter clockwise direction when the surface cleaning head 102 is pushed in the forward direction and will rotate in the clockwise direction when the surface cleaning head 102 is pulled in the rearward direction. When the drive handle 112 extends forwardly from the surface cleaning head 102, the second brush 120b will rotate in the counter clockwise direction when the surface cleaning head 102 is pushed in the rearward direction and will rotate in the clockwise direction when the surface cleaning head 102 is pulled in the forward direction.

[0311] Optionally, as exemplified in FIG. 30, a single clutch assembly may be used such that one or more brushes that is not engaged by the clutch rotates in the same direction. For example, one or more brushes may not be driven by a separate motor or mechanical linkage. Instead, the clutch may be used to determine a direction of rotation and/or speed of rotation of a first wheel as discussed elsewhere herein and one or more additional brushes may be slaved to the first wheel by a mechanical drive member. For example, as exemplified, the surface cleaning apparatus 100 can further include a pulley 180 operatively coupling, e.g., the second brush drive gear 162b to the first brush drive gear 162a. Accordingly, when the second brush drive gear 162b is driven to rotate in the clockwise or counter clockwise direction, the first brush gear 162a will be correspondingly driven to rotate in the clockwise or counter clockwise direction. In this way, the first and second brushes 120a, 120b can be always rotate in the same direction.

[0312] Optionally, the clutch assembly 170 can include a first clutch assembly operatively connected to the first brush 120a and a second clutch assembly operatively connected to the second brush 120b. It will be appreciated that the first clutch assembly may be part of the drive system of the front wheels 110a in addition to, or in the alternative to, the second clutch assembly being part of the drive system of the rear wheels 110b. The clutch assemblies can be configured to drive the first and second brushes 120a, 120b to rotate in the same direction, opposite directions, or to drive one of the first and second brushes 120a, 120b to rotate and stop or disengage the other of the first and second brushes 120a, 120b from rotating. The clutch assemblies can further be configured such that the brushes 120a, 120b can operate in a first mode in the forward direction (e.g., same direction, opposite direction, one rotates the other stationary) and a second mode in the rearward direction and/or in the first

mode when the drive handle extends rearwardly from the surface cleaning head 102 and the second mode when the drive handle extends forwardly from the surface cleaning head 102, wherein the mode comprises one or more of the direction of rotation the speed of rotation, stopping a brush from rotation, lifting a brush off of a target surface or disengaging a brush from a drive system so as to be freely rotatable.

Collection Chamber with Dumping Door

[0313] In accordance with another aspect of this disclosure, which may be used alone or in combination with any other aspect, the dirt collection chamber 132 includes one or more openable dumping doors 136. It will be appreciated that the or each door may be openable when the dirt collection chamber 102 is in the head 102 or when the dirt collection container 132 is removed from the head 102.

[0314] The or each door 136 is selectively or concurrently moveable between an open position in which an opening into the collection region is open and a closed position in which the opening into the collection region is closed. The openable dumping door 136 may be a wall or part of a wall of the collection chamber 132 that is selectively openable. The door 136 may selective close an opening that provides direct access to the collection region within. Alternatively, the door 136 may selectively close a pathway that extends between the dirt collection region and an exterior of the collection chamber, e.g., the dirt flow path that opens into the collection region. The door 136 may be a lid, covering an opening through an upper end of the collection chamber. A direct access dumping door allows the user to more easily remove the debris than if a user was required to dump through the dirt flow path openings.

[0315] Referring to FIGS. 3B-5, in the illustrated example the large and small particulate collection areas 142a, 142b can be removed and emptied concurrently. The collection container 132 is removably mounted to the surface cleaning apparatus 100 in a cavity 196 in the surface cleaning head 102. As shown, the optional handle 134 extends rearwardly from the collection container 132 when mounted to the surface cleaning head 102 and extends through an opening 198 in the rear end 105 of the surface cleaning head 102. The user may grip the handle 134 and lift the collection container 132 from the cavity 196 of the surface cleaning head 102, open the door 136, and concurrently dump the contents of the large and small particulate collection areas 142a, 142b. The common collection area 142 shown in FIG. 3B can be similarly emptied.

[0316] In the example of FIGS. 4-5, the openable door 136 for dumping of the contents of the collection container 132 is a lateral sidewall of the collection chamber 132.

[0317] A collection chamber 132 may include a dumping door whether or not it is removeable in a closed configuration. For example, as discussed subsequently, a non-return flap may close an inlet to the dirt collection container when it is removed from the head 102. The exemplary collection chamber 132 of FIGS. 4 and 5 does not include a non-return flap and is therefore removeable in an open configuration in which portions of the dirt flow paths into the collection regions remain open, but still includes a dumping door 136 to allow for easier emptying.

Closed Removeable Collection Chamber

[0318] In accordance with another aspect of this disclosure, which may be used alone or in combination with any

other aspect, the collection chamber **132** is removeable in a closed configuration in which the collection region **142** is enclosed. A closed removal configuration inhibits debris (e.g., dust) from exiting and disturbing a user, such as while the user carries the collection chamber to a disposal location. **[0319]** As exemplified in FIGS. **8** and **9**, a non-return flap **135** falls across the dirt flow path **138a** as the collection chamber **132** is lifted away from the main body **101** of the head **102** to close the dirt flow path **138a**. The flap **135** may form the dumping door **136**. Also, or alternatively, the collection chamber **132** may include a separate dumping door **136**.

One Collection Region Emptiable Through Another

[0320] In accordance with another aspect of this disclosure, which may be used alone or in combination with any other aspect, the surface cleaning apparatus **100** includes a collection region **142** that is emptiable through another collection region **142**. This aspect may be used if dirt collection containers that have a dumping door and/or empty through an inlet opening as discussed previously.

[0321] For example, the small debris collection region may empty through the large debris collection region. Accordingly, a dumping opening may be an opening in a wall of the large debris collection area, and the large and small debris collection areas are separated by a screen to prevent the large debris from entering the small debris collection areas but allow the small debris to move into the large debris collection area to be dumped through the dumping opening.

[0322] Referring to FIGS. **8-10**, the exemplary collection container **132** is shown including large and small particulate collection areas **142a**, **142b**, wherein the large particulate collection area **142a** is positioned upward of the small particulate collection area **142b** (i.e., when the surface cleaning head **102** is on a horizontal surface) and extending rearwardly of the small particulate collection area **142b**. In the example shown, the large and small particulate collection areas **142a**, **142b** are part of a single dirt collection container **132** that can be emptied concurrently when the dirt collection container is removed from the head **102**.

[0323] The exemplary small particulate collection area **142b** of FIGS. **8-10** is separated from the large particulate collection area **142a** by a porous member **130** (e.g., a screen) sized such that large debris cannot pass through to the small particulate collection area **142b**, but small debris can pass through the screen **130** to the large particulate collection area **142a** when the collection container **132** is turned upside-down for dumping. Alternately, the screen may open (e.g., pivot open, when the collection container **132** is inverted for emptying).

[0324] As shown, an optional handle **134** is provided on the door **136**, which is positioned at the top side **106** of the surface cleaning head **102** when the collection container **132** is mounted in the cavity **196**. The collection container **132** can include a latch or other release to secure the door **136** closed while the user transports the collection container **132** using the handle **134**. The user may then actuate the latch to open the door **136** and empty the contents of the collection container **132** through the opening.

Separately Removeable Large and Small Debris Collection

[0325] In accordance with another aspect of this disclosure, which may be used alone or in combination with any

other aspect, the surface cleaning apparatus **100** includes separate large and small debris collection regions. The large and small dirt collection regions may be part of a single dirt collection container or separate dirt collection containers.

[0326] The large and small debris collection regions may be independently removeable or concurrently removable and/or independently emptiable or concurrently emptiable. For example, if a user is primarily collecting large debris, the user may dump the large debris more frequently than the small debris.

[0327] The exemplary large and small particulate collection areas **142a**, **142b** of FIGS. **18B** and **18C** can be emptied separately. In the illustrated example, the collection container **132** includes a large particulate collection subcontainer **132a** in which the large collection area **142a** is located and a small particulate collection subcontainer **132b** in which the small debris collection region **142b** is located. The large and small collection subcontainers **132a**, **132b** are individually removeable from the main body **101** of the head **102**.

[0328] The large and small particulate collection subcontainers **132a**, **132b** can be liners for the large and small particulate collection areas **142a**, **142b**, respectively, and can be disposable and replaceable, such as paper, or reusable and washable, such as fabric or plastic. For example, the large and small particulate collection subcontainers **132a**, **132b** may be nested in and removable from respective cavities **196a**, **196b** in a larger collection container **132**.

[0329] The user may remove the large or small particulate collection subcontainers **132a**, **132b** from the collection container **132** for individual emptying or disposal while the collection container **132** remains mounted to the surface cleaning head **102** in the cavity **196**. Alternatively, the user may first remove the collection container **132** from the cavity **196** to transport the large and small particulate collection subcontainers **132a**, **132b** together to a refuse bin and subsequently remove the large and/or small particulate collection subcontainers **132a**, **132b** from the collection container **132** for individual emptying, or empty one or both of the subcontainers **132a**, **132b** while they remain within the larger container **132**.

[0330] It will be appreciated that instead of mounting or nesting the large and small particulate collection containers **132a**, **132b** in or to the collection container **132**, each of the large and small particulate collection containers **132a**, **132b** can be removably mounted to the surface cleaning head **102** directly in the main body **101** of the head **102** and individually or concurrently removable from the head **102**.

Expandable Collection Region

[0331] In accordance with another aspect of this disclosure, which may be used alone or in combination with any other aspect, the debris collection region **142** is expandable. The volume of the region **142** is expandable to contain more debris. Accordingly, part or all of one or more walls of a dirt collection region **142** may be expandable or moveable.

[0332] For example, the collection container **132** may include or consist of a flexible portion, such as plastic or fabric bag. A flexible collection container **132** may expand under the weight of the collected debris and thus may provide additional debris collection volume.

[0333] A flexible container **132** can form a common collection area **142** or two or more discrete collection areas **142a**, **142b**. For example, flexible container **132** may have

large and small collection areas **142a**, **142b** and one or both may be expandable. Alternately, as discussed previously, a head **102** may have two or more collection containers and one or more of these may be expandable.

[0334] Optionally, the flexible portion may be supported by a rigid portion. The flexible portion **200** may be removable from the main body **101** of the head **102** with the rigid portion **202** to remove a collection region from the head **101**. Alternately, the flexible portion **200** may be removable from the main body **101** of the head **102** while the rigid portion **202** remains with the head **101**.

[0335] Referring to FIGS. 43-45B, shown therein is surface cleaning apparatus **100** having an example collection container **132** having a flexible portion **200** and a rigid portion **202**. The rigid portion **202** can provide structural support for the flexible portion **200**. The flexible portion **200** may be collapsed to enclose a smaller volume, and extended to enclose a larger volume. The flexible portion may be extended manually or automatically (e.g., by the pressure of additional debris being directed into the collection region by a brush).

[0336] The exemplary rigid portion **202** of FIGS. 43-45B includes a divider splitting the rigid portion **202** into segments of the first and second paths **138a**, **138b** to the common collection area **142**. As shown, the collection chamber **132** is removably mounted to the surface cleaning head **102** and forms the rear end **105** of the surface cleaning head **102**. The rigid portion **202** of the collection chamber **132** is received in between the top side **106**, bottom side **107**, and opposed sides **108** of the surface cleaning head **102**.

[0337] The rigid portion **202** can be coupled to the surface cleaning head **102** by any means, for example, a friction fit or one or more engagement members. For example, the rigid portion **202** may include one or more biased members (e.g., pins) that are biased to an extended position wherein the biased members extend from the outer perimeter of the rigid portion **202**. The biased members can be receivable in one or more corresponding bores in the inner perimeter of the surface cleaning head **102** along the top surface **106**, bottom surface **107**, and/or one or both laterally opposed sides **108**. As shown, the collection container **132** includes a release **204** on the handle **134** that can be operated by the user in connecting and disconnecting the collection container **132** and the surface cleaning head **102**. The user may grasp the handle **134** and depress the release **204**, such as using their thumb, to move the biased members to a retracted position. Once retracted, the rigid portion **202** may be removed from or inserted between the top side **106**, bottom side **107**, and opposed sides **108** of the surface cleaning head **102**. Alternately, the biased members can be moved by movement of a handle from (e.g., a flat storage position as exemplified in FIG. 6) and a raised removal position (e.g., as exemplified in FIG. 44). Any other coupling suitable to releasably connect the collection area **132** and surface cleaning head **102** may be used.

[0338] The flexible portion **200** may include, e.g., a simple bag, such as a plastic or paper bag. The bag may be collapsed by being, e.g., crumpled upon itself or neatly folded.

[0339] Referring to FIGS. 46-51, shown therein is surface cleaning apparatus **100** having an example collection container **132** wherein the flexible portion **200** is configured with an accordion fold that can be compressed toward the surface cleaning head **102** in the forward direction (see e.g.,

FIGS. 46 and 47) and, as the collection container **132** fills with debris, can expand in the rearward direction (see e.g., FIGS. 48 and 49).

Brush with Curved Bristles

[0340] In accordance with another aspect of this disclosure, which may be used alone or in combination with any other aspect, the surface cleaning apparatus **100** includes one or more brushes with curved bristles. Some or all of the bristles of a brush may be curved. The bristles may be curved to assist in sweeping debris into a dirt container.

[0341] The bristles may be curved in a direction of expected rotation. Accordingly, the direction of curvature may depend on the intended direction of rotation of a brush. If a head has two brushes which rotate in opposite directing, then the bristles of each brush may be curved in a different direction. Curved bristles may form a shape that is conducive to scooping up debris.

[0342] Optionally, the curved bristles may be flexible so as to flex against the curve when encountering the target surface, and the resultant force generated by the bristles returning to their original state after disengaging from the target surface may be increased by the curve thus assisting in sweeping debris forward.

[0343] Bristles **124** can be curved in the clockwise or counter clockwise directions. Bristles **124** can be curved in the direction of rotation of the brush **120**. For example, if the bristles **124** are of the first brush **120a**, which rotates in a clockwise direction for collecting large particulate matter, the bristles **124** can be curved in the clockwise direction (see e.g., FIGS. 53A-53D). Bristles **124** curved in the clockwise direction may facilitate the collection of large debris by acting as a scoop to sweep up the large debris and guide it along the first path **138** without permitting the large debris to fall off the distal end of the bristles **124** during rotation of the brush **120a**.

[0344] As another example, if the bristles **124** are of the second brush **120b**, which rotates in the clockwise or counter clockwise direction (depending on the configuration of the surface cleaning head **102**) for collecting small particulate matter, the bristles can similarly be curved in the same clockwise or counter clockwise direction. Bristles **124** curved in the direction of rotation of the second brush **120b** may facilitate collection of small debris. The bristles **124** may flex against the curve as they are dragged across the surface to be cleaned during rotation of the brush **120b**. Accordingly, the resultant force generated by the bristles **124** returning to their original state after disengaging from the surface to be cleaned may be increased by the curve, and may thus assist in sweeping small debris up the ramp **146** and into the small particulate collection area **142b**.

[0345] Referring now to FIG. 54, the brushes **120** can include both straight and curved bristles **124**. As shown, the curved bristles **124** are curved in the clockwise direction to facilitate collection of large debris when the brush **120** rotates in the clockwise direction, and the straight bristles **124** enable collection of small debris when the brush **120** rotates in the counter clockwise direction.

Scrubbing a Target Surface

[0346] In accordance with another aspect of this disclosure, which may be used alone or in combination with any other aspect, the surface cleaning apparatus **100** is switchable between a debris collection mode and a scrubbing mode. A scrubbing mode is used to clean a surface, such as

by removing dried on dirt or a stain or assisting in collecting fine particulate matter that may not be swept up by the head.

[0347] In a scrubbing mode, the speed of rotation of a brush may be faster than in a debris collection mode. For example, the brush **120** may have a first rotational speed for debris pick up and a second rotational speed for scrubbing. The scrubbing rotational speed being faster than the debris pick up speed. The scrubbing speed may be at least 50% faster, at least 100% faster or at least 150% faster.

[0348] Alternately, or in addition, in a scrubbing mode, the brush may be brought closer to the target surface than in a debris collection mode. For example, the brush **120** may have a first height between the axis of rotation and the target surface in a debris collection mode and a second height between the axis of rotation and the target surface in a scrubbing mode. The second height may be less than the first height, such as 10% less, 15% less or 20% less. The first height may be the diameter of the brush **120**. The first height may be between the diameter of the brush and 10% less than the diameter of the brush **120**. The first height may be between the diameter of the brush and 15% less than the diameter of the brush **120**.

[0349] In a scrubbing mode, the apparatus **100** may dispense a cleaning fluid such as water and/or soap onto the target surface and/or onto the brush **120**. As discussed elsewhere herein, a module comprising or consisting of a reservoir or reservoir and pump may be attached to the head **102** for the scrubbing mode.

[0350] It will be appreciated that a scrubbing brush may be provided for use in the scrubbing mode. The scrubbing brush may use, e.g., microfibers and/or short and/or densely packed bristles.

Suction Assembly

[0351] In accordance with another aspect of this disclosure, which may be used by itself or in combination with one or more other aspects of this disclosure, the surface cleaning apparatus **100** can further include a suction assembly. The suction assembly may be used to pick up finer debris than the brush(es) or to assist the brush(es) to pick up finer debris. Accordingly, the suction assembly may improve collection efficiency of the surface cleaning apparatus **100**, such as small particulate matter that is too fine for the brushes to collect.

[0352] The suction assembly may be provided in an air flow path that is separate from a debris flow path and may extend from a dirty air inlet in head **102** to a clean air outlet. In such an embodiment, the suction assembly includes one or more dirt collection members. Alternately, the suction assembly may be provided in the debris flow path, upstream and/or downstream from a dirt collection region. In such an embodiment, the suction assembly may include one or more dirt collection members.

[0353] The suction assembly includes an air flow path extending between an air inlet and an air outlet, with an air moving member (e.g., a clean or dirty air motor and fan assembly) in the air flow path.

[0354] The air treatment member of the suction assembly, if provided, may be a filter bag, cyclone, momentum separator or filter positioned in the air flow path to remove dirt from an air flow in the air flow path.

[0355] Referring to FIGS. **60-64**, surface cleaning apparatus **100** is shown having an example suction assembly **208**. The suction assembly **208** is arranged to draw air through an

opening **212** in the bottom end of the surface cleaning apparatus **100** (e.g., through a slot in the bottom surface **107**). In some examples, the apparatus **100** includes more than one suction assembly **208** or and air flow path of a suction assembly **208** includes more than one air inlet. The opening **212** to the air flow path of the suction assembly **208** may be forward and/or rearward of a brush in a direction of motion of the head **102**. If the head **102** includes more than one brush **120**, then the opening **212** may be between brushes, forward of the brushes, and/or rearward of the brushes in a direction of motion of the head. The exemplary suction assembly **208** of FIGS. **60-64** is positioned forward of the first brush **120a** at the front end **104** of the surface cleaning head **102**.

[0356] In any example, the suction assembly can be operated in any suitable way. For example, the suction assembly may activate automatically in response to a condition, such as when the surface cleaning head **102** is moved in the forward and/or rearward directions. The suction assembly may additionally or alternately have a manual activation means, such as a power switch. The suction assembly may be actuated by a clutch mechanism, a sensor which detects a target surface, the rotation or direction of rotation of a particular brush or the position of a handle

[0357] In some examples, the suction assembly **208** is integrated into the assembly **100**. In such a case, the suction assembly may consist of an air moving member that is in a debris path. If the suction assembly is downstream of a dirt collection region (e.g., a dirt collection region has an air outlet, which may comprise a porous member such as a screen) then the suction assembly may comprise one or more air treatment members that may be removable for emptying and/or cleaning.

[0358] Alternatively, in some examples, the suction assembly **208** is removeable from the apparatus **100** (e.g., a removable module as discussed subsequently) or for independent operation to clean a surface and as such may be any vacuum cleaner known in the art. Accordingly a removeable suction assembly **208** can be used independently as a vacuum, such as a stick vacuum or hand vacuum.

[0359] The exemplary suction assembly **208** of FIGS. **60-64** is a hand vacuum having a handle **210**. The exemplary suction assembly **208** of FIGS. **60-64** is mounted to the front end **104** of the surface cleaning head **102** in a slot **212** and is removable from the slot **212** for use as a hand vacuum separate from the surface cleaning apparatus **100**. The slot **212** extends through the surface cleaning head **102** to the bottom surface **107**. When the hand vacuum is positioned in the slot **212**, a suction inlet **214** of the hand vacuum is positioned in the slot **212**. The suction inlet **214** may be proximate the bottom surface **107** of the surface cleaning head **102**. When the hand vacuum is positioned in the slot **212**, the suction inlet **214** can be used to lift small particulate matter, such as fine and/or light weight debris (e.g., drywall dust, sawdust, silt) through the slot **212** and suction inlet **214**.

[0360] In the illustrated example, the hand vacuum includes a power switch **216** provided proximate the handle **210**. The user may manually activate/deactivate the hand vacuum **216** using the power switch **216** when used in conjunction with the surface cleaning apparatus **100** and when used as a separate hand vacuum, such as for spot

cleaning. It will be appreciated that the hand vacuum may be actuated during a particular mode of operation as discussed previously.

[0361] As exemplified, the hand vacuum includes a dirt collection region within or discrete from the air treatment member. The dirt collection region in the hand vacuum may be emptied by any means known in the hand vacuum art. The hand vacuum can be emptied directly into a garbage bin or can be emptied into the collection container 132, such as by opening the door 136 and emptying the dirt into the collection area 142a. The collection chamber 132 of the surface cleaning apparatus 100 can then be emptied as described previously. In some examples, the suction assembly 208 dumps dirt automatically into the debris collection region 142 of the head 102.

[0362] Referring to FIGS. 65-69, surface cleaning apparatus 100 is shown having another example suction assembly 208 positioned rearward of the second brush 120b at the rear end 105 of the surface cleaning head 102. The suction assembly 208 is a hand vacuum 208 mounted to the rear end 105 of the surface cleaning head 102 in the slot 212. By positioning the hand vacuum at the rear end 105, particulate matter may be sequentially collected. That is, when the surface cleaning head 102 is moved in the forward direction, the first brush 120a can collect large debris, the second brush 120b can collect small debris, and the hand vacuum can collect any remaining debris, such as small particulate matter that may be too fine for the first and second brushes 120a, 120b to collect. This may avoid the small particulate collection area 142b and the collection chamber of the hand vacuum 208 from being rapidly filled with larger debris.

[0363] Referring to FIGS. 70-74, surface cleaning apparatus 100 is shown having another example hand vacuum mounted to the surface cleaning head 102 in the slot 212 positioned intermediate of the first and second brushes 120a, 120b. As shown, the slot 212 extends through the collection container 132. The hand vacuum 208 is thus removable together with the collection container 132 for emptying or may be separated from the collection container 132 prior to emptying the collection container 132. As shown, the slot 212 extends through the first path 138a. The first path 138a is divided into two branches that pass on the lateral sides of the slot 212 leading from the first brush 120a to the large particulate collection area 142a. As shown, the slot 212 extends through the small particulate collection area 142b. The second path 138b leads to two small particulate collection areas 142b provided on the lateral sides of the slot 212. In any example, the second path 138b may also lead into the slot 212 such that a portion of the small particulate matter collected by the second brush 120b can be drawn into the hand vacuum 208. This may increase the storage capacity of the small particulate collection areas 142b.

[0364] Alternately, one or more of the dirt containers may have a porous dirt outlet (e.g., screen) and, in operation, the inlet of the hand vac may be in fluid flow communication downstream of the porous dirt outlet.

[0365] Optionally, the suction assembly 208 may include a power source (e.g., onboard, such as a battery or capacitor, or a power cord to an external power supply). The power source powers the air moving member. The power source may also power one or more powered components of the surface cleaning apparatus 100.

[0366] If the suction assembly 208 is a removeable vacuum, and when the removeable vacuum is docked with

the surface cleaning apparatus 100 one or more further components of the surface cleaning apparatus 100 may be electrically coupled to the docked vacuum to draw power therefrom.

[0367] Alternately, or in addition, the on board energy storage member of a removable vacuum can be charged when docked with the surface cleaning apparatus 100 (e.g., in the slot 212 of the surface cleaning head 102) such as by an on board energy storage member provided in the head 102 or when the head is connected to household mains.

[0368] Alternately, or in addition, the on board energy storage member of a removable vacuum can be separately charged when removed from the surface cleaning apparatus 100.

Downwardly Extending Fence

[0369] In accordance with another aspect of this disclosure, which may be used by itself or in combination with one or more other aspects of this disclosure, the surface cleaning apparatus 100 includes a fence projecting down from the bottom end of the apparatus 100 adjacent a dirt inlet (e.g., extending from an edge of the dirt inlet). The fence may be positioned rearward (in a direction of travel of the head 102) of an operating brush. The fence improves collection efficiency of the surface cleaning apparatus 100 by inhibiting debris from passing rearward of the operating brush as the head travels forwardly.

[0370] The downwardly extending fence may be a passive (non-moving) brush or flange, optionally a flexible flange, which may extend downwardly to a location proximate but above the target surface or it may contact the target surface and, optionally flex in a rearward direction as the head travels in a forward direction.

[0371] Optionally, the fence may be a displaceable fence. A displaceable fence can be selectively extended and retracted, e.g., to be extended (e.g., lowered) when the head 102 is moved in one direction and retracted (e.g., raised) when the head 102 is moved in an opposite direction.

[0372] Referring to FIGS. 19 and 20A, the surface cleaning apparatus 100 includes an example displaceable fence 218. The displaceable fence 218 is a barrier extending from the bottom surface 107 of the surface cleaning head 102 configured to engage the surface to be cleaned. The barrier may be a flexible material such as bristles, rubber, or the like. As shown, the displaceable fence 218 is positioned to, when the head 102 is moved forward, block (or reduce the amount of) debris passing under the surface cleaning head 102 and bypassing the first brush 120a. The debris may build up against the displaceable fence 218 such that the first brush 120a can collect the built-up debris that might have otherwise bypassed by the brush 120a. The displaceable fence 218 can move from the extended position shown when the surface cleaning head 102 is moved in the forward direction to a retracted position when the surface cleaning head 102 is moved in the rearward direction. In the retracted position, the displaceable fence 218 is raised out of the way of debris passing under the surface cleaning head 102.

[0373] The displaceable fences 218 can be moved by any of the mechanical or electrical actuation means as described herein. Positioning of the displaceable fences 218 can be manually adjusted by the user and/or automatically adjust, such as using a sensor, at the initiation of a user activated switch, moving the head in a particular direction or the position of the handle.

Dispensing Reservoir

[0374] In accordance with another aspect of this disclosure, which may be used by itself or in combination with one or more other aspects of this disclosure, the surface cleaning apparatus **100** includes a dispensing reservoir. A dispensing reservoir allows the apparatus **100** to be used in additional ways. The dispensing reservoir may dispense a material using gravity and/or a dispensing member. For example, if the material is a particulate solid (e.g., grass seed or fertilizer), then the dispensing reservoir may use a hooper and/or spreader. If the material is a liquid (e.g., a cleaning fluid, water or a liquid fluid) then a pump may be used.

[0375] The dispensing reservoir may be in fluid communication with the head and therefore deliver the material into the head from which it may be spread by, e.g., one or more brushes or a dispenser (e.g., a spreader or spray nozzles) provided as part of the head. Alternately or in addition, the dispensing reservoir may include a dispenser (e.g., a spreader or spray nozzles). Accordingly, the apparatus **100** may dispense a material across a surface, using a sweeping brush **120** to distribute the material (e.g., grass seed or fertilizer) or a scrubbing brush to dispense cleaning fluids (e.g., water and/or soap) onto the target surface.

[0376] Referring to FIGS. **75-78**, the surface cleaning apparatus **100** shown has an example module **220** removably mounted to the top side **106** of the surface cleaning head **102**. The module **220** can be, for example, a cleaning module or a seed/fertilizer module. It will be appreciated that the module may be removably mounted to any surface of the head and may be securable thereto by a locking member.

[0377] As exemplified, the module **220** has a reservoir **222**. If the module **220** is a cleaning module, then the reservoir **222** can contain water and/or cleaning fluid. If the module **220** is a seed/fertilizer module, then the reservoir **222** can contain seeds and/or fertilizer. As shown, the exemplary reservoir **222** has a hopper that is sloped toward a reservoir outlet **224** in communication with a dispensing inlet **226** in the top side **106** of the surface cleaning head **102**.

[0378] One or both of the reservoir outlet **224** and the dispensing inlet **226** may be openable/closable (e.g. by a flap or valve) and operable to selectively connect and disconnect communication of the reservoir outlet **224** in flow communication with the head **102**. For example, the reservoir outlet **224** and dispensing inlet **226** may be communicatively connected when the surface cleaning head **102** is moved in the forward and/or rearward direction and may be communicatively disconnected when the surface cleaning head **102** is stationary. The reservoir outlet **224** and dispensing inlet **226** can be sized to restrict and/or operated to control the passage of the contents of the reservoir **222** therethrough.

[0379] As exemplified in FIG. **78**, a seed/fertilizer module has a reservoir outlet **224** that is in communication with the dispensing inlet **226** in the top side **106** of the surface cleaning head **102** above the brush **120**. The inlet **226** may include a port into the brush chamber **121**, e.g., into an upper end of the brush chamber **121**. In use, seeds and/or fertilizer may pass from the reservoir **222** of the module **220** to the interior of the surface cleaning head **102**. The seeds and/or fertilizer can then land on the bristles **124** of the brush **120**, which can facilitate dispersal of the seeds and/or fertilizer on the target surface. The brush **120** can further be configured to rotate such that the seeds and/or fertilized may not be swept up by the brush **120** after dispersal. For example,

when the surface cleaning head **102** is moved in the forward direction, the brush **120** may rotate in the clockwise direction to disperse the seeds and/or fertilizer by dropping them rearward of the brush **120**. Alternately or in addition, the debris flow path **138** may be closed (e.g., a flap or valve) to prevent the brush **120** from sweeping the falling seed and/or fertilizer rearwardly into the large collection area **142a**. When the surface cleaning head **102** is moved in the rearward direction, the brush **120** may rotate in the counter clockwise direction to disperse the seeds and/or fertilizer by tossing them forward of the brush **120**. Optionally, the front end **104** (e.g., wall **149**) of the surface cleaning head **102** may be moved away from the brush **120** or removed to provide less or no obstruction when the brush **120** tosses the seeds and/or fertilizer forwardly.

[0380] Similarly, when the module **220** is a cleaning module, water and/or cleaning solution may pass from the reservoir **222** of the module **220** to the interior of the surface cleaning head **102**, and the bristles **124** of the brush **120** can spray the water and/or cleaning solution on the surface to be cleaned. The bristles **124** can then scrub the surface in a scrubbing mode of operation. The scrubbing mode of operation may be suitable for use with rigid bristles **124** in accordance with any of the examples described herein, such as with retracted bristles, doubled-up bristles, and/or the second brush **120b**. Alternately, if the brush is a scrubbing brush that absorbs water (e.g., a microfiber brush), the brush may absorb the fluid and then apply the fluid to the target surface.

[0381] In any example, the reservoir **222** of the cleaning module **220** may direct the material onto the surface to be cleaned (i.e., rather than into a brush chamber **121** or otherwise into the debris flow path **138**). For example, the reservoir **222** of the cleaning module **220** may be connectable to a spray nozzle (downstream of a pump) to direct a pressurized spray of the water and/or cleaning solution onto the surface.

[0382] Optionally, to avoid overflowing during movement of the surface cleaning head **102**, the reservoir **222** of the cleaning module **220** may be closed. In such examples, the cleaning module **220** may further include a water/solution inlet or hose connection for filling the reservoir **222** or a closeable lid, which may be sealable.

Integrated Lighting

[0383] In accordance with another aspect of this disclosure, which may be used by itself or in combination with one or more other aspects of this disclosure, the surface cleaning apparatus **100** includes one or more lights to illuminate the target surface. Lights may aid the user in identifying debris on the surface to be cleaned, such as under low light conditions or debris that may otherwise blend with the surface. The lights may also assist a sensor determine the type of targeted surface the head is on and/or the type of debris being collected. With or without a light, the sensor may send a signal to a controller that issues a signal to an electric (e.g., motor) or electro-mechanical (e.g., solenoid) member to adjust any one or more ways that a head may operate as discussed herein.

[0384] The light may be mounted to any suitable part of the surface cleaning apparatus **100**. In some examples, the light is mounted to the head **102** to be adjacent the target surface. Referring to FIG. **57**, the exemplary surface cleaning apparatus **100** includes a light **228** mounted to the

surface cleaning head **102**. The exemplary light **228** of FIG. **57** is mounted to an upper end **106** of the head **102**. Mounting the light **228** to an upper part of the head **102** allows the light to illuminate further forward from the head **102** than a light mounted to a lower end of the head **102**, such as to illuminate a portion of the target surface that the head **102** is immediately approaching to allow a user to determine how to clean that portion of the target surface. However, it will be appreciated that a light may also or alternatively be mounted to a lower part of the head **102** to illuminate a portion of the target surface near and/or under the head **102**. In any such case, the light is directed downwardly and optionally also forwardly.

[0385] The exemplary light **228** of FIG. **57** is mounted to the head **102**, however in some examples a light may also or alternatively be mounted to another portion of the apparatus **100**. In some examples, the apparatus **100** includes a light mounted to the drive handle **112**. A light mounted to the drive handle **112** may be able to illuminate further from the head **102** than a light mounted to the head **102** due to the increased elevation of the light above the target surface. In embodiments in which the handle **112** is moveable between first and second positions, a light mounted to the drive handle **112** moves with the handle **112** to change the illumination pattern based on the position of the handle **112** such as to adapt the illumination pattern for different modes of operation.

[0386] A light may be directed in any suitable direction. In some examples, the apparatus **100** may include a light directed forwardly and/or rearwardly. A forwardly directed light may be mounted to the apparatus **100** at a front end **104**. A rearwardly directed light may be mounted to the apparatus **100** at a rear end **105**. Referring to FIG. **57**, the exemplary light **228** is mounted to the front end **104** of the head **102** and directed forwardly to aid the user in identifying debris on the target surface when moving in the forward direction. The apparatus **100** may also or alternatively include a light mounted to the rear end **105** and directed rearwardly. In some examples, the apparatus **100** includes a rearward directed light and a forward directed light, such as where the apparatus **100** is operable in a sweeping operation while moving forwardly and is also operable in a sweeping operation while moving rearwardly.

[0387] Optionally, the apparatus **100** includes a light that moves between a first position directed in a first direction and a second position directed in a second direction, such as between forward and rearwardly directed positions. Any suitable repositioning system may be used. In some examples, the light is moved between positions by repositioning the drive handle **112**, such as by repositioning the drive handle **112** to change between modes of operation as described elsewhere herein. In such a case, the light may be redirectable (e.g., mounted on a swivel, to point in a direction of motion when the handle position is pivoted from being rearward of the rear end of the head and forward of the forward end of the head). Optionally, this may reduce the number of lights needed, such as by using one light that is moveable between forward and rearward directions rather than one light directed forward and another light directed rearwardly.

[0388] Any suitable type of light may be used. The lights may be, for example, light emitting diodes. The lights may emit directly or through a conveyance, for example the lights may include a light pipe to direct the light source to the

target surface. The light may be generated at a first location on the surface cleaning apparatus **100** and carried to another location on the surface cleaning apparatus **100** to be emitted. For example, the light may be generated at the rearward end of the head **102** and carried, such as by light pipes, to the forward end of the head **102** (e.g., to reduce the bulk of components carried at the forward end of the head **102**).

[0389] The light may be powered by any suitable way. For example, one or more replaceable or rechargeable on board energy storage members, such as one or more batteries and/or capacitors, may be coupled to the lights to power the light. In examples in which batteries are the power source, the light(s) may be run on the same on board energy storage members used to power any electrical components of the surface cleaning apparatus **100** such as noted above.

[0390] The on board energy storage members may be mechanically charged through the rotary motion of the wheels **110** when the surface cleaning head **102** is moved by the user.

[0391] In examples in which an on board energy storage members is used to power one or more motors of the surface cleaning apparatus **100**, the motor(s) may additionally or alternatively serve as generator to charge the on board energy storage members.

Magnetic Debris Collection

[0392] In accordance with another aspect of this disclosure, which may be used by itself or in combination with one or more other aspects of this disclosure, the surface cleaning apparatus **100** includes one or more magnets to pick up magnetic debris. Magnetic debris may be heavier than non-magnetic debris, and a magnet may help to collect the debris. Also, or alternatively, a user may wish to collect magnetic debris separately from non-magnetic debris, such as for separate disposal or reuse. For example, the apparatus **100** may be used to clean a shop floor or to clean a lawn following a construction project wherein nails or other metallic components may be hidden in the grass, for example.

[0393] Referring to FIG. **7**, the exemplary apparatus **100** includes a magnet **230**. The exemplary magnet **230** of FIG. **7** is mounted to a bottom end **107** of the surface cleaning head **102**. Mounting the magnet **230** at the bottom end **107** of the head **102** positions the magnet to pass closely over the target surface to be cleaned to improve magnetic collection. However, it will be appreciated that a magnet may be mounted elsewhere, such as on a front end **104** or a rear end **105**.

[0394] A magnet may be mounted forward and/or rearward of one or more brushes **120** and/or dirt inlet **128** in a direction of travel of the head **102**.

[0395] The magnet **230** may be mounted forward of a brush to pick up magnetic debris before it reaches the brush, such as to allow the magnetic debris to be collected separately from non-magnetic debris. For example, magnetic debris (e.g., nails or screws) may be undesirable in a compost or garden where non-magnetic debris (e.g., leaves or dirt) may otherwise be disposed of.

[0396] The magnet **230** may be mounted rearward of a brush, such as to pick up magnetic debris which the brush is unable to pick up such as heavy debris that the bristles do not lift. Mounting the magnet **230** rearward of a brush may allow the brush to collect as much magnetic debris as it can to reduce the load that the magnet **230** carries, such as where the brush can direct the debris to a collection chamber while

the magnet **230** holds the debris until an active removal step (e.g., manual removal by the user or manual or automatic switching off an electromagnet to release the debris).

[0397] The magnet **230** may be mounted forward of one type of brush and rearward of another type of brush, such as to allow the user to select which collection chamber may contain magnetic debris. Accordingly, the magnet **230** is mounted forward of a small debris brush and rearward of a large debris brush. The exemplary magnet **230** of FIG. 7 is mounted rearward of the front large debris brush **120a** and forward of the rear small debris brush **120b**.

[0398] Optionally, the magnet **230** may be mounted downstream of a dirt inlet, such as to draw magnetic debris towards a brush that extends through the dirt inlet. The magnet **230** may be mounted in a brush chamber in which the brush is mounted. The magnet **230** may be provided to help the brush pick up magnetic debris, which may be heavier than most non-magnetic debris that the brush is meant to pick up.

[0399] The exemplary apparatus **100** of FIG. 7 includes a single magnet **230**, however it will be appreciated that an apparatus **100** may include more than one magnet **230** and may include magnets in different positions, such as a magnet forward of a brush and a magnet rearward of the same brush.

[0400] Any suitable size and/or shape of magnet may be used. The exemplary magnet **230** of FIG. 7 includes a rectangular bottom surface extending from one lateral side **108** to the other lateral side. If more than one magnet is included, magnets may be the same size and shape as one another or may be of different sizes and/or shapes and/or strengths, such as a larger magnet forward of a brush to pick up larger magnetic debris (e.g., nails) and a smaller magnet rearward of a brush to pick up smaller magnetic debris (e.g., metal filings).

[0401] A magnet may be arranged in any suitable way. For example, a magnet may be an elongated shape with the long dimension extending generally traverse to a direction of movement. The exemplary magnet **230** of FIG. 7 extends with a longest dimension generally traverse to the forward-rearward direction of the head **102**. If more than one magnet is included, the magnets may be arranged the same way or in different ways, such as a first magnet extending traverse to the direction of motion and a second magnet extending generally parallel to the direction of motion of the head.

[0402] Any suitable type of magnet may be used, such as permanent magnets or selectively operable electromagnets. If more than one magnet is included, the magnets may be the same type as one another or may be of different types, such as a permanent magnet forward of a brush and an electromagnet rearward of a brush. The magnet may be removably mountable to the head **102**.

Mower Adaptation

[0403] In accordance with another aspect of this disclosure, which may be used by itself or in combination with one or more other aspects of this disclosure, the surface cleaning apparatus **100** is selectively reconfigurable as a mower. A rotatable mower blade may be interchanged with a brush. The mower blade may be driven by the same drive system as the brush.

[0404] In accordance with this aspect, one or more brushes **120** of the surface cleaning apparatus **100** can be interchanged with a mower blade. Referring to FIG. 79, the exemplary brush **120** may be interchanged with the exem-

plary reel blade **240**. For example, the surface cleaning apparatus **100** can function as a lawn mower, and can include one or more mower blades (e.g., a reel or cylindrical blade) sized for use with the surface cleaning apparatus **100**. One or both brushes **120** can be replaced with a mower blade to allow the surface cleaning apparatus **100**, such as to allow the apparatus **100** to cut debris (e.g., grass or weeds) from the target surface. The brush **120** and/or blade may be mounted to the head **102** via an easy-release fastener to allow for easy interchanging. For example, the brush **120** and/or blade may be mounted via wing screws, clips, or other easy-release fasteners.

[0405] It will be appreciated that the debris path may be adjusted as discussed previously so as to provide a larger or smaller brush chamber so as to accommodate both a sweeping brush **120** and a mower blade.

[0406] The blade may be driven by the same drive system as drives the brush. The blade may be driven in the same way as the brush (i.e., the same rotational speed and/or direction), or the drive system may drive the blade differently from the brush. For example, the drive system may include a mode for the blade (e.g., a mower mode) which is adapted for cutting and a separate mode for the brush (e.g., a brush mode) which is adapted for sweeping. Switching between the modes may involve mechanically adjusting the drive system (e.g., via moving the handle **112** such as described elsewhere herein) or electronically adjusting the drive system (e.g., changing a setting on a motor control system) or the mower blade may be provided with different gearing than a sweeping brush. Switching between modes may be manual (e.g., pushing a button or moving a lever) or automatic (e.g., a sensor sensing that the blade is mounted to the drive system rather than the brush).

[0407] The surface cleaning apparatus **100** may collect the debris, such as collecting cut grasses and weeds, e.g., via the debris flow path **138** and the collection region **142**. Alternatively, in some examples, one or more debris flow path is closed or selectively closable when the surface cleaning apparatus **100** is in a mowing configuration. For example, the surface cleaning apparatus **100** may be used to trim a lawn and allow the trimmings to fall back to the target surface, e.g., to fertilize the lawn. In this way, the surface cleaning apparatus **100** may be used to trim the target surface without removing debris from the target surface.

[0408] It will be appreciated that, if an expandable dirt collection container is not used with a sweeping blade, the collection container may be replaced by a flexible container when a mower blade is attached (e.g., the expandable container of the embodiment of FIG. 50 may replace the container of the embodiment of FIG. 45B when a mower blade is attached).

[0409] While the above description describes features of example embodiments, it will be appreciated that some features and/or functions of the described embodiments are susceptible to modification. For example, the various characteristics which are described by means of the represented embodiments or examples may be selectively combined with each other. Accordingly, what has been described above is intended to be illustrative of the claimed concept and non-limiting. It will be understood by persons skilled in the art that other variants and modifications may be made without departing from the scope of the invention as defined in the claims appended hereto. The scope of the claims should not be limited by the preferred embodiments and

examples, but should be given the broadest interpretation consistent with the description as a whole.

CLAUSE SETS

Clause Set A

- [0410] 1. A surface cleaning apparatus comprising:
- [0411] (a) a surface cleaning head having a first brush which is rotatable about a first rotational axis in a clockwise direction and a second brush which is rotatable about a second rotational axis in a counterclockwise direction;
- [0412] (b). a first path from the first brush to a large particulate collection area; and,
- [0413] (c). a second path from the first brush to a small particulate collection area
- [0414] wherein, when the surface cleaning head is placed on a horizontal surface and moved in a forward direction with the brush in position to clean the surface, the rotational axis extends generally horizontally and generally transverse to the forward direction.
- [0415] 2. The surface cleaning apparatus of clause 1 wherein the large particulate collection area and the small particulate collection area are a common volume that receive particulate matter from each of the brushes.
- [0416] 3. The surface cleaning apparatus of clause 1 wherein the large particulate collection area and the small particulate collection area are discrete collection volumes.
- [0417] 4. The surface cleaning apparatus of clause 3 wherein the large particulate collection area and the small particulate collection area are concurrently removable from the surface cleaning apparatus.
- [0418] 5. The surface cleaning apparatus of clause 3 wherein the large particulate collection area and the small particulate collection area are separately removable from the surface cleaning apparatus.
- [0419] 6. The surface cleaning apparatus of clause 1 wherein each of the first and second brushes are operable at a common speed.
- [0420] 7. The surface cleaning apparatus of clause 1 wherein the first brush rotates at a first speed, the second brush rotates at a second speed, and the first speed is less than the second speed.
- [0421] 8. The surface cleaning apparatus of clause 7 wherein the first brush has a larger diameter than the second brush.
- [0422] 9. The surface cleaning apparatus of clause 1 wherein each of the first and second brushes are located at a front end of the surface cleaning head.
- [0423] 10. The surface cleaning apparatus of clause 9 wherein the first brush has a larger diameter than the second brush.
- [0424] 11. The surface cleaning apparatus of clause 10 wherein the first brush rotates clockwise and the second brush rotates counter clockwise.
- [0425] 12. The surface cleaning apparatus of clause 11 wherein, when the surface cleaning head is moved in a forward direction, the first brush rotates and the second brush is not driven to rotate and, when the surface cleaning head is moved in a rearward direction, the second brush rotates and the first brush is not driven to rotate.
- [0426] 13. The surface cleaning apparatus of clause 11 wherein, when the surface cleaning head is moved in a

forward direction on a surface, the first brush rotates and the second brush is disengaged from the surface.

[0427] 14. The surface cleaning apparatus of clause 13 wherein, when the surface cleaning head is moved in a rearward direction, the second brush rotates and the first brush is disengaged from the surface.

[0428] 15. The surface cleaning apparatus of clause 1 wherein the first brush is located at a front end of the surface cleaning head and the second brush is located at a rear end of the surface cleaning head.

[0429] 16. The surface cleaning apparatus of clause 15 wherein one of the first and second brushes has a larger diameter than the other of the first and second brushes.

[0430] 17. The surface cleaning apparatus of clause 15 wherein the first brush has a larger diameter than the other of the second brush.

[0431] 18. The surface cleaning apparatus of clause 16 wherein the first brush rotates clockwise and the second brush rotates counter clockwise.

[0432] 19. The surface cleaning apparatus of clause 15 wherein, when the surface cleaning head is moved in a forward direction, the first brush rotates and the second brush is not driven to rotate and, when the surface cleaning head is moved in a rearward direction, the second brush rotates and the first brush is not driven to rotate.

[0433] 20. The surface cleaning apparatus of clause 15 wherein, when the surface cleaning head is moved in a forward direction on a surface, the first brush rotates and the second brush is disengaged from the surface and, when the surface cleaning head is moved in a rearward direction, the second brush rotates and the first brush is disengaged from the surface.

[0434] 21. The surface cleaning apparatus of clause 15 wherein the first brush has a larger diameter than the other of the second brush and, when the surface cleaning head is moved in a forward direction on a surface, the first brush rotates and the second brush is disengaged from the surface.

Clause Set B

[0435] 1. A surface cleaning apparatus comprising a surface cleaning head, the surface cleaning head comprising:

[0436] (a) a first brush which is located at a front end of the surface cleaning head and wherein the first brush is rotatable in one of a clockwise and counterclockwise direction; and,

[0437] (b) a second brush which is located at a rear end of the surface cleaning head and wherein the second brush is rotatable in the other of the clockwise and counterclockwise direction.

[0438] 2. The surface cleaning apparatus of clause 1 further comprising a first path from the first brush to a large particulate collection area and a second path from the first brush to a small particulate collection area.

[0439] 3. The surface cleaning apparatus of clause 1 wherein the large particulate collection area and the small particulate collection area are a common volume that receive particulate matter from each of the brushes.

[0440] 4. The surface cleaning apparatus of clause 1 wherein the large particulate collection area and the small particulate collection area are discrete collection volumes.

[0441] 5. The surface cleaning apparatus of clause 4 wherein the large particulate collection area and the small particulate collection area are concurrently removable from the surface cleaning apparatus.

[0442] 6. The surface cleaning apparatus of clause 4 wherein the large particulate collection area and the small particulate collection area are separately removable from the surface cleaning apparatus.

[0443] 7. The surface cleaning apparatus of clause 1 wherein each of the first and second brushes are operable at a common speed.

[0444] 8. The surface cleaning apparatus of clause 1 wherein the first brush rotates at a first speed, the second brush rotates at a second speed, and the first speed is less than the second speed.

[0445] 9. The surface cleaning apparatus of clause 7 wherein the first brush has a larger diameter than the second brush.

[0446] 10. The surface cleaning apparatus of clause 1 wherein, when the surface cleaning head is moved in a forward direction, the first brush rotates and the second brush is not driven to rotate and, when the surface cleaning head is moved in a rearward direction, the second brush rotates and the first brush is not driven to rotate.

[0447] 11. The surface cleaning apparatus of clause 1 wherein, when the surface cleaning head is moved in a forward direction on a surface, the first brush rotates and the second brush is disengaged from the surface and, when the surface cleaning head is moved in a rearward direction, the second brush rotates.

[0448] 12. The surface cleaning apparatus of clause 1 further comprising a brush motor assembly which, in a first operating mode, rotates the first brush in a first direction when the surface cleaning head is moved forwardly and which, in a second operating mode, rotates the second brush in an opposite direction when the surface cleaning head is moved rearwardly.

[0449] 13. The surface cleaning apparatus of clause 12 wherein the brush motor assembly comprises a first brush motor operatively connected to the first brush and a second brush motor operatively connected to the second brush.

[0450] 14. The surface cleaning apparatus of clause 13 wherein, when the surface cleaning head is moved forwardly, the first brush motor is energized and the second brush motor is deenergized and when the surface cleaning head is moved rearwardly, the second brush motor is energized and the first brush motor is deenergized.

[0451] 15. The surface cleaning apparatus of clause 12 further comprising a drive handle that is rotatable from a first position to a second position wherein, when the handle is in the first position, the handle is positioned rearward of the rear end of the surface cleaning head whereby the handle is operable by a user to move the surface cleaning head forwardly, and when the handle is in the second position, the handle is positioned forward of the front end of the surface cleaning head whereby the handle is operable by a user to move the surface cleaning head rearwardly, and movement of the handle adjusts the brush motor assembly between the first and second modes.

[0452] 16. The surface cleaning apparatus of clause 1 further comprising wheels and a clutch assembly and the wheels are drivingly connected to the first and second brushes by a mechanical linkage that includes the clutch assembly wherein, when the surface cleaning head is moved forwardly, the clutch assembly is in a first position and the first brush rotates in the one of the one of a clockwise and counterclockwise direction and, when the surface cleaning head is moved rearwardly, the clutch assembly is in a second

position and the second brush rotates in the other of the clockwise and counterclockwise direction.

[0453] 17. The surface cleaning apparatus of clause 16 further comprising a drive handle that is rotatable from a first position to a second position wherein, when the handle is in the first position, the handle is positioned rearward of the rear end of the surface cleaning head whereby the handle is operable by a user to move the surface cleaning head forwardly, and when the handle is in the second position, the handle is positioned forward of the front end of the surface cleaning head whereby the handle is operable by a user to move the surface cleaning head rearwardly, and movement of the handle moves the clutch assembly between the first and second positions.

[0454] 18. The surface cleaning apparatus of clause 17 wherein the clutch assembly comprises a first clutch operatively connected to the first brush and a second clutch operatively connected to the second brush.

[0455] 19. The surface cleaning apparatus of clause 16 wherein a change in direction of rotation of the wheels moves the clutch between the first and second positions.

[0456] 20. The surface cleaning apparatus of clause 1 wherein a user-controlled actuator is operable to selectively control the direction of rotation of the first brush and the second brush.

[0457] 21. The surface cleaning apparatus of clause 1 wherein, when the surface cleaning head is moved in a forward direction on a surface, the first brush rotates and the second brush is disengaged from the surface and, when the surface cleaning head is moved in a rearward direction, the second brush rotates and the first brush is disengaged from the surface.

[0458] 22. The surface cleaning apparatus of clause 1 wherein one of the first brush and the second brush has a larger diameter than the other of the first brush and the second brush, and when the surface cleaning head is moved in a direction in which the larger of the first and second brushes is a leading brush, the larger of the first and second brushes rotates and the second brush is disengaged from the surface and, when the surface cleaning head is moved in a rearward direction, the second brush rotates and the first brush is disengaged from the surface.

[0459] 23. A surface cleaning apparatus comprising a surface cleaning head having a first end and a second opposed end, the surface cleaning head comprising:

[0460] (a) a first brush which is located at a first end of the surface cleaning head;

[0461] (b) a second brush which is located at a second opposed end of the surface cleaning head; and,

[0462] (c) a drive handle that is rotatable from a first position to a second position wherein, when the handle is in the first position, the handle is positioned outward of the first end of the surface cleaning head whereby the handle is operable by a user to move the surface cleaning head in a first direction, and when the handle is in the second position, the handle is positioned outward of the second end of the surface cleaning head whereby the handle is operable by a user to move the surface cleaning head a second opposed direction,

[0463] wherein the first brush has a larger diameter than the second brush, and

[0464] wherein, when the surface cleaning head is moved in the first direction on a surface, the first brush rotates and the second brush is disengaged from the

surface and, when the surface cleaning head is moved in the second direction, the second brush rotates, and

[0465] wherein movement of the handle from the second position to the first position disengages the second brush from the surface.

[0466] 24. The surface cleaning apparatus of clause 23 wherein the first brush is rotatable in a clockwise direction and the second brush is rotatable in a counterclockwise direction.

Clause Set C

[0467] 1. A surface cleaning apparatus comprising a surface cleaning head having a first end and a second opposed end, the surface cleaning head comprising:

[0468] (a) a first brush which is located at a first end of the surface cleaning head and which has a first diameter;

[0469] (b) a second brush which is located at a second opposed end of the surface cleaning head and has a second diameter, wherein the diameter of the first brush is larger than the diameter of the second brush; and,

[0470] (c) a drive handle that is rotatable from a first position to a second position wherein, when the handle is in the first position, the handle is positioned outward of the first end of the surface cleaning head whereby the handle is operable by a user to move the surface cleaning head in a first direction, and when the handle is in the second position, the handle is positioned outward of the second end of the surface cleaning head whereby the handle is operable by a user to move the surface cleaning head a second opposed direction,

[0471] wherein, when the surface cleaning head is moved in the first direction on a surface, the first brush rotates and the second brush is inoperable to clean the surface and, when the surface cleaning head is moved in the second direction, the second brush rotates, and

[0472] wherein movement of the handle from the second position to the first position renders the second brush inoperable.

[0473] 2. The surface cleaning apparatus of clause 1 wherein the first brush is rotatable in a clockwise direction and the second brush is rotatable in a counterclockwise direction.

[0474] 3. The surface cleaning apparatus of clause 1 wherein, when the surface cleaning head is moved in the second direction, the first brush is inoperable to clean the surface.

[0475] 4. The surface cleaning apparatus of clause 1 wherein movement of the handle from the second position to the first position disengages the second brush from the surface.

[0476] 5. The surface cleaning apparatus of clause 1 wherein, when the surface cleaning head is moved in the second direction and the handle is moved from the first position to the second position the first brush is disengaged from the surface whereby the first brush is inoperable to clean the surface.

[0477] 6. The surface cleaning apparatus of clause 1 further comprising wheels and a clutch assembly, and the wheels are drivingly connected to the first and second brushes by a mechanical linkage that includes the clutch assembly wherein, when the surface cleaning head is moved in the first direction, the clutch assembly is in a first position and the wheels are drivingly connected to the first brush and

not the second brush whereby the first brush rotates, and when the surface cleaning head is moved in the second direction, the clutch assembly is in a second position and the wheels are drivingly connected to the second brush whereby the second brush rotates, and movement of the handle moves the clutch assembly between the first and second positions.

[0478] 7. The surface cleaning apparatus of clause 6 wherein, when the clutch assembly is in a second position, the wheels are not drivingly connected to the first brush.

[0479] 8. The surface cleaning apparatus of clause 6 wherein the clutch assembly comprises a first clutch operatively connected to the first brush and a second clutch operatively connected to the second brush.

[0480] 9. The surface cleaning apparatus of clause 1 further comprising a brush motor assembly which, in a first operating mode, rotates the first brush when the surface cleaning head is moved in the first direction and which, in a second operating mode, rotates the second brush when the surface cleaning head is moved in the second direction, and movement of the handle adjusts the brush motor assembly between the first and second modes.

[0481] 10. The surface cleaning apparatus of clause 9 wherein the brush motor assembly comprises a first brush motor operatively connected to the first brush and a second brush motor operatively connected to the second brush.

[0482] 11. The surface cleaning apparatus of clause 10 wherein, when the surface cleaning head is moved in the first direction, the first brush motor is energized and the second brush motor is deenergized and when the surface cleaning head is moved in the second direction, the second brush motor is energized and the first brush motor is deenergized.

[0483] 12. The surface cleaning apparatus of clause 1 wherein, when the surface cleaning head is moved in the first direction on a surface, the second brush is disengaged from the surface.

[0484] 13. The surface cleaning apparatus of clause 12 wherein, when the surface cleaning head is moved in the second direction, the first brush is disengaged from the surface.

[0485] 14. The surface cleaning apparatus of clause 12 further comprising a lift off mechanism that is moveable between a lowered position in which the second brush engages the surface and a raised position in which the second brush is disengaged from the surface, and movement of the handle between the first and second positions moves the lift off mechanism between the lowered and raised positions.

[0486] 15. A surface cleaning apparatus comprising a surface cleaning head having a first end, a second opposed end and wheels, the surface cleaning head comprising:

[0487] (a) a first brush which is located at a first end of the surface cleaning head and which has a first diameter;

[0488] (b) a second brush which is located at a second opposed end of the surface cleaning head and has a second diameter, wherein the diameter of the first brush is larger than the diameter of the second brush; and,

[0489] (c) a drive handle that is rotatable from a first position in which that handle is operable by a user to move the surface cleaning head in a first direction and a second position in which the handle is operable by a user to move the surface cleaning head a second opposed direction,

- [0490] wherein, when the surface cleaning head is moved in the first direction on a surface, the first brush rotates and the second brush is inoperable to clean the surface and, when the surface cleaning head is moved in the second direction, the second brush rotates, and
- [0491] wherein a change in direction of rotation of the wheels renders the second brush inoperable.
- [0492] 16. The surface cleaning apparatus of clause 15 further comprising a clutch assembly, and the wheels are drivingly connected to the first and second brushes by a mechanical linkage that includes the clutch assembly wherein, when the surface cleaning head is moved in the first direction, the clutch assembly is in a first position and the wheels are drivingly connected to the first brush and not the second brush whereby the first brush rotates, and when the surface cleaning head is moved in the second direction, the clutch assembly is in a second position and the wheels are drivingly connected to the second brush whereby the second brush rotates, and a change in direction of rotation of the wheels moves the clutch between the first and second positions.
- [0493] 17. The surface cleaning apparatus of clause 16 wherein the clutch assembly comprises a first clutch operatively connected to the first brush and a second clutch operatively connected to the second brush.
- [0494] 18. The surface cleaning apparatus of clause 15 further comprising a brush motor assembly which, in a first operating mode, rotates the first brush when the surface cleaning head is moved in the first direction and which, in a second operating mode, rotates the second brush when the surface cleaning head is moved in the second direction, and a change in direction of rotation of the wheels adjusts the brush motor assembly between the first and second modes.
- [0495] 19. The surface cleaning apparatus of clause 18 wherein the brush motor assembly comprises a first brush motor operatively connected to the first brush and a second brush motor operatively connected to the second brush.
- [0496] 20. The surface cleaning apparatus of clause 15 wherein the first brush is rotatable in a clockwise direction and the second brush is rotatable in a counterclockwise direction.

Clause Set D

- [0497] 1. A surface cleaning apparatus comprising a surface cleaning head, the surface cleaning head comprising:
- [0498] (a) a first brush which is rotatable about a first rotational axis in a clockwise direction, wherein the first brush has a first core and a plurality of first bristles which extend outwardly from the first core and at least some of the bristles are curved in the clockwise direction; and,
- [0499] (b) a second brush which is rotatable about a second rotational axis in a counterclockwise direction, wherein the second brush has a second core and a plurality of second bristles which extend outwardly from the second core and at least some of the bristles are curved in the counterclockwise direction,
- [0500] wherein, when the surface cleaning head is placed on a horizontal surface and moved in a forward direction with the brush in position to clean the surface, the rotational axis extends generally horizontally and generally transverse to the forward direction.
- [0501] 2. The surface cleaning apparatus of clause 1 wherein the first bristles comprise individual resilient bristles.
- [0502] 3. The surface cleaning apparatus of clause 1 wherein the first bristles are curved from a radial inner end of the first bristles at a location adjacent the first core to a radial outer end of the first bristles and the second bristles are curved from a radial inner end of the second bristles at a location adjacent the second core to a radial outer end of the second bristles.
- [0503] 4. The surface cleaning apparatus of clause 1 wherein the first bristles have a radial inner portion extending outwardly from the first core and a radial outer portion extending outwardly from the radial inner portion and the radial inner portion is generally linear and the radial outer portion is generally curved.
- [0504] 5. The surface cleaning apparatus of clause 1 wherein the second bristles have a radial inner portion extending outwardly from the second core and a radial outer portion extending outwardly from the radial inner portion and the radial inner portion is generally linear and the radial outer portion is generally curved.
- [0505] 6. The surface cleaning apparatus of clause 5 wherein the first bristles have a radial inner portion extending outwardly from the first core and a radial outer portion extending outwardly from the radial inner portion of the first bristles and the radial inner portion of the first bristles is generally linear and the radial outer portion of the first bristles is generally curved.
- [0506] 7. The surface cleaning apparatus of clause 1 wherein the first brush is located at a first end of the surface cleaning head and the second brush is located at a second opposed end of the surface cleaning head.
- [0507] 8. A surface cleaning apparatus comprising a surface cleaning head, the surface cleaning head comprising:
- [0508] (a) a first brush which is rotatable about a first rotational axis in a clockwise direction, wherein the first brush has a first core and a plurality of first bristles which extend outwardly from the first core and at least some of the bristles are curved in the clockwise direction; and,
- [0509] (b) a second brush which is rotatable about a second rotational axis in a counterclockwise direction, wherein the second brush has a second core and a plurality of second bristles which extend outwardly from the second core,
- [0510] wherein, when the surface cleaning head is placed on a horizontal surface and moved in a forward direction with the brush in position to clean the surface, the rotational axis extends generally horizontally and generally transverse to the forward direction.
- [0511] 9. The surface cleaning apparatus of clause 8 wherein the first bristles comprise individual resilient bristles.
- [0512] 10. The surface cleaning apparatus of clause 8 wherein the first bristles are curved from a radial inner end of the first bristles at a location adjacent the first core to a radial outer end of the first bristles.
- [0513] 11. The surface cleaning apparatus of clause 8 wherein the first bristles have a radial inner portion extending outwardly from the first core and a radial outer portion extending outwardly from the radial inner portion and the radial inner portion is generally linear and the radial outer portion is generally curved.

[0514] 12. The surface cleaning apparatus of clause 8 wherein the first brush is located at a first end of the surface cleaning head and the second brush is located at a second opposed end of the surface cleaning head.

[0515] 13. A surface cleaning apparatus comprising a surface cleaning head, the surface cleaning head comprising:

[0516] (a) a first brush which is rotatable about a first rotational axis in a clockwise direction, wherein the first brush has a first core and a plurality of first bristles which extend outwardly from the first core; and,

[0517] (b) a second brush which is rotatable about a second rotational axis in a counterclockwise direction, wherein the second brush has a second core and a plurality of second bristles which extend outwardly from the core and at least some of the bristles are curved in the counterclockwise direction,

[0518] wherein, when the surface cleaning head is placed on a horizontal surface and moved in a forward direction with the brush in position to clean the surface, the rotational axis extends generally horizontally and generally transverse to the forward direction.

[0519] 14. The surface cleaning apparatus of clause 13 wherein the second bristles comprise individual resilient bristles.

[0520] 15. The surface cleaning apparatus of clause 13 wherein the second bristles are curved from a radial inner end of the second bristles at a location adjacent the second core to a radial outer end of the second bristles.

[0521] 16. The surface cleaning apparatus of clause 13 wherein the second bristles have a radial inner portion extending outwardly from the second core and a radial outer portion extending outwardly from the radial inner portion and the radial inner portion is generally linear and the radial outer portion is generally curved.

[0522] 17. The surface cleaning apparatus of clause 13 wherein the first brush is located at a first end of the surface cleaning head and the second brush is located at a second opposed end of the surface cleaning head.

Clause Set E

[0523] 1. surface cleaning apparatus comprising a surface cleaning head having a first end, a second end and a brush assembly wherein, in a first mode of operation, the brush assembly rotates in one of a clockwise and counterclockwise direction and directs large particulate matter to a large particulate collection area and wherein, in a second mode of operate, the brush assembly rotates in the other of the clockwise and the counterclockwise direction and directs small particulate matter to a small particulate collection area that is discrete from the large particulate collection area wherein, when the surface cleaning head is placed on a horizontal surface and moved in a first direction with the brush assembly in position to clean the surface, a rotational axis of the brush assembly extends generally horizontally and generally transverse to the first direction.

[0524] 2. The surface cleaning apparatus of clause 1 wherein the small particulate collection area is exterior to a volume of the large particulate collection area.

[0525] 3. The surface cleaning apparatus of clause 1 wherein the large particulate collection area and the small particulate collection area are concurrently removable from the surface cleaning apparatus.

[0526] 4. The surface cleaning apparatus of clause 1 wherein the large particulate collection area and the small particulate collection area are separately removable from the surface cleaning apparatus.

[0527] 5. The surface cleaning apparatus of clause 1 wherein, when the surface cleaning head is placed on a horizontal surface with the brush assembly in position to clean the surface, and with the large particulate collection area and the small particulate collection area in position in the surface cleaning head, an inlet to the large particulate collection area is located at a higher elevation than an inlet to the small particulate collection area.

[0528] 6. The surface cleaning apparatus of clause 5 wherein the large particulate collection area and the small particulate collection area comprise a one-piece assembly.

[0529] 7. The surface cleaning apparatus of clause 1 wherein, when the surface cleaning head is placed on a horizontal surface with the brush assembly in position to clean the surface, and with the large particulate collection area and the small particulate collection area in position in the surface cleaning head, an inlet to the large particulate collection area is provided on a first side of the large particulate collection area and faces the first direction and an inlet to the small particulate collection area is provided on a second side of the small particulate collection area and faces the second direction.

[0530] 8. The surface cleaning apparatus of clause 7 wherein, when the surface cleaning head is placed on a horizontal surface with the brush assembly in position to clean the surface, and with the large particulate collection area and the small particulate collection area in position in the surface cleaning head, an inlet to the large particulate collection area is located at a higher elevation than an inlet to the small particulate collection area.

[0531] 9. The surface cleaning apparatus of clause 7 wherein the large particulate collection area and the small particulate collection area comprise a one-piece assembly.

[0532] 10. The surface cleaning apparatus of clause 8 wherein the large particulate collection area and the small particulate collection area comprise a one-piece assembly.

[0533] 11. The surface cleaning apparatus of clause 1 wherein the brush assembly comprises a first brush which is rotatable about a first rotational axis in a clockwise direction and which is provided at the first end, and a second brush which is rotatable about a second rotational axis in a counterclockwise direction and which is provided at the second end.

[0534] 12. The surface cleaning apparatus of clause 1 wherein the large particulate collection area and the small particulate collection area comprise a collection assembly that is positioned between the first and second brushes.

[0535] 13. The surface cleaning apparatus of clause 11 wherein the first brush rotates at a first speed, the second brush rotates at a second speed, and the first speed is less than the second speed.

[0536] 14. The surface cleaning apparatus of clause 13 wherein the first brush has a larger diameter than the second brush.

[0537] 15. The surface cleaning apparatus of clause 1 wherein the brush assembly comprises a first brush which is rotatable about a first rotational axis in a clockwise direction and a second brush which is rotatable about a second

rotational axis in a counterclockwise direction and each of the first and second brushes are located at the first end of the surface cleaning head.

[0538] 16. The surface cleaning apparatus of clause 15 wherein the first brush has a larger diameter than the second brush.

[0539] 17. The surface cleaning apparatus of clause 1 wherein the small particulate collection area has a closure member which is moveable between an open position in which an inlet to the small particulate collection area is open and a closed position in which the inlet is closed.

[0540] 18. The surface cleaning apparatus of clause 17 wherein the small particulate collection area is removable from the surface cleaning apparatus, and the closure member is moved to the closed position upon removal of the small particulate collection area from the surface cleaning apparatus.

[0541] 19. The surface cleaning apparatus of clause 1 wherein, when the surface cleaning head is placed on a horizontal surface with the brush assembly in position to clean the surface, the small particulate collection area has a lid and the lid is openable.

[0542] 20. The surface cleaning apparatus of clause 19 wherein the small particulate collection area is removable from the surface cleaning apparatus, and the lid is openable after the small particulate collection area is removed from the surface cleaning apparatus.

Clause Set F

[0543] 1. A surface cleaning apparatus comprising a surface cleaning head having a first end, a second end and a brush assembly wherein, in a first mode of operation, the brush assembly rotates in a clockwise direction and in a second mode of operation, the brush assembly rotates in a counterclockwise direction wherein, when the surface cleaning head is placed on a horizontal surface and moved in a first direction with the brush assembly in position to clean the surface, a rotational axis of the brush assembly extends generally horizontally and generally transverse to the first direction and wherein the collection assembly has a volume that is expandable and receives particulate matter in at least one of the first and second modes of operation.

[0544] 2. The surface cleaning apparatus of clause 1 wherein the collection assembly comprises a collection container that is removably mounted to the surface cleaning apparatus.

[0545] 3. The surface cleaning apparatus of clause 2 wherein the collection container is a flexible bag and the collection assembly further comprises a bag receiver.

[0546] 4. The surface cleaning apparatus of clause 1 wherein the brush assembly comprises a first brush which is rotatable about a first rotational axis in the clockwise direction and which is provided at the first end, and a second brush which is rotatable about a second rotational axis in the counterclockwise direction and which is provided at the second end.

[0547] 5. The surface cleaning apparatus of clause 4 wherein the collection assembly is positioned between the first and second brushes.

[0548] 6. The surface cleaning apparatus of clause 4 wherein the first brush rotates at a first speed, the second brush rotates at a second speed, and the first speed is less than the second speed.

[0549] 7. The surface cleaning apparatus of clause 6 wherein the first brush has a larger diameter than the second brush.

[0550] 8. The surface cleaning apparatus of clause 1 wherein the brush assembly comprises a first brush which is rotatable about a first rotational axis in the clockwise direction and a second brush which is rotatable about a second rotational axis in the counterclockwise direction and each of the first and second brushes are located at the first end of the surface cleaning head.

[0551] 9. The surface cleaning apparatus of clause 8 wherein the first brush rotates at a first speed, the second brush rotates at a second speed, and the first speed is less than the second speed.

[0552] 10. The surface cleaning apparatus of clause 9 wherein the first brush has a larger diameter than the second brush.

[0553] 11. The surface cleaning apparatus of clause 1 wherein the collection assembly comprises a large particulate collection area and a small particulate collection area, and the large particulate collection area and the small particulate collection area are discrete collection volumes.

[0554] 12. The surface cleaning apparatus of clause 11 wherein the small particulate collection area is provided in the large particulate collection area.

[0555] 13. The surface cleaning apparatus of clause 11 further comprising a first path from the first brush to a large particulate collection area and a second path from the first brush to a small particulate collection area.

[0556] 14. The surface cleaning apparatus of clause 13 wherein the first brush rotates at a first speed, the second brush rotates at a second speed, and the first speed is less than the second speed.

[0557] 15. The surface cleaning apparatus of clause 14 wherein the first brush has a larger diameter than the second brush.

1. A surface cleaning apparatus comprising a surface cleaning head having a front end, a rear end and a brush that is rotatably mounted about a rotation axis wherein, when the surface cleaning head is placed on a horizontal surface and moved in a forward direction with the brush in position to clean the surface, the rotational axis extends generally horizontally and generally transverse to the forward direction, wherein the brush is rotatable in a clockwise direction and wherein the brush is also rotatable in a counterclockwise direction.

2. The surface cleaning apparatus of claim 1 further comprising:

- (a) a first path from the brush to a large particulate collection area wherein, when the brush rotates in one of the clockwise and counterclockwise directions, the brush conveys large particulate matter towards the large particulate collection area; and,
- (b) a second path from the brush to a small particulate collection area wherein, when the brush rotates in the other of the clockwise and counterclockwise directions, the brush conveys small particulate matter towards the small particulate collection area.

3. The surface cleaning apparatus of claim 2 wherein, when the brush rotates in the clockwise direction, the brush conveys large particulate matter towards the large particulate collection area and, when the brush rotates in the counterclockwise direction, the brush conveys small particulate matter towards the small particulate collection area.

4. The surface cleaning apparatus of claim 3 wherein the large particulate collection area and the small particulate collection area are positioned rearward of the brush and the second path comprises a ramp extending rearwardly to the small particulate collection area.

5. The surface cleaning apparatus of claim 1 further comprising:

- (a) a first path from the brush to a collection area wherein, when the brush rotates in one of the clockwise and counter clockwise directions, the brush conveys large particulate matter towards the collection area; and,
- (b) a second path from the brush to the collection area wherein, when the brush rotates in the other of the clockwise and counter clockwise directions, the brush conveys small particulate matter towards the collection area.

6. The surface cleaning apparatus of claim 5 wherein, when the brush rotates in the clockwise direction, the brush conveys large particulate matter towards the collection area and, when the brush rotates in the counter clockwise direction, the brush conveys small particulate matter towards the collection area.

7. The surface cleaning apparatus of claim 6 wherein the large particulate collection area and the small particulate collection area are positioned rearward of the brush and the second path comprises a ramp extending rearwardly to the collection area.

8. The surface cleaning apparatus of claim 1 wherein the brush rotates at a first speed in one of the clockwise and counter clockwise directions and the brush rotates at a second speed in the other of the clockwise and counter clockwise directions and the first speed is less than the second speed.

9. The surface cleaning apparatus of claim 1 wherein, when the brush rotates in the clockwise direction, the brush rotates at a first speed and, when the brush rotates in the counter clockwise direction, the brush rotates at a second speed and the first speed is less than the second speed.

10. The surface cleaning apparatus of claim 1 wherein the brush rotates in one of the clockwise and counter clockwise directions when the surface cleaning head is moved forwardly and the brush rotates in the other of the clockwise and counter clockwise directions when the surface cleaning head is moved rearwardly.

11. The surface cleaning apparatus of claim 10 further comprising a brush motor, the brush motor rotates in a first direction when the surface cleaning head is moved forwardly and the brush motor automatically rotates in an opposite direction when the surface cleaning head is moved rearwardly.

12. The surface cleaning apparatus of claim 10 further comprising wheels, the brush is drivably connected to at least one of the wheels by a mechanical linkage and the mechanical linkage comprises a clutch wherein, when the surface cleaning head is moved forwardly, the clutch is in a first position and the brush motor rotates in a first direction and, when the surface cleaning head is moved rearwardly, the clutch is in a second position and the brush motor rotates in an opposite direction.

13. The surface cleaning apparatus of claim 12 further comprising a drive handle that is rotatable from a first position to a second position wherein, when the handle is in the first position, the handle is positioned rearward of the rear end of the surface cleaning head whereby the handle is operable by a user to move the surface cleaning head forwardly, and when the handle is in the second position, the handle is positioned forward of the front end of the surface cleaning head whereby the handle is operable by a user to move the surface cleaning head rearwardly, and movement of the handle moves the clutch between the first and second positions.

14. The surface cleaning apparatus of claim 12 wherein a change in direction of rotation of the wheels moves the clutch between the first and second positions.

15. The surface cleaning apparatus of claim 1 wherein a user-controlled actuator is operable to selectively control the direction of rotation of the brush.

16. The surface cleaning apparatus of claim 1 further comprising a suction motor and a suction slot is located rearwardly of the brush.

17. The surface cleaning apparatus of claim 1 further comprising a collection container that is removably mounted to the surface cleaning apparatus.

18. The surface cleaning apparatus of claim 1 further comprising a collection container that has a volume that is expandable.

19. The surface cleaning apparatus of claim 17 wherein the collection container is a flexible bag and the surface cleaning apparatus further comprises a bag receiver.

20. The surface cleaning apparatus of claim 1 further comprising a cleaning module that is removably attachable to the surface cleaning apparatus, the cleaning module comprising a liquid reservoir wherein, when the cleaning module is mounted to the surface cleaning apparatus, the surface cleaning apparatus is operable to direct a cleaning fluid at a surface to be cleaned.

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